



SCC

SCAP Compliance Checker
Version 5.12 User Manual
for Microsoft Windows

**Naval Information
Warfare Center**



ATLANTIC

Developed by:
NIWC Atlantic (formerly SPAWAR Atlantic)
P.O. Box 190022
North Charleston, SC 29419-9022
scc.fct@navy.mil
<https://www.niwcatlantic.navy.mil/scap/>

Distribution A: Approved for Public Release (16 Feb 2021) NIWCLANT SPR# 2021-57

Table of Contents

1. Introduction	1
1.1 Background	1
1.2 Platforms Supported	2
1.3 SCAP Content Included	2
1.4 Changelog	4
2. Requirements	9
2.1 Minimum Hardware/OS Requirements	9
2.2 Scanning Requirements	9
3. Install/uninstall	11
3.1 Windows Software Installation	11
3.2 Windows Software Uninstall	13
3.3 Install/Remove software via zip file	13
3.4 Install Details	14
4. GUI Based Usage	16
4.1 Launching the Graphical User Interface	16
4.2 Installing & Configuring Content	17
4.3 Performing a Scan	22
4.4 Editing Options	27
4.5 Viewing Results	41
4.6 SCAP Scanning with Manual and/or Hybrid Questions	42
4.7 XCCDF Tailoring	46
4.9 Cisco IOS / IOS XE Offline (Show Tech) File Scanning	48
4.10 Standalone OVAL Usage	50
4.11 Standalone OCIL Usage	51
4.12 Post Scanning Report Generation	53
4.13 File and Help Menus	56
4.13 Configuration Profiles	57
5. Command Line Usage	59
5.1 Basic Command Line Usage	59
5.2 Command Line Configuration Parameters	60
5.3 Command Line Scanning Parameters	66
5.4 Option Descriptions and Datatypes	70
5.5 Generating Post Scan Reports from the Command Line	81
5.6 Multiple Computer Deployment	85
5.7 Manual and Hybrid Question Answer Files	86
6. Hybrid Tests	89
6.1 Hybrid Scope	90
6.2 Hybrid Target	91
6.3 Hybrid Value(s)	91
6.4 XML Example using target 'ALL'	92
6.5 More Complex XML Examples	92
6.6 What happens if I do not enter any data into Hybrid tests?	97
7. Understanding Scan Results	98
7.1 Understanding Scan Reports	98
7.2 Navigating the Results Directory	101
7.3 Viewing Screen, Error or Debug Logs	103
8. Running SCC as a Service	105
8.1 Installing the SCC Service	105
8.2 Configuring the SCC Service	105
8.3 Update Service Configuration	106
9. Running SCC as a non-Administrator	107
9.1 Installing SCC	107
9.2 Install directory non-writable by non-Administrators	107

9.3 GUI usage as non-administrator.....	107
9.4 Command line usage as non-administrator.....	107
9.5 Uninstalling.....	108
10. Windows Update Agent (WUA) Tests	109
10.1 How does SCC perform WUA tests?	109
APPENDIX A - FREQUENTLY ASKED QUESTIONS.....	110
A.1 Why can't I install a DISA STIG Manual XCCDF into SCC?	110
A.2 How can I scan CENTOS Linux, Rocky Linux, Alma Linux, Debian Linux etc. with an existing SCAP benchmark?	110
A.3 I just installed new/updated SCAP content from DISA, why am I no longer seeing Manual Questions?	110
A.4 Can I scan Linux/Solaris/Mac from Windows?	111
A.5 Is SCC officially SCAP validated?	111
A.6 Who should I report SCAP content issues to?	111
A.7 Why does NIWC's 'Enhanced' content have a different version from DISA SCAP content?	111
A.8 Does SCC provide any remediation functionality?	112
A.9 Where can I learn more about creating my own SCAP content?.....	112
A.10 Why is SCC default SCAP content and application updates URL's hosted on a .com?	112
A.11 Can I create my own offline SCAP content repository for my isolated network?.....	113
A.12 Can SCC run directly from a CD-ROM?.....	115
A.13 Can SCC be run as a non-Administrator or non-root user?	115
A.14 How can I use RunAs (Secondary Logon) with SCC?	115
A.15 What type of network traffic does a remote Windows SCC scan generate?	116
A.16 Where can I download the DOD Root Certificates for Windows (InstallRoot)?	116
A.17 Why does SCC review all cached profiles to confirm that User profile settings are accurate	116
A.18 How do I make local users compliant with user level GPOs?	117
APPENDIX B - KNOWN ISSUES.....	118
B.1 Potential out of memory crashes with very large OVAL XML content files	118
B.2 Unable to scan RHEL8 systems via SSH with OS application white listing enabled (SCC failed to launch).....	118
B.3 Host Key Check Failed when scanning RHEL7/8 and Ubuntu systems via SSH when changing between SCC 5.4 and 5.4.2 or later	118
B.4 Account lockout issues when scanning Ubuntu remotely via SSH with correct credentials	118
B.5 Command prompt closes when run with UAC.....	118
B.6 Maximum directory/file length error when copying SCC results.....	119
B.7 Reviewing Domain Controllers with numerous user accounts	119
B.8 Issues with WMI scanning if GPO is configured to block execution of all .bat files	120
B.9 Issues with remote scanning (Classic and WMI) if scanning PC is not fully patched	120
APPENDIX C - TROUBLESHOOTING	121
C.1 Troubleshooting UNIX SSH Remote Scanning.....	121
C.2 Troubleshooting Windows Remote Scanning	122
APPENDIX D – SCC AND SCAP.....	129
D.1 SCAP Validations & Capabilities	129
D.2 Standards Supported	129
D.3 SCAP Implementation	130
D.4 OVAL Probes Supported by SCC 5.12 for Windows	139
APPENDIX E - REFERENCES & DEFINITIONS.....	142
E.1 References	142
E.2 Definitions.....	143
APPENDIX F - LICENSES	146
F.1 End User License Agreement.....	146
APPENDIX G - TECHNICAL SUPPORT & Feedback	147
G.1 Technical Support	147

G.2 Tutorials.....	147
G.3 Software Releases	147
G.4 Provide Feedback on SCC.....	147
APPENDIX H - CREDITS AND FUNDING.....	148
H.1 Credits and Funding	148
H.2 SCC Funding for FY26 and Beyond.....	148

1. INTRODUCTION

The Security Content Automation Protocol (SCAP) Compliance Checker (SCC) is a SCAP 1.3 Validated Authenticated Configuration Scanner, with support for SCAP versions 1.0, 1.1, 1.2, 1.3 and 1.4, and an Open Vulnerability Assessment Language (OVAL) adopter, capable of performing compliance verification using SCAP content, and authenticated vulnerability scanning using OVAL content.

1.1 Background

1.1.1 About this Manual

This User Manual is intended to explain all of the features and functionality of the SCC application, along with some basic information regarding the SCAP standards. As SCC is used by thousands of people across hundreds of government agencies, a single Standard Operating Procedure (SOP) is not feasible. Each agency may need to create their own SOP based on their intended usage of SCC.

For DOD Usage, and integration with the Security Technical Implementation Guides (STIG) Viewer, please refer to DISA's (Defense Information Systems Agency) documentation, which is located at: <https://www.cyber.mil/stigs/srg-stig-tools/>

1.1.2 What is SCC?

SCC is an XML interpreter of SCAP content, meaning SCC performs automated security configuration checks based on the content that is installed and enabled. The end user can install SCAP content into SCC, and enable one or more SCAP content streams to perform compliance checking.

1.1.3 Who can use SCC?

Starting with SCC version 5.4, the application can be freely distributed and used by anyone. Refer to Appendix F for our legal disclaimer in our End User License Agreement.

1.1.4 Who funds the development and support of SCC?

Starting with FY23 SCC has been funded by groups of end users. See Appendix H for details about historical funding, current funding, and future funding efforts.

1.1.5 What is SCAP and SCAP Content?

At a very high level, SCAP is a set of XML standards, primarily XCCDF and OVAL, which include policy settings and technical instructions to perform automated checking.

SCAP Content is a collection of XML files, usually bundled in a zip file, which defines the checks to be evaluated on a target system or targeted systems. This bundle, or 'stream', instructs what checks to perform, provides all text fields such as titles, references, descriptions, and to some extent, how to perform them. SCAP validated scanners such as SCC ingest the stream and perform the checks listed therein.

1.2 Platforms Supported

- Microsoft Windows 10 (x64)
- Microsoft Windows 11 (x64)
- Microsoft Windows Server 2012 (x64)
- Microsoft Windows Server 2012 R2 (x64)
- Microsoft Windows Server 2016 (x64)
- Microsoft Windows Server 2019 (x64)
- Microsoft Windows Server 2022 (x64)
- Microsoft Windows Server 2025 (x64)
- Cisco IOS
- Cisco IOS XE
- Platforms supported by remote SSH based scanning (with the UNIX Remote Scanning Plugin installed)
 - Apple macOS
 - macOS Ventura/13 (arm64)
 - macOS Sonoma/14 (arm64)
 - macOS Sequoia/15 (arm64)
 - Red Hat Distributions
 - Red Hat Enterprise Linux 7 (x64)
 - Red Hat Enterprise Linux 8 (x64 & aarch64)
 - Red Hat Enterprise Linux 8 (x64 & aarch64)
 - Red Hat Enterprise Linux 9 (x64)
 - SUSE Linux
 - Enterprise SUSE Linux 12 (x64)
 - Enterprise SUSE Linux 15 (x64 & aarch64)
 - Oracle Linux
 - Oracle Linux 7 (x64)
 - Oracle Linux 8 (x64 & aarch64)
 - Oracle Linux 9 (x64)
 - Debian Distributions
 - Ubuntu 18.04 LTS (amd64)
 - Ubuntu 20.04 LTS (amd64& arm64)
 - Ubuntu 22.04 LTS (amd64)
 - Raspberry Pi OS (arm64)
 - Solaris 11 (x86 & SPARC)

Note 1: *There are separate SCC installers per architecture (Windows, Linux (RPM, DEB), Solaris, MacOS)*

Note 2: *'Supported' is defined as the application has been designed to run on the Operating System and architecture, and has been tested in our lab to execute as expected. Content may not be provided, but end users could obtain content from other sources, or write their own, and install and run in the application. See below for the list of content included in the installer.*

1.3 SCAP Content Included

1.3.1 SCAP Streams

- NIWC Manual Question "Enhanced" DISA STIG SCAP content obtained from:
<https://www.niwcatlantic.navy.mil/scap/scap-content-repository/>
 - Adobe Acrobat Reader DC Continuous
 - Google Chrome
 - Microsoft Edge
 - Internet Explorer 11
 - Mozilla Firefox
 - Microsoft Office 365 ProPlus
 - Microsoft .NET Framework
 - Windows Defender AV
 - Windows Firewall
 - Windows 2012 and 2012 R2 Domain Controller
 - Windows 2012 and 2012 R2 Member Server
 - Windows Server 2016
 - Windows Server 2019
 - Windows Server 2022
 - Windows 10
 - Windows 11
 - Cisco IOS XE Router RTR
 - Cisco IOS XE Router NTM
- NIWC Developed DISA STIG SCAP content obtained from:
<https://www.niwcatlantic.navy.mil/scap/scap-content-repository/>
 - Microsoft Internet Information Server (IIS) 8.5
 - Microsoft Internet Information Server (IIS) 10.0
 - Microsoft OneDrive
 - Microsoft DNS Server
 - Microsoft SQL Server 2016 (also supports 2017, 2019)
 - Microsoft SQL Server 2022

1.4 Changelog

Changelog for 5.12

Below is an abbreviated list of the primary changes from version 5.11 to 5.12. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Updated all content based on DISA Q3 2025 releases
 - Minor bug fixes
 - Updated internal dependencies
- Windows
 - Created SCAP content for Windows DNS Server
 - Created SCAP content for SQL Server 2022
- Linux
 - Added support for RHEL/Oracle Linux 9 on aarch64
- macOS
 - Added support for macOS Ventura/13 (M series CPU)
 - Added support for macOS Sonoma/14 (M series CPU)
 - Added support for macOS Sequoia/15 (M series CPU)
- Solaris
 - N/A

Changelog for 5.11

Below is an abbreviated list of the primary changes from version 5.10.2 to 5.11. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Added support for upcoming SCAP 1.4
 - Added support for OVAL 5.12 and OVAL 5.12.1
 - Added support for new shellcommand test
 - Added support for new 'merge' OVAL function
 - Digitally signed all content with a NIWC code signing CAC
 - Updated all content based on DISA Q2 2025 releases
 - Added support for Deviations
 - Added support for DISA's CKLB (JSON) formatted results
- Windows
 - Improved capabilities of OVAL wuaupdatesearcher (Windows Update Agent) tests
 - Added support for offline cab file, wsusscn2.cab (see section 10 for details)
 - Created SCAP benchmark for checking for missing updates
- Linux
 - N/A
- Solaris
 - N/A

Changelog for 5.10.2

Below is an abbreviated list of the primary changes from version 5.10.1 to 5.10.2. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Updated SCAP content based on DISA STIG Manuals from 2025 Quarter 1 release
 - Updated all internal dependencies.
 - Minor improvements to GUI to better handle large monitors with screen resolution > 3000 wide
- Windows
 - Fixed issues running on systems that have a different version of openssl installed to Windows\System32
- Linux
 - Added DISA SCAP content for Ubuntu 22.04
 - Dropped support for 32 bit Raspbian 11
- Solaris
 - N/A

Changelog for 5.10.1

Below is an abbreviated list of the primary changes from version 5.10 to 5.10.1. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Updated SCAP content based on DISA STIG Manuals from 2024 Quarter 4 release
 - Decreased default thresholds for pass/fail details to prevent large and slow to create reports and XCCDF
 - Fixed report generation slowdown when creating reports with numerous failed SQL records (MS SQL Server content)
 - Updated all internal dependencies.
- Windows
 - Fix issues scanning some SQL Server instances when Microsoft ODBC version 18 drivers are present
 - Update Windows 'ntuser' tests to exclude user profiles that have not be logged into via Windows Explorer
- Linux
 - Fixed SCC GUI launch issues on Ubuntu 20 aarch64
- Solaris
 - Fixed issues running SCC on Solaris 11 x86 due to missing dependency

Changelog for 5.10

Below is an abbreviated list of the primary changes from version 5.9 to 5.10. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Updated SCAP content based on DISA STIG Manuals from 2024 Quarter 3 release
 - Improved XCCDF and CKL results to be more concise and easier to understand when viewed with STIG Viewer, eMASS etc.
 - Added Result Analysis to HTML and Text reports to help explain why a specific check failed
 - Fixed Tailoring issue with remote SSH scanning
 - Updated all internal dependencies.
- Windows
 - Fix issues scanning some SQL Server instances when Microsoft ODBC version 18 drivers are present

- Fixed Cmdlet parameters that contain braces {} should not surround values with single quotes
- Linux
 - N/A
- Solaris
 - N/A

Changelog for 5.9

Below is an abbreviated list of the primary changes from version 5.8 to 5.9. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Add feature of "hybrid" checks which take end user provided answers and use as OVAL state for pass/fail of specific STIG rules
 - Only used with MS SQL Server content for SCC 5.9, may expand to other platforms in the future.
 - Refer to Section 6 of this user manual for full details
- Windows
 - Added support for SQL Server 2016 -> 2022
 - Created SCAP benchmarks from the SQL 2016 STIGS, applicable to SQL Server 2016 -> 2022
 - Removed support for 32 bit Windows 10
- Linux
 - Fixed 'type' issues with with selinuxsecuritycontext_tes
 - Added new debian_evr_string simple datatype
 - Removed support for RHEL6 and Ubuntu 16
- MacOS
 - Removed support for MacOS
- Solaris
 - Removed support for Solaris 10

Changelog for 5.8

Below is an abbreviated list of the primary changes from version 5.7.2 to 5.8 Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Several improvements to XCCDF Tailoring
 - Much faster load times on Tailoring form
 - Redesigned tailoring form for easier use
 - Added support for selecting different check systems, which works in combination with updated NIWC Enhanced SCAP content to allow any automated rule to be converted to a manual check, if automated check doesn't match end user requirements.
 - Added support for refine-values to allow any rule severity/weight to be changed based on end user requirements
 - New Configuration Profiles, which allows GUI and CLUI methods to quickly switch between sets of SCC configurations
 - Added new feature to make individual content streams be applicable to all targets, configurable via GUI and CLUI
 - Added new command line parameter to make SCC not update/save/modify it's options file during a scan

- Windows
 - Fixed Windows classic remote scan related to residual data for remote classic scans with trustees when scanning multiple computers
 - Fixed issues with Windows sid_object and filters against the trustee_sid
- Linux
 - Added support for Ubuntu 22.04 (AMD64)
 - Fixed reporting ipv6 network addresses
- MacOS
 - N/A
- Solaris
 - N/A

Changelog for 5.7.2

Below is an abbreviated list of the primary changes from version 5.7.1 to 5.7.2. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Fixed Manual Questions not being applied to certain targets via WMI, primarily when scanning by IP address only
 - Fixed issues with DNS lookup of ipv6 based targets for SSH based scanning
 - Improved error handling and error reporting on WMI and SSH scanning
 - Updated reports to group by automated/manual when sorting by severity
- Windows
 - Fixed application crash when scanning large numbers of computers via WMI, and many of the targets are offline.
 - Updated WMI scanning to only perform DNS lookup of targets when WMI Alternate Credentials are enabled.
- Linux
 - Fixed reporting ipv6 network addresses
- MacOS
 - Fix

Changelog for 5.7.1

Below is an abbreviated list of the primary changes from version 5.7 to 5.7.1. Please refer to the release notes for a complete list of updates.

- For All Platforms
 - Fixed data interoperability issue with SCC's CKL file and eMASS
 - Disabled the creation of XCCDF Results for Manual Questions rules, if Manual Question processing option is disabled.
 - Updated upgrade feature on Windows to upgrade any Manual Question results from a previous installation
 - Improved functionality of Manual Question GUI, adding searching and sorting
 - Minor improvements to Manual Question autoanswer text file templates
- Windows
 - Fixed issue running on Windows when no network cards have IP Addresses
- Linux
 - Fixed issue running GUI at 1024 x 768 resolution
- MacOS
 - Fixed minor issues with incorrectly identifying NIST content as containing manual checks.

Changelog for 5.7

Below is an abbreviated list of the primary changes from version 5.6 to 5.7. Please refer to the 5.7 release notes for a complete list of updates.

- For All Platforms
 - Updated all SCAP content bundled with SCC to be enhanced with manual checks, providing 100% coverage of DISA STIG Manuals, using SCAP standard of Open Checklist Interactive Language (OCIL)
 - Updated default SCAP content repository to NIWC's repository located at <https://www.niwcatlantic.navy.mil/scap/scap-content-repository/>
 - Updated SCC GUI to provide interface for manual questions, refer to section 4.6 for details
 - Added manual question autoanswer files for command line automation, refer to section 5.7 for details
 - Added creation of DISA's Checklist (CKL) report output, for SCAP content that has been enhanced with manual questions.
 - Improved error reporting in HTML/Text/XML
 - Improved result details in XCCDF XML results, which should be supported by DISA STIG Viewer in the future
 - Now officially SCAP 1.3 validated by an independent lab
 - Added funding popup on GUI and text based message on cscs --config as SCC will be crowdfunded in FY24
- For Windows
 - Added prompting to upgrade configuration from previous installation on first launch.
 - Added NIWC developed SCAP content for
 - Microsoft Internet Information Server (IIS) 8.5
 - Microsoft Internet Information Server (IIS) 10.0
 - Microsoft Office 365 ProPlus
 - Microsoft OneDrive
- For all UNIX
 - N/A
- For Linux
 - Added upgrade methods via RPM and DPKG methods, which imports SCC's options
 - Dropped support for rasbian 8
- For Solaris
 - N/A
- For Mac OS X
 - N/A

2. REQUIREMENTS

2.1 Minimum Hardware/OS Requirements

SCC can run on virtually any Windows based computer, however, below are some minimum specifications.

HARDWARE	MINIMUM/RECOMMENDED
CPU	Intel/AMD x86 or x64 based processor. Recommend at least a 1.5 Ghz dual core or newer.
RAM	1.0 GB Minimum, 2.0 GB or more is recommended. SCC uses about 250 MB to startup, and may use up to 1 GB or more during certain scans, especially WMI based remote scans from 64 bit Windows.
Disk space installation	The base install of SCC requires approximately 150 MB of disk space. Reviews may generate 100+ MB of results depending on content and logging options enabled.
Operating System	Windows 2012 or later

2.2 Scanning Requirements

SCC has three scanning modes for Windows. Local, Classic Remote and WMI Remote. Each have slightly different requirements.

SCANNING REQUIREMENT	LOCAL	CLASSIC REMOTE	WMI REMOTE
Local Administrative Rights	X	X	* See Note Below
Manage Auditing and Security Logon User Right	X	X	X
Windows Management Instrumentation (WMI) Service	X	X	X
Domain Admin Rights		X	X
File and Printer Sharing for Microsoft Networks		X	X
Server Service		X	X
Remote Procedure Call (RPC) Service System dependency for Remote Registry and WMI		X	X
Firewall Exceptions If the client firewall is blocking WAN/LAN connectivity to the file shares, RPC or WMI, SCC will not be able to perform a remote review. Classic reviews also require firewall rules to allow Remote Registry access, and DCOM.		X	X
"Access this computer from the network" User Right for the account running SCC (and not denied via "Deny access to this computer from the network")		X	X
Remote Registry Service		X	
DCOM (if content which uses WuaUpdateSearcher)		X	

Disable UAC remote restrictions (non-domain, alternate credentials only)			** See Note Below
Ability to execute Windows 'makecab.exe'			X

* WMI Note: Windows WMI mode can be run with SCC running as an administrator but can also be run as a non-administrator, and perform scans with alternate credentials, stored in SCC. Refer to section 4.4.1.7 WMI Remote Scanning Options for more information.

** WMI Note 2: When attempting to scan computers not in a Windows Active Directory domain, you will need to disable UAC remote restrictions. Refer to B.10 for more information.

3. INSTALL/UNINSTALL

To obtain a copy of the SCAP Compliance Checker software please refer to the Technical Support section of this manual.

It is recommended to verify the SHA256 checksums of the downloaded zip files with the SCC checksum file from <https://www.niwcatlantic.navy.mil/scap/> before proceeding to install.

3.1 Windows Software Installation

Note: Installer should be run with an Administrator account.

3.1.1 Graphical Installation Method

1. Double click on the installer (SCC_5.12_Windows_Setup.exe).
2. Verify that the installer has been signed by a DOD trusted cert: CS.NIWC-ATLANTIC.001
Note: DOD Root certificates need to be installed to verify the certificate trust path.
3. Review the User Account Control prompt, and click Yes.
4. Read the License Agreement page, click "I accept the agreement", and click Next.
5. Choose the Destination Folder (or leave at the default) and click Next.
6. Select Components (or leave default)
 - a. Determine if pre-bundled SCAP content is desired.
 - b. Determine if installing as a service is desired. Refer to the section titled 'Running SCC as a Service' for additional information.
7. Select a desktop shortcut if desired
8. Read the Ready to Install page and click Install.

3.1.2 Command Line Installation Method

The Setup program accepts optional command line parameters. These can be useful to system administrators, and to other programs calling the Setup program.

`/HELP, /?`

Shows a summary of this information. Ignored if the UseSetupLdr [Setup] section directive was set to no.

`/SP-`

Disables the 'This will install... Do you wish to continue?' prompt at the beginning of Setup.

`/SILENT, /VERYSILENT`

Instructs Setup to be silent or very silent. When Setup is silent the wizard and the background window are not displayed but the installation progress window is. When a setup is very silent this installation progress window is not displayed. Everything else is normal so for example error messages during installation are displayed and the startup prompt is (if you haven't disabled it with DisableStartupPrompt or the '/SP-' command line option explained above).

`/SUPPRESSMSGBOXES`

Instructs Setup to suppress message boxes. Only has an effect when combined with '/SILENT' or '/VERYSILENT'.

The default response in situations where there's a choice is:

- Yes in a 'Keep newer file?' situation.

- No in a 'File exists, confirm overwrite.' situation.

- Abort in Abort/Retry situations.

- Cancel in Retry/Cancel situations.

- Yes (=continue) in a

- DiskSpaceWarning/DirExists/DirDoesntExist/NoUninstallWarning/ExitSetup Message/ConfirmUninstall situation.

- Yes (=restart) in a FinishedRestartMessage/UninstalledAndNeedsRestart situation.

5 message boxes are not suppressible:

- The About Setup message box.

- The Exit Setup? message box.

- The FileNotInDir2 message box displayed when Setup requires a new disk to be inserted and the disk was not found.

- Any (error) message box displayed before Setup (or Uninstall) could read the command line parameters.

- Any message box displayed by [Code] support function MsgBox.

`/DIR="x:\dirname"`

Overrides the default directory name displayed on the Select Destination Location wizard page. A fully qualified pathname must be specified. May include an "expand:" prefix which instructs Setup to expand any constants in the name. For example: '/DIR=expand:{pf}\My Program'.

`/TYPE=type name`

Overrides the default setup type. Options are Standard and Custom, selecting Custom is required to select components as listed below

`/COMPONENTS="comma separated list of component names"`

Overrides the default component settings. Using this command line parameter causes Setup to automatically select a custom type. If no custom type is defined, this parameter is ignored.

Only the specified components will be selected; the rest will be deselected.

If a component name is prefixed with a "*" character, any child components will be selected as well (except for those that include the dontinheritcheck flag). If a component name is prefixed with a "!" character, the component will be deselected.

The available components are:

- Content\DISA_STIG_SCAP_Content

- Other\SCC_Service

This parameter does not change the state of components that include the fixed flag.

Example:

Deselect all components, then select the "SCC Service" component:

`/COMPONENTS="Other\SCC_Service"`

Examples

Install SCC with default options silently with no GUI shown


```
SCC_5.12_Windows_Setup.exe /VERYSILENT
```

Install SCC to a given directory with default options silently with no GUI shown

```
SCC_5.12_Windows_Setup.exe /VERYSILENT /DIR="C:\SCC"
```

Install SCC with the Service and default content silently with no GUI shown

```
SCC_5.12_Windows_Setup.exe /VERYSILENT /TYPE=custom  
/COMPONENTS="Content\NIST_USGCB_SCAP_Content, Content\DISA_ST  
IG_SCAP_Content, Other\SCC_Service"
```

3.2 Windows Software Uninstall

Note: The uninstaller will not delete any files added after the installation, such as end user installed content, so the installation directory may exist and need to be manually removed.

3.2.1 Standard Method

1. Click Start -> Control Panel -> Programs -> Uninstall a program
2. Double click on SCAP Compliance Checker 5.12
3. Follow the prompts

3.2.2 Silent Uninstall

To perform an automated, silent uninstallation, run the uninstaller via command line with a /SILENT or /VERYSILENT flag

```
"C:\Program Files\SCAP Compliance Checker 5.12"\unins000.exe  
/SILENT
```

3.3 Install/Remove software via zip file

3.3.1 'Install' software via zip file

To install, simply extract file scc-5.12_Windows.zip to any directory.

3.3.2 Removal of software installed via zip file

To remove software 'installed' from the generic zip file, simply remove the installation directory and all sub-directories and files.

3.4 Install Details

3.4.1 Files Installed by the SCC

FILE	DESCRIPTION
bin	SCC dependencies shared for 32 and 64 bit.
csc.exe	Launcher program for the command line version of SCC
csc-remote.exe	Launcher program for the command line version of SCC, used for remote WMI reviews.
scc.exe	Launcher program for the graphical version of SCC
scc.sig32	Use for verifying the integrity of 32 bit installations
scc.sig64	Use for verifying the integrity of 64 bit installations
lib32	32 bit version of SCC dependencies
lib64	64 bit version of SCC dependencies
options.xml	Default options file used by SCC
hosts.txt	Sample host file
unins000.exe	Uninstaller for this application
Documentation\ReleaseNotes.txt	Summary of changes for this version of the software.
Documentation\SCC_Help.chm	Compiled, searchable help file
Documentation\SCC_UserManual.pdf	PDF version of the User Manual
Documentation\TermsOfUse.txt	Text file containing the Usage, which is displayed during the installation.
Documentation\ThirdPartyLicenses.txt	Contains list of third party libraries used in SCC and their respective licenses.
Documentation\ThirdPartyLicenses	Directory containing text formatted 3rd party licenses, referenced in ThirdPartyLicenses.txt
Local	Location in which SCC writes temporary files during execution.
RemotePlugin	Location in which remote UNIX and Windows scanning plugins may reside, empty by default.
Resources\Compiled*	Folder containing compiled library files for SCC use
Resources\Content*	Parent content folder for SCAP, SCAP 1.2, OVAL, OVAL External Variables, and OCIL content folders
Resources\Content\External_Variables	Contains any External Variables files associated with an OVAL content stream
Resources\Content\OVAL_Content	Contains any OVAL vulnerability content included with the installer or installed by the end user with the Install OVAL Content feature.
Resources\Content\OCIL_Content	Contains any stand alone OCIL content included with the installer or installed by the end user with the Install OCIL Content feature.

Resources\Content\SCAP_Content	Contains any SCAP 1.0 or SCAP 1.1 content included with the installer or installed by the end user
Resources\Content\SCAP12_Content	Contains any SCAP 1.2 content included with the installer or installed by the end user
Resources\Content\TrustedPublicCerts	Contains known/trusted certificates to verify digital signatures in SCAP 1.2 content
Resources\Content\XCCDF_Tailoring	Contains XCCDF Tailoring files which can be used with SCAP 1.2 datastreams
Resources\DB	Database utilized when processing SCAP 1.2 data streams
Resources\DefaultFiles	Contains default files used by the SCC
Resources\Graphics*	Images and icons used with SCC.exe
Resources\Schema*	Files used to validate the SCAP XML content
Resources\Thresholds*.xml	Contains the default and any user customized compliance thresholds
Resources\Transforms*	Files used to create post scan HTML and text reports from the OVAL and XCCDF XML results
C:\Users\All Users\Microsoft\Windows\Start Menu\Programs\SCAP Compliance Checker 5.12	Start menu icons created during the installation process

3.4.2 Registry Keys Created During SCC Installation

REGISTRY KEY	DESCRIPTION
HKLM\Software\Microsoft\Windows\CurrentVersion\Uninstall\SCAP Compliance Checker 5.12	Standard uninstall information which allows the Add/Remove Programs to remove the SCAP Compliance Checker software.

3.4.3 Files Created During Software Execution

FILE	DESCRIPTION
<User Defined Directory>\Sessions Refer to Data Directory option in "Editing Options" for details.	XML, HTML and Text based results created during a review. Also contains Screen, Debug and Error logs that are specific to a scan session.
<User Defined Directory>\ApplicationLogs Refer to Data Directory option in "Editing Options" for details.	SCC Application Logs (not related to any SCAP scan session) including Screen, Error and Debug logs that could be created during a review depending on user preferences.
<User Defined Directory>\Config Refer to Data Directory option in "Editing Options" for details.	Contains scan session database and host credential database.
<User Defined Directory>\options.xml	Configuration settings from the SCC.exe
<SCC Install>\Local	Temporary files created during SCC execution

4. GUI BASED USAGE

Section 4 of this document explains the basic Graphical User Interface (GUI) usage of SCC to perform SCAP based compliance scanning, standalone OVAL and OCIL scanning, and editing options.

Below is a quick overview of how SCC works.

1. Open the SCC GUI.
2. View available SCAP content included with SCC.
3. Install any additional SCAP content into SCC.
4. Enable SCAP content and select the desired profile from each SCAP content stream.
5. Select Scan Type
6. Scan Computer(s) with enabled SCAP content.
7. View reports.

4.1 Launching the Graphical User Interface

To start the application with a GUI, click:

Start -> All Programs -> SCAP Compliance Checker 5.12 -> SCAP Compliance Checker (SCC) 5.12

1. Review information on the User Account Control pop-up
2. Verify that the SCC application has been signed by a DOD trusted cert: CS.NIWC-ATLANTIC.001
Note: DOD Root certificates need to be installed to verify the certificate trust path.
3. Click "Yes"

4.2 Installing & Configuring Content

SCC's installer contains the latest publicly available SCAP content from DISA, NIWC and NIST, which was available at that time. However, new and updated content may need to be installed by end users, especially if the current SCC release is several months old.

4.2.1 Updating SCAP Content with SCC

If the computer and the user running SCC have Internet access, SCC has the ability to update the DISA and NIWC developed STIG SCAP content.

To check for updated SCAP content:

- Click Help -> Check For Content Updates

To view more information on any content stream:

- Right click
 - View Content Details

To install content:

- Either Left click on each check box to enable
- Or Right click
 - Enable All
 - Disable All
- Then click "Install Checked Content"

* See Editing Options -> Update Options for more information on content updates and repositories.

Refer to SCC's FAQs if you are interested in making your own offline copy of DISA's content repository, or to learn why SCC's default content repository XML file is located on github instead of cyber.mil.

4.2.2 Installing Content Manually

The steps below will guide you through installing content within SCC. Note that the steps are the same SCAP 1.0, 1.1, 1.2 and 1.3 data streams.

To obtain NIWC Enhanced DISA STIG SCAP content which contain automation and Manual Questions: <https://www.niwcatlantic.navy.mil/scap/scap-content-repository/>

OR

To obtain DISA STIG SCAP content which does not contain any Manual Questions, please download "benchmarks" from <https://www.cyber.mil/stigs/SCAP>

1. Launch the SCC GUI
2. Content should be visible by default in the main window. If content is not visible, due to existing scan logs, click the "Show Content" button in the left column, "Select Content" pane.
3. Click on the "Install" button in the "Content" pane

- a. Select from the following options:
 - i. Validate XML on content install (Yes/No)
 - ii. Overwrite existing content (Yes/No)
 - iii. Enable content on install (Yes/No)
4. Browse to the content file you would like to install
 - a. For .zip files, the following is supported
 - Zip of all SCAP 1.0/1.1 files
 - Zip of all SCAP 1.2/1.3 files
 - Zip of all standalone OVAL files (not part of a SCAP stream)
 - For more information on installing OVAL External Variables, refer to Section 4.9 Standalone OVAL Usage
 - Zip of all standalone OCIL files (not part of a SCAP stream)
 - Mixing of content types (SCAP/SCAP1.2/OVAL/OCIL) is not supported
 - b. For .xml files, the following is supported
 - XML file that is a SCAP 1.2/1.3 datastream
 - XML file that is a valid OVAL file
 - XML file that is a valid OCIL file
5. After installation is complete, enable the content and choose the desired profile.

Note: DISA STIG "Manual" files are not SCAP content. They contain an XCCDF XML file, but do not contain any OVAL XML. They are intended for performing a manual review of the system.

4.2.3 Enabling/Disabling Content

To enable/disable a single Content stream:

1. **Left** click the check box to the left of the content stream name

To enable all Content streams:

1. **Right** click on any row
2. Click "Select All"

To disable all Content streams:

1. **Right** click on any row
2. Click "Clear All"

NOTE: *If multiple benchmarks are included in a SCAP 1.2 data stream, SCC splits each benchmark on the SCAP Content tab. This allows the end-user to enable/disable a specific benchmark within a larger data stream.*

4.2.4 Selecting a Profile

A profile is a collection of rules and is designed to allow the same set of SCAP content (XML) to perform different sets of checks based on end user need. SCAP content can contain one or more different profiles. By default, SCC enables the first profile found.

For USGCB, there is only one profile in the content, but for other content such as DISA, the end user will need to select the appropriate profile, according to the sensitivity of the computer being scanned. For DISA STIGS, below is the normal list of available profiles in each SCAP content stream.

- o MAC 1 Public
- o MAC 1 Sensitive
- o MAC 1 Classified
- o MAC 2 Public
- o MAC 2 Sensitive
- o MAC 2 Classified
- o MAC 3 Public
- o MAC 3 Sensitive
- o MAC 3 Classified
- o Disable Slow Rules
- o CAT I Only

How many checks and results are impacted by changing the profile is completely dependent on the intent of the SCAP Content Author (not SCC). The checks in all profiles could be all the same, or they could differ greatly.

To select a profile:

1. **Left** click on the Stream name to populate the "Stream Details" window on the right
2. Select the desired profile from the Profile dropdown.

4.2.5 Selecting a Profile to use with All SCAP Content

To select a single profile, and apply it to all SCAP content:

1. **Right** click on the row to delete
2. Click "Set All Profiles"
3. A new form will open, showing a dropdown of all profiles found in all content.
4. Left click to select the desired profile
5. Click Save to save and close the form

4.2.6 Deleting Content

To delete a single content row

1. **Right** click on the row to delete
2. Click "Delete Selected Content"

To delete several content rows

1. Select as many content rows as you would like to delete, with **Left click and Shift** or **Left click and Ctrl**.
2. Right click on any of the selected rows
3. Click "Delete Selected Content"

To delete ALL Content

1. **Right** click on any row
2. Click "Delete All"

4.2.7 Viewing Content Details

To view additional information about the SCAP Content:

1. **Left** click on the Content Name
2. View the "Content Details" information on the right pane, it contains the following fields:

ITEM	DESCRIPTION
Title	The XCCDF "title" field, directly from the content.
Datastream	The SCAP 1.2/1.3 Datastream " data-stream id" field. This can be useful if there are multiple datastreams with the same benchmark ID.
Profile	A drop-down containing 1 or more profiles from the XCCDF content.
Release Info	The XCCDF "release-info" field, directly from the content.
Status	The XCCDF "status" field, directly from the content.
Validation	The end result of SCC's attempt to validate the content, either at install or scan time.
Signature	The end result of SCC's attempt to validate the digital certificate of SCAP 1.2/1.3 datastream (if found)
Platform	The CPE-Dictionary CPE Item Title, directly from the XML content
Publisher	The XCCDF "publisher" field, directly from the XML content.
Description	The XCCDF "description" field, directly from the XML content.
Notice	The XCCDF "notice" field, directly from the XML content.

4.2.7.1 Saving SCAP Prose Reports

Once you have populated the Stream Details pane, a human readable 'prose' version of the XCCDF and OVAL files to either HTML or Text format may be produced. To use, a profile must be selected on the SCAP Content form. If no profile is selected, the buttons will be disabled.

The Prose report is a human readable representation of the SCAP content, very similar to the All Setting reports, but does not contain any scan data. It is meant to show the XCCDF rules and OVAL definitions in a logical tree structure format.

4.2.7.2 XCCDF Tailoring

This is a feature for advanced users wanting to modify how SCAP 1.2 and SCAP 1.3 benchmarks perform checks. Refer to the XCCDF Tailoring section 4.8 for details.

4.2.7.3 Manual Questions

Starting with SCC 5.7, if SCAP content that contains OCIL manual questions is installed, SCC provides the ability to answer manual questions from the DISA STIG. To use, select the desired SCAP content, click on "Manual Questions" in the content details pane, and answer the non-automated items from the STIG Manual, and those results, along with the automated checks will be included in SCC results. See section 4.6 "SCAP Scanning with Manual Questions" for more information.

4.2.7.4 Applicability

By default, SCC only performs scan using SCAP content if the content is applicable to a given target. This option forces SCC to perform a scan of a specific content stream against all targets.

4.3 Performing a Scan

After installing and enabling the desired content and profile, the application is ready to perform compliance scanning.

1. Launch the SCC GUI
2. Select Scan Type
 - o Local Scan
 - o Single Remote Scan (Windows)
 - o Multiple Remote Scan (Windows)
 - o Cisco IOS / IOS XE Offline Scan
 - o Cisco IOS / IOS XE Remote SSH Scan
 - o Remote UNIX SSH Scan
 - o Windows WMI and UNIX SSH Remote Scan
3. Select Content Stream(s) and their respective applicable Profiles
4. Start Scan
5. View Reports

4.3.1 Select Scan Type

The SCC can review the local computer or remote computers over LAN/WAN connections. Select one of the following options:

4.3.1.1 Local Scan

This option instructs SCC to scan the computer in which the SCC software is installed.

4.3.1.2 Single Remote Computer (Windows)

This option instructs SCC to scan a single remote Windows computer over the LAN/WAN.

- Choose remote scan mode
 - o Classic
 - o Scan to be performed with remote API calls, no agent on remote computer
 - o WMI
 - o Scan to be performed by command line agent temporarily installed to target computer, and started and monitored via WMI
- A single NetBIOS computername or IP Address should be entered into the text field provided.

4.3.1.3 Multiple Remote Computers (Windows)

This option instructs SCC to scan a list of remote Windows computer over the LAN/WAN.

- Choose remote scan mode
 - o Classic

- o Scan to be performed with remote API calls, no agent on remote computers
- o WMI
 - o Scan to be performed by command line agent temporarily installed to target computers, and started and monitored via WMI
 - o Scans are performed in parallel decreasing scan times for large numbers of computers.
- Host File
 - o Create a new host file
 - o Create a host list by querying Active Directory for the current domain
 - o Options exist for validating this file
 - o Validate host list using NSLookup to ensure computername is in DNS
 - o Validate host list using NSLookup and Ping to ensure computername is in DNS and computer is online
 - o Choose an existing hosts file
 - o This list should be in the form of a text file with a single computer listed per line.
- Organizational Unit (OU)
 - o Select one or more OU's from Active Directory (shift + left click to select more than one OU)
 - o List of hosts created each time a scan is performed.
- Entire Domain
 - o Scan all computers in the domain, list of hosts created each time a scan is performed.

4.3.1.4 Cisco IOS / IOS XE Config File

Refer to section 4.9 "Cisco IOS Config Scanning" for usage.

4.3.1.5 Cisco IOS / IOS XE Remote SSH

This option instructs SCC to scan a list of remote CiscoIOS Routers/Switches over the LAN/WAN. Refer to Edit/Select SSH Hosts, for instructions on selecting hosts and credentials.

4.3.1.5.1 Edit/Select SSH Hosts (Cisco)

Note: If the host credential manager has not been used before, a popup window will appear prompting for a new master password and confirming the new password.

This opens a credential manager in which the end user can

- o Add a new host
 - o DNS name or IP Address
 - o Description (optional)
 - o SSH Port
 - o Authentication Type
 - o Enable: No additional authentication
 - o Enable: Use same password as username

- o Enable: Use 'Enable' password. This requires selecting a credential that has an 'Enable' password
 - o Enable: Use 'Enable' username and 'Enable' password. This requires selecting a credential that has both an 'Enable' username and an 'Enable' password.
 - o Select an existing credential, or add a new one
 - o Test SSH connection to remote device using selected credential
- o Add a new credential
 - o Username
 - o Nickname (optional)
 - o Password
 - o 'Enable' username (optional depending on how device is configured), this field is used if Authentication Type is "Use 'Enable' username and 'Enable' password.
 - o 'Enable' password (optional depending on how device is configured), this field is used if Authentication Type is "Use 'Enable' username and 'Enable' password, or Use 'Enable' password.
- o Import an existing host file, formatted as text with a single DNS name or IP Address per line
 - o Each host in this file will have it's SSH connection tested after import
- o Double click a host to edit it
- o Right click on the host list to:
 - o Edit host
 - o Enable Selected Hosts (If SSH test passes)
 - o Enable All hosts (If SSH test passes)
 - o Disable Selected Hosts
 - o Disable Hosts with Failed SSH Connections
 - o Disable All Hosts
 - o Test Selected SSH Connections
 - o Test All SSH Connections
 - o Delete SSH hostkey/SHA256 Fingerprint
 - o Delete Selected Hosts
 - o Delete All Hosts
- o Double click on a credential to edit it
- o Right click on credentials to:
 - o Edit credential
 - o Delete selected credential
 - o Delete all credentials

4.3.1.6 UNIX Remote SSH

This option instructs SCC to scan a list of remote UNIX/Linux/Mac systems over the LAN/WAN. Refer to Edit/Select SSH Hosts, for instructions on selecting hosts and credentials.

Note: If the UNIX SSH Scanning Plugin has not been installed, an information message will appear in the left window, with a browse dialog to select the plugin, which can usually be obtained from the same location you obtained the SCC installers.

4.3.1.6.1 Edit/Select SSH Hosts (UNIX)

This opens a credential manager in which the end user can

- o Add a new host
 - o DNS name or IP Address
 - o Local System Name (Read only, queried from remote system)
 - o Description (optional)
 - o SSH Port
 - o Operating System
 - o SCC Autodetect is default
 - o If autodetect fails, manual selection can be used.
 - o Authentication Type
 - o SSH as 'root'
 - o SSH as non-root user, then 'su' with root password
 - o **Note:** If the remote host is Solaris, Ubuntu or Mac OS X, this method may not work, due to the implementation of 'su' in those operating systems. The recommended method is one of the 'sudo' options below.
 - o SSH as non-root user, then 'sudo', no password
 - o SSH as non-root user, then 'sudo', with user's password
 - o Select an existing credential, or add a new one
 - o Test SSH connection to remote device using selected credential
 - o Tests to make sure the hostname/ip is valid
 - o Checks remote server's hostkey
 - o Tests authentication
 - o Determines remote 'local system name'
 - o Tests running commands as root level user
 - o Determines remote Operating System
 - o Checks freespace on remote /tmp and remote base SCC directory (user configurable, see scanning options for more information)
 - o SHA256 fingerprint
 - o The first time SCC SSH's to a remote UNIX host, it will download the remote system's hostKey, and display the SHA256 checksum of the hostkey for the end user to inspect.
 - o If the remote hostKey changes in the future, SCC will warn the user about a potential man in the middle attack and disable the host.
 - o If the new hostkey is to be trusted, SCC will allow you to re-trust and re-enable the host.
- o Add a new credential
 - o SSH Username
 - o Nickname (optional)
 - o SSH and/or sudo password
 - o Enter the password associated with the username, it could be used for just the SSH connection, but also for sudo depending on the authentication type selected for the host.
 - o Private Key (required if password is not present)
 - o Note that SCC will use the private key first if both an SSH password and private key are present.
 - o Passphrase (if required by private key)
- o Import an existing host file, formatted as text with a single DNS name or IP Address per line
 - o Each host in this file will have it's SSH connection tested after import

- o Double click a host to edit it
- o Right click on the host list to:
 - o Edit host
 - o Enable Selected Hosts (If SSH test passes)
 - o Enable All hosts (If SSH test passes)
 - o Disable Selected Hosts
 - o Disable Hosts with Failed SSH Connections
 - o Disable All Hosts
 - o Test Selected SSH Connections
 - o Test All SSH Connections
 - o Delete SSH hostkey/SHA256 Fingerprint
 - o Delete Selected Hosts
 - o Delete All Hosts
- o Double click on a credential to edit it
- o Right click on credentials to:
 - o Edit credential
 - o Delete selected credential
 - o Delete all credentials

4.3.1.7 Windows WMI and UNIX Remote SSH Scan

This is a Windows only feature that combines WMI scanning and UNIX SSH Remote scanning into a single scan. Refer to each individual method for instructions on selecting/enabling hosts.

Note: If the UNIX SSH Scanning Plugin has not been installed, an information message will appear in the left window, with a browse dialog to select the plugin, which can usually be obtained from the same location you obtained the SCC installers.

4.3.2 Select Content

Select content as described in section 4.2.

4.3.3 Performing Analysis

To perform a scan, click the '**Start Scan**' button.

To cancel a review, click the '**Cancel Scan**' button.

4.4 Editing Options

The SCC application has many end user customizable options, although the installation defaults are those most frequently used. After using SCC a few times, the end user may want to adjust some of these options, depending on their personal preferences.

1. Launch the SCC GUI
2. Click Options ->Show Options

4.4.1 Scan Options

4.4.1.1 SCAP Processing

OPTION	DESCRIPTION
Run all content regardless of applicability	This option will ignore the content's CPE-OVAL results and continue processing the content against the system. This option can be used to run content that is not normally applicable to the target system (e.g. Red Hat SCAP content on a Debian system). Note that this option alters the standard SCAP rules for gathering certain objects which can result in incomplete results and/or false positives.
Process Manual & Hybrid Questions if found in SCAP content	This option will ignore any manual questions in the content and cause any XCCDF rule that is a manual question to be excluded from XCCDF results. This will disable the creation of DISA CKL reports, as all rules are required for CKL reports. This will remove any warnings that SCC would normally print if manual question results were not found for a given content stream.
Attempt to download external OVAL and XCCDF Tailoring files	This option allows the user to disable the SCAP 1.2 requirement of attempting to download OVAL and/or XCCDF Tailoring files from the internet, if specified in the content. If this option is enabled (default) and the SCAP 1.2 datastream lists a http reference for the OVAL or XCCDF Tailoring component, SCC will attempt to download it, and store it locally. Once it has been downloaded, it does not attempt again. This feature is not currently used by any production NIST or DISA SCAP content, but the feature is required for SCC to obtain SCAP 1.2 validation.
Force OVAL results to 5.10.1 for SCAP 1.2 interoperability	SCC by default saves results in OVAL 5.12.1. However, this option could be enabled by the user for certain usage (primarily SCAP 1.2 validation) or tools that import OVAL results (but only support OVAL 5.10.1).
Save results when content is not applicable for SCAP 1.3 interoperability	This option is designed to force SCC to create XML results even when content is not applicable to the target. It could be used for troubleshooting debugging purposes, but is primarily intended to only be used as part of NIST SCAP 1.3 validation, which has a test case which mandates SCAP validated tools create results when content is not

	applicable.
--	-------------

4.4.1.2 OVAL Processing

This set of options allows SCC to process currently available content in an efficient and accurate manner, however it does not comply with the letter of the law when it comes to the OVAL standard.

OPTION	DESCRIPTION
Ignore remote file systems during OVAL file scans	This option will ignore remote file systems, such as Windows shares, and UNIX NFS mount points. This option could be specified in the SCAP content as well, but in all of the publicly available SCAP content to date, the content authors have not specified to skip scanning of remote file systems. If this option is disabled, and the SCAP content does not specify to exclude remote file systems, SCC will scan all drives/mount points on the system, and will likely cause the application to slow down, dramatically in certain cases, and the results will potentially include issues from the server hosting the remote files. Until SCAP content is updated to ignore remote file systems, it is recommended to keep this option enabled in SCC.
Treat the OVAL 'equals' operation as 'case insensitive equals'	This option allows end users to override the instructions in the OVAL content and to ignore case of files and directories. Ideally this should be specified in the OVAL content, and enabling this option will cause warnings that results may not be as intended by the content author.
Domain Controllers: Process the OVAL accesstoken_test by user right	The Windows accesstoken_test will collect user right information for all user accounts even if the accounts have no rights. Under certain circumstances (domain controllers), this could result in the collection of thousands of user accounts which may lead to extremely large result files and/or memory errors. If this option is enabled, then user right information will be collected only for user accounts that actually have user rights assigned.
Enable OVAL item creation threshold	In certain circumstances, a combination of content issues, or system configuration can cause large numbers of OVAL items to be created. This causes two primary issues, the first being SCC's memory and CPU usage during the scan will increase, potentially to the point of crashing. Secondly, if SCC is able to complete the scan, the resulting XML files will be too large to create any Text or HTML reports from. This option caps the number of OVAL items created, on a per OVAL test basis, to the number specified in the form. This option can be updated by the user depending on their preference. If SCC runs out of memory and crashes even with this option enabled, it is recommended to lower the threshold by a sizable amount and re-run. If SCC reaches the threshold for a single test, the end result of the test will be 'error' as SCC will skip processing any additional items, and will not be able to make a final determination of compliance with regards to

	pass/fail, and the end user will likely need to perform the check manually to determine true compliance. This should not be a common occurrence, and the content author may need to be contacted, to determine if the test can be written in a method which does not create such a large volume of results. This option is enabled by default. By default, the OVAL Item Creation Threshold is set at 50,000. This field is guarded by input validation and will only allow a user to enter an integer between 0 and 999999. Any input outside of those values will result in an error and the option will not be allowed to be saved.
--	--

4.4.1.3 Manual Question Processing

OPTION	DESCRIPTION
Sort Questionnaires by DISA CAT (CAT I, CAT II etc..)	This option sorts Manual Questions, both in the SCC GUI and in the Manual Question text based autoanswer templates to be sorted by severity, displaying CAT I items first, followed by CAT II etc... If this option is disabled, questions will be displayed in the order they were found in the DISA STIG Manual.
Regenerate Templates	This option allows the end user to regenerate all Manual Autoanswer Templates. This does not update any existing completed autoanswer file(s).
Backup Manual Question Results with datetime/username to a Backups sub-directory	This option only applies to SCC GUI when saving manual question results from SCC's Manual Question editing form. When enabled, this option causes SCC to create a backup of the Manual Question results to a subfolder of Completed_Files each time "Save Document and Close" is pressed. This creates a backup file with the date/timestamp and the username of the person logged into the computer. These files can be used to manually restore a backup, or show changes with diff utilities. Refer to Section 4.6.2.2 for more information.
Ignore XCCDF rule 'version' changes when upgrading manual questions	Due to the mass re-versioning of DISA STIG Manual rules for July 2024, which put a new rule version on every single rule regardless of other changes to the rule, this option has been created and enabled by default. If this option is disabled, any manual question that does not match the rule and version from the current DISA STIG will be excluded when upgrading manual questions.

4.4.1.4 Manual Question AutoAnswer Completed Files Directory

OPTION	DESCRIPTION
Obtain Manual Question AutoAnswer Files from the default directory	This option sets SCC to obtain Manual Question autoanswer files from <SCC Install>Resources/Content/Manual_Questions/Completed_Files/

Obtain Manual Question Autoanswer files from a Custom Directory	This option sets SCC to obtain Manual Questions autoanswer files from any end user specified directory. This also instructs the SCC GUI to save Manual Questions to this custom directory.
---	--

4.4.1.5 WMI/SSH Remote Scanning Options

OPTION	DESCRIPTION
Maximum local threads for performing remote WMI or SSH scan	This option allows the end user to determine the number of local threads for parallel scanning with WMI and SSH based scanning. Each thread takes about 30 MB of RAM. More threads will speed up large remote scans, but could cause slowdowns or issues on the local computer.
Maximum minutes to allow SCC to start on remote system	This option allows the end user to specify a maximum amount of time to wait for SCC to be launched on the remote computer. This may fix issues on certain slow remote systems, especially with certain anti-virus products and features enabled that slow down extraction of zips and cabs.

4.4.1.6 SSH Remote Scanning Options

OPTION	DESCRIPTION
SSH: Remote Base Directory, SCC will make a sub-dir of 'scc-remote'	Directory to copy a temporary copy of SCC for remote SSH based scans. This directory must exist, and should exist on it's own partition if possible. Default is /opt. SCC will create a subdirectory called 'scc-remote' in the specified directory, which it will install a temporary copy of SCC, and a results subdirectory for scanning logs and results. The <remote base directory>/scc-remote directory will be removed after each remote SSH scan.
Uninstall Remote UNIX SSH Plugin	This allows for the removal of the remote plugin from the scanning computer.
SSH: Allow SCC to temporarily whitelist itself on target RHEL 8/9 and Oracle Linux 8/9	For RHEL 8/9 and Oracle Linux 8/9 targets: This option allows for remotely scanning STIG compliant RHEL 8 and Oracle Linux 8 computers, which requires enabling application whitelisting with fapolicy daemon. As this option temporarily change the configuration of the target system, this option is disabled by default. Below is an overview of what SCC does, should the end user want to enable this feature: 1. From the scanning computer, after the temporary files are copied to the target computer in the user specified directory (default is /opt/scc-remote), the scanning computer runs the following command to whitelist csc-remote via SSH: <pre>fapolicy-cli -f add /opt/scc-remote/csc-remote</pre>

	<pre>fapolicyd-cli --update</pre> 2. Then csc-remote is called and it whitelists all of its known binaries with the same method as #1 3. csc-remote scans the computer 4. When csc-remote completes, the scanning computer then removes the directory and all files/subdirectories with the following via SSH: <pre>fapolicyd-cli -f delete /opt/scc-remote/</pre> <pre>fapolicyd-cli --update</pre> To summarize, SCC uses the system command (if found) of fapolicyd-cli to whitelist a known list of binaries that comprise the SCC application before a scan starts, and when the scan completes, the whitelist is then removed from the target computer's fapolicy database.
--	---

4.4.1.7 WMI Remote Scanning Options

OPTION	DESCRIPTION
WMI: Use Alternate Credentials	This optional feature allows SCC to connect to target computers specifying alternate credentials. Please note that with this mode enabled, the command line interface of --wmi will prompt the user for the master password, in the same method as --ssh scanning, so any scripted usage of --wmi should have this option disabled. - o This feature allows SCC to scan from one active directory domain to another, and to scan systems that are not in any active directory domain. - o This feature also allows for a non-administrator to launch SCC and perform remote WMI based reviews if they have a domain admin enter scanning credentials. The password stored in SCC is not reversible to the end user, and during the scan the end user should not have any admin level access to the target computer.

4.4.2 Content Options

4.4.2.1 Installing Newer Versions of Existing SCAP Content

OPTION	DESCRIPTION
Do Nothing, leave older content as-is when installing newer versions	This option leaves any older version of content as installed, and if it's enabled, it remains enabled.
Disable older versions of matching content	This option leaves the older version of content as installed, but disables it.
Archive older versions of matching	This option moves the older version of content to <install>\Resources\Content\Archived_Content

content	
Delete older versions of matching content	This option will delete any older version of matching content.

4.4.2.2 XML Schema Validation

OPTION	DESCRIPTION
Perform XML Schema Validation on Input Files during Installation	This option validates that the XML content is syntax error free before installing new content into SCC.
Perform XML Schema Validation on Input Files before scanning	This options validates that the XML content is syntax free before SCC uses it for each scan.
Perform XML Schema Validation on Output Files	This options validates that the XML result files are syntax and error free after creation.

4.4.2.3 XML Digital Signatures

OPTION	DESCRIPTION
Perform XML Digital Signature Validation before scanning	This option will validate signed XML content files prior to execution.
Cancel Scan(s) on XML Digital Signature Validation Failure	This option will automatically cancel a scan if the signed XML file(s) fail XML digital signature validation.

4.4.2.4 Content Developer Mode

OPTION	DESCRIPTION
Run content without performing integrity checks (not recommended for production)	This option is reserved for content development. Enabling this option allows SCC to use SCAP content without performing integrity checks, which could lead to risky content being run, including shell commands and SQL queries. This should not be enabled on any production system.

4.4.3 Reporting Options

4.4.3.1 Select Reports

REPORT	DESCRIPTION
All Settings	This report contains detailed pass and fail results from each check performed. It is a large report and is not intended for printing
All Settings Summary	This report contains a summary of pass and fail results from each check performed.
Non-Compliance	Non-compliance reports contain detailed results from each failed check. It is a large report and is not intended for printing

Non-Compliance Summary	This report contains a summary of failed checks
------------------------	---

4.4.3.2 Report File Types

FORMAT	DESCRIPTION
HTML	HTML formatted reports for viewing with a web browser
Text	Plain Text reports for viewing with a text editor such as Notepad or Wordpad.

4.4.3.3 Report Sorting

FORMAT	DESCRIPTION
Sort HTML/Text reports by XCCDF rule Severity, and group by automated/manual	With this option enabled, SCC's reports will be sorted by severity and then groups by Automated/Manual. If this option is disabled, SCC's reports will be sorted in the same order as the STIG manual, with no groupings.

4.4.3.4 Collected Item Thresholds

REPORT	DESCRIPTION
Only display a user defined number of collected items	This option prevents the All-Settings and Non-Compliance HTML and/or Text reports from getting extremely large if SCAP content contains file/user test that collect items for thousands or millions of files/directories and users. This does not impact the accuracy of the check, it just limits the number of detailed finding to include in reports.
'pass' threshold	This is the number of collected items that pass to include for each test. Default is 50
'fail' threshold	This is the number of collected items that do not pass to include for each test. Default is 100

4.4.3.5 XML Results

OPTION	DESCRIPTION
Save XCCDF XML files	This option allows the user to disable saving the XCCDF XML files after the review. It should always be enabled unless drive space is limited. If this option is not enabled, multiple computer summary reports cannot be created.
Save OVAL XML files - Full with System Characteristics	This option allows the user to enable saving the OVAL XML files, which contain the detailed results from each review, and can be very helpful in debugging problems, or recreating reports after scans occur.
Save OVAL XML files - Full without System Characteristics	This option allows the user to save a slightly less verbose version of OVAL results, which exclude the System Characteristics, and is required by SCAP 1.2.
Save OVAL XML files - Thin	This option allows the user to save even less data to OVAL results, which exclude the System Characteristics and Test results, and is required by SCAP 1.2.
Do Not Save OVAL	This option allows the user to not save OVAL XML results after each scan

Results	is complete.
Save OCIL Results	This option allows the user to disable saving OCIL XML files (if content includes OCIL content).
Save NIST ARF XML files	This option creates the NIST SCAP 1.2/1.3 Asset Reporting Format (ARF), which may be useful if an AFT results consumer is being used, or for testing official SCAP capabilities.
Save DISA Checklist CKL files with 'Enhanced' content	When running NIWC Enhanced SCAP content, which contains Manual Questions, create the DISA Checklist 'CKL' report, which can be used by systems such as eMASS, and eliminates the need to use the STIG Viewer to import SCC results, answer manual questions and export CKL results. Creating CKL results from 'standard' DISA SCAP content is not supported, as DISA SCAP benchmarks do not contain all of the rules from the STIG Manual, only those that are automated
Save DISA Checklist CKLB files with 'Enhanced' content	When running NIWC Enhanced SCAP content, which contains Manual Questions, create the DISA Checklist 'CKLB' report, which can be used by systems such as eMASS, and eliminates the need to use the STIG Viewer to import SCC results, answer manual questions and export CKLB results. Creating CKLB results from 'standard' DISA SCAP content is not supported, as DISA SCAP benchmarks do not contain all of the rules from the STIG Manual, only those that are automated.
--> Create a single CKL/CKLB per target system (combine multiple benchmarks results)	Create a single CKL or CKLB file which contains results from multiple benchmarks. One resulting file per target computer.
Save Failed CPE XML result files	This option enables saving of Common Platform Enumeration (CPE) results for SCAP streams that are not applicable to the target system. This option should only be enabled for debugging why a SCAP stream is not performed against a target system. Enabling it will create numerous small XML files, which are not required for any other reporting purpose.

4.4.3.6 Summary Viewer

OPTION	DESCRIPTION
Save Summary Viewer	This option saves an HTML report that is created at the end of each scan, and provides an easy way to see all of the HTML/Text/XML results created by SCC during that scan session.
Summary Viewer Sorting	The summary viewer HTML report can be sorted by three fields. This report is primarily useful if more than one computer or content stream is used. Below is the default sort order and a description of each: 1. Session: The date/timestamp from when the scan started. 2. Stream : The SCAP content or OVAL datastream name being used 3. Host : The hostname for the target computer(s) being scanned.

Note: The Summary Viewer can also be sorted manually by clicking on any column of the report after it's generated.
--

4.4.3.7 Scan Sessions

OPTION	DESCRIPTION
Save Scan Session Information (directory, scores, filenames)	This option enables creating a scan session database which resides in the root of the SCC results directory, and allows for easy viewing and searching and deleting of existing SCC scan sessions.
Double Clicking any log, report or XML file in the session viewer in SCC opens	This option instructs SCC to open SCC created files with a file viewer internal to SCC. If this option is disabled, SCC will attempt to open the file with the Operating System default file viewer based on the file type, although on many linux distributions, this may not work well, as SCC is run as root, and HTML and PDF viewers may not be allowed to run as root.

4.4.4 Logging Options

4.4.4.1 Logging Options

OPTION	DESCRIPTION
Save Screen Log	This option saves the analysis log printed to the "Status" screen to a text file for viewing after the review.
Save Debug Log	This option saves a large amount of additional information related to what occurred during a review. This option is disabled by default and should only be used when attempting to resolve errors in the application, as it will slow down the application and potentially use large amounts of disk space. Please refer to Appendix A.8 Debugging for additional information on SCC debug logs and their intended usage.
Enable Verbose 'TRACE' Level Debug	This option saves even more debug at a very verbose level that is only be recommended to debug crash type issues, as it may generate GB of text and slowdown SCC in general. This option is only available if Save Debug Log is enabled.
Suppress Warnings	This option will prevent warnings from being reported. As warnings are not critical, this option may be desired for certain users. This option may be useful in conjunction with 'ignore remote file systems' and 'ignore case' listed in the File Scanning section.

4.4.5 Output Options

4.4.5.1 Configuration Save Location

OPTION	DESCRIPTION
Save Configuration to the User's Home Directory	This option dynamically sets the base directory in which SCC saves it's configuration (5.12_options.xml) on a per-user basis.

	<i>Ex: C:\users\TestUser\SCC\Config\5.12_options.xml</i>
Save Configuration to the Running Application Directory	This option sets the based directory in which SCC saves it's configuration (options.xml) to the location SCC is running/installed. <i>Ex: C:\Program Files\SCAP Compliance Checker 5.12\options.xml</i>

4.4.5.2 SCC Security Configuration Directory Options

This is a new option for SCC 5.3, and currently contains the SSH Host Credential Database, SSH trusted keys, and other security related items related to SSH. In the future it could contain other security related configuration.

If SSH based scanning is not used, this directory may not exist, and the SSH Host Credential Database is not created until it is used.

OPTION	DESCRIPTION
Save SCC's Security Configuration to the User Home Directory	This option dynamically sets the base directory in which SCC saves it's security configuration to each user's home directory. This option is recommended, both for increased security, and to prevent it's accidental removal, as the host credential database is designed to work across multiple installations of SCC. <i>Ex: C:\users\TestUser\SCC\Config</i>
Save SCC's Security Configuration to the Running Application Directory	This option sets the based directory in which SCC saves it's security configuration to the location SCC is running/installed. Note: This setting is not recommended, due to potential security issues, and loss of credential data if SCC is uninstalled/reinstalled into the same directory. <i>Ex: C:\Program Files\SCAP Compliance Checker 5.12\Config</i>
Save SCC's Security Configuration to a Custom Directory	This option allows the end user to specify any custom directory to save SCC's security configuration. This can be a local path such as C:\SomeDirectory or a network share but only with the UNC path of \\ServerName\ShareName\SomeDirectory. Using a mapped drive such as Z:\SomeDirectory is known to cause errors and is not officially supported. This field is guarded by input validation and will only allow a user to enter an absolute directory path that exists. Any input outside of those values will result in an error and the option will not be allowed to be saved.

4.4.5.3 SCC Results Directory Options

OPTION	DESCRIPTION
Save Results to the User Home Directory	This option dynamically sets the base directory in which SCC saves all Logs and Results on a per-user basis. <i>Ex: C:\users\TestUser\SCC</i>
Save Results to the Running Application Directory	This option sets the based directory in which SCC saves all Logs and Results to the location SCC is running/installed. <i>Ex: C:\Program Files\SCAP Compliance Checker 5.12</i>
Save Results to a Custom Directory	This option allows the end user to specify any custom directory to save all SCC Results and Log. This can be a local path such as C:\SomeDirectory or a network share but only with the UNC path of \\ServerName\ShareName\SomeDirectory. Using a mapped drive such as Z:\SomeDirectory is known to cause errors and is not officially supported. This field is guarded by input validation and will only allow a user to enter an absolute directory path that exists. Any input outside of those values will result in an error and the option will not be allowed to be saved.

4.4.5.4 SCC Results Subdirectory Options

OPTION	DESCRIPTION
Create 'ApplicationLogs' subdirectory	This option creates a directory to store SCC application logs (those not related to performing a SCAP content scan session)
Create 'Sessions' subdirectory	This creates a base directory to store all scan sessions.
> Create 'Date/Timestamp' subdirectories	This option automatically creates a subfolder with the 'Date/Timestamp' within the results directory. (This option is highly recommended, as it's required for new scan session viewing feature.)
>> Create 'Results' subdirectory	This option automatically creates a subfolder of 'Results' within the Sessions directory
>> Create 'Logs' subdirectory	This option automatically creates a subfolder of 'Logs' within the Sessions directory, as a sibling of the 'Results' directory. These 'scan' logs will be directly related to the HTML/Txt/XML files in the 'Results' directory.
>>> Create 'SCAP', 'OVAL', 'OCIL' and 'ApplicationLog' directories	This option automatically creates a subfolder based on the content 'SCAP', 'OVAL', 'OCIL' within the results directory, and ApplicationLogs within the Logs directory
>>>> Create 'Target Name' Subdirectories	This option automatically creates a subfolder with the Target Name' within the results directory. (This option is disabled by default.)
>>>>> Create	This option automatically creates a subfolder with the Content Name'

'Content Name' Subdirectories	within the results directory. (This option is disabled by default.)
>>>>> Create 'XML' Subdirectory	This option automatically creates a subfolder called XML

4.4.5.5 SCC Report Filename Options

OPTION	DESCRIPTION
Target Name	This option adds the target host name to resulting report filename. <i>Ex: Computer1_All-Settings_Mozilla_Firefox.html</i>
Target DNS Domain	This option adds the target dnsname to resulting report filename. <i>Ex: Computer1.yourdomain.gov_All-Settings_Mozilla_Firefox.html</i>
SCC Version	This option adds the SCC version to the report filename: <i>Ex: Computer1_SCC-5.12_All-Settings_Mozilla_Firefox.html</i>
Content Version	This option adds the content version to the report filename: <i>Ex: Computer1_SCC-5.12_All-Settings_Mozilla_Firefox-001.015.html</i>
Date/Timestamp	This option adds a date/timestamp to the report filename: <i>Ex: Computer1_SCC-5.12_2021-02-17_125008_All-Settings_Mozilla_Firefox-001.015.html</i>

4.4.5.6 Permission Options

OPTION	DESCRIPTION
Allow SCC to set restricted permissions on SCC created Logs and Results	This option allows SCC to set restricted permissions on the Logs and Results (XML, Text, HTML) created by SCC. This can be useful especially if results are set to write back to the application install, or some other location where non-privileged users have read access. On Windows: SCC sets the permissions to be the user running SCC, Administrators, and System Disabling this option defaults back to the OS defaults.

4.4.6 SFTP Report File Transfer Options

SCC has the ability to copy results after each scan via SFTP to a centralized server for easier data collection. This option is not related to SSH based scanning, refer to section 4.3 for SSH based scanning of UNIX systems.

Note: *SSHv2 is supported, SSHv1 is not.*

4.4.6.1 SFTP File Transfer Options

OPTION	DESCRIPTION
Enable File Transfers	Transfer any report/log that is enabled
Delete Local Results After Transfer	This option will delete the local results off of the machine after SCC has successfully transferred the files to the SSH server

4.4.6.2 SFTP Server Information

OPTION	DESCRIPTION
Hostname/IP	Enter the DNS hostname or IP Address of the SFTP server to copy result to. This field is guarded by input validation and will only allow a user to enter a hostname or ip address. Any input outside of those values will result in an error and the option will not be allowed to be saved.
Port	Enter the port which the SFTP server is listening (normally 22) This field is guarded by input validation and will only allow a user to enter a port number between 0 and 65536. Any input outside of those values will result in an error and the option will not be allowed to be saved.
Directory	By default, this is set as the user's home directory (e.g. /home/<username>) but can be changed here to reflect the desired directory you would like the reports to be transferred to. This field is guarded by input validation and will only allow a user to enter an absolute directory path that exists. Any input outside of those values will result in an error and the option will not be allowed to be saved.

4.4.6.3 SFTP Connection Type

OPTION	DESCRIPTION
Connection with Username/Password	Select this option if you plan to authenticate with username/password combination
Connection with Private Key/Passphrase	Select this option if you plan to authenticate with a private key. Note: SCC only supports private keys that are secured with a private key passphrase
Username	Required for either Username/Password or Private key authentication This field is guarded by input validation and will only allow a user to enter a proper username (eg, nothing that starts with ";"). Any input outside of those values will result in an error and the option will not be allowed to be saved
Password	Only required if Username/Password authentication has been selected
Private Key	Only required if Private Key/Passphrase authentication has been selected
Private Key Passphrase	Only required if Private Key/Passphrase authentication has been selected

4.4.7 Update Options

4.4.7.1 HTTP Connection Options

OPTION	DESCRIPTION
Use System Proxy	SCC should use the system configured proxy (or not) that has been configured as part of the OS and is used Internet Browsers
Use Environment Variable Defined Proxy	Primarily a UNIX/Linux method, but SCC looks for http_proxy or https_proxy environment variables
No Proxy	SCC connects without any proxy
Custom Proxy	Enter the proxy to be used by SCC, in the format of http://<proxy server>:<port> or https://<proxy server>:<port>

4.4.7.2 SCC Application Updates

OPTION	DESCRIPTION
Periodically Check for SCC Application Updates	Allow SCC to check for application updates when SCC launches. This feature does not download or install/upgrade SCC, it only notifies the user that an updated version is available.
Frequency (Days)	Number of days between SCC Application Checks.
SCC Update URL	Internet (or could be updated to an Intranet location) to query for SCC release information.

4.4.7.3 SCAP Content Updates

OPTION	DESCRIPTION
Periodically Check for SCAP Content Updates	Allow SCC to check for application updates when SCC launches. This feature allows the user to download and install updated content into SCC.
Frequency (Days)	Number of days between SCC Application Checks.
Include pre-release (Draft/Test) versions of SCAP Content	Internet (or could be updated to an Intranet location) to query for SCAP Content information.

4.4.7.4 SCAP Content Repository Options

OPTION	DESCRIPTION
Add Repository URL	Press the Add Repository URL button to add a URL to the repository list.
Edit	Edit an existing URL by right clicking on it, and clicking Edit.
Delete	Delete repository URL by right clicking on it, and clicking Delete.
Delete All	Delete all repository URL's by right clicking and selecting Delete All.

4.4.8 Service Options

Refer to Section 8 - SCC as a Service

4.5 Viewing Results

4.5.1 Viewing Scan Results

After the SCC software completes the review, reports, XML files and any scan related logs can be viewed within SCC by clicking:

'View Results' button on bottom left of main screen

or

Results -> View Results

Scan Sessions:

This section displays scan Sessions, which corresponds to each click of 'Start Scan' or each review performed by command line. Each session may contain many computers, with many different SCAP content streams. When opening this form, the most recent scan session is automatically selected. Left click to select any scan session, or right click for more options. After selecting a session, see Results and Reports below.

Results:

This section displays the high level results of a SCAP scan; it contains the hostname, scap content name, score, error count and warning count. Left click any row to select different Host/content combinations.

Reports/XML/Logs:

This section displays the reports, XML and Logs (if created) based on the Host/Content combination selected in the Results pane. Double click to open the report in SCC, or right click for more options.

4.5.2 Viewing Application Logs

If something unexpected occurs within the SCC GUI, which is not part of a SCAP scan, application logs can be useful in determining the cause.

Results -> View Application Logs

Application Logs:

This form displays any application logs (if any exist, such as screen, debug or error). Double click to open the log in SCC, or right click for more options.

4.6 SCAP Scanning with Manual and/or Hybrid Questions

Starting with SCC 5.7, SCC includes NWIC 'Enhanced' STIG SCAP content by default, which converts non-automated requirements from the STIG Manual into manual questions in SCC, which are included as OCIL.

4.6.1 Answering Manual Questions in SCC's GUI

1. Click on the desired content in the SCAP content tree.
2. Click on 'Manual Questions' on the Content Details pane.
3. A new Manual Questions form should appear.
4. Click on any row in the tree on the left side of the page, the first question will be selected by default.
5. Select the desired answer from the available options.
6. Enter any comments in the Comments text field.
7. Click 'Save Answer and Comments'.
8. This should take you to the next unanswered question. Repeat until all questions are answered.

4.6.2 Answering Hybrid Questions in SCC's GUI

Starting with SCC 5.9, SCC includes the ability to take end user entered values in a very specific format refer to **Section 6** of the User Manual which explain Hybrid questions in detail and provide several usage examples.

- Review system documentation and determine what is authorized for the STIG rule in question
- Determine which targets the authorized value(s) is/are applicable to

4.6.2.1 Using SCC's variable builder

1. Click "New" to create a new hybrid variable
 - a. Select Scope from the dropdown
 - b. Enter the target based on the format provided in the user interface
 - c. Enter authorization for authorized values, such as your SSP document date/version/page, ISSM or whoever approved the setting.
 - d. Enter the authorized_value(s) as a single word, CSV delimited set of words, or one word per line
 - e. Enter any private 'notes' if desired. This field is only used for end users to self-document how/why variables exist. Not included in any reports.
 - f. Click Save
2. Editing an existing variable row
 - a. Either select an existing row with a single left click, and then click Edit, or double click on the row.
 - b. Make changes as desired and click Save
3. Deleting an existing variable row
 1. Select a row by single left clicking and then clicking Delete
4. Moving rows up or down
 0. Select a row by single left clicking and clicking Move Up or Move Down

Note: Each row will be processed in order priority, and only the first value found that is applicable is applied to a target, so a target of 'ALL' (if needed) should only be used last in a list.

4.6.2.2 Creating hybrid variables using text editing

1. Enter the values as needed in the format of

```
<hybrid_variables>
  <hybrid_variable>
    <scope></scope>
    <target></target>
    <authorization></authorization>
    <authorized_values>
      <authorized_value></authorized_value>
    </authorized_values>
    <notes></notes>
  </hybrid_variable>
</hybrid_variables>
```
2. Enter as many hybrid_variable XML blocks as needed for your target(s) and values, note that each hybrid_variable will be processed in order priority, and only the first value found that is applicable is applied to a target, so a target of 'ALL' (if needed) should only be used last in a list.

Manual editing example:

A very simple example of this would be, if the value of 'dbo' was the required setting for all database scanned, then the following should be entered:

```
<hybrid_variables>
  <hybrid_variable>
    <scope>DATABASE</scope>
    <target>ALL</target>
    <authorization>My SSP, My
ISSM</authorization>
    <authorized_values>
      <authorized_value>dbo</authorized_value>
    </authorized_values>
  </hybrid_variable>
</hybrid_variables>
```

4.6.2.3 Saving the hybrid variable

1. Click Save Authorized Values

4.6.3 Save and Close

When you have completed all of the questions, or all that you want to change for now, click 'Save Document and Close'. This will save your answers and they will be applied to all future scans.

4.6.4 Automatic Backups of Manual Questions Results

If you have the option enabled to automatically backup Manual Question results, a new backup of the text based results will be stored to a subdirectory called Backups, each time you press Save Document and Close Form. If your Manual Question autoanswer results file in the Completed_Results directory somehow gets corrupt or lost, you can manually restore a backup version.

Backups will be in the format of
<BenchmarkID>_<BenchmarkVersion>.txt_<datetime>_<username>

Example:

MOZ_Firefox_Windows_6.3.2_Autoanswer.txt_2023-02-16_084507_testUser1

To restore a backup file, rename the file from:

MOZ_Firefox_Windows_6.3.2_Autoanswer.txt_2023-02-16_084507_testUser1
to MOZ_Firefox_Windows_6.3.2_Autoanswer.txt

Then copy the renamed file from the Backups directory to the Completed_Files directory.

4.6.5 Interaction with Manual Question Autoanswer Files

Starting with SCC 5.7, SCC supports a text based method for answering manual questions, via autoanswer text files. SCC's OCIL interface reads from and updates these files, so an end user can update an autoanswer file, or answer questions via SCC GUI and the results should be the same. Refer to section 5.7 for more information on SCC's new text based manual question autoanswer support.

4.6.6 Upgrading Manual Question Results

Starting with SCC 5.7 (on Windows) SCC will prompt to upgrade configuration and manual question results on first launch. If the manual question results are configured to be stored in the application installation directory (default), SCC will copy the files from 5.7 to 5.7.1 etc. If the manual question results directory has been configured to a custom directory, SCC 5.7.1 will use that custom directory, as a side effect of the configuration upgrade.

For Linux: When using package based installation, with the RPM or DEB upgrade method selected, SCC will install to the same directory as previous versions, upgrade the configuration to the current version, and have all of the manual questions results available.

4.6.7 Converting Manual Question Results to Newer SCAP Content

Starting with SCC 5.7.1, SCC now automatically converts manual question results to the latest SCAP content. Only questions that match the same XCCDF rule and version are updated, so if any change is made to an XCCDF rule in the updated STIG manual, answers will not be converted and those questions will need to be re-answered.

4.7 XCCDF Tailoring

SCAP 1.2 and SCAP 1.3 specifications allow for a separate 'tailoring' XML file which allows users to customize certain portions of the XCCDF stream, without modifying the source XML file.

Customizations include:

- o Creation of a new profile
 - o All modifications will be reported under a new profile, with the original profile name followed by "_tailored"
- o Selecting and deselecting rules and groups
 - o This allows users to disable rules that may cause incorrect results, or take too long to complete.
 - o Results will be marked as 'not selected'
- o Modifying refine-rules, values and refine-values
 - o This allows users to modify certain rules to meet their organizational requirements. Note that not all rules will have modifiable values, and OVAL XML content may need to be edited.

SCC supports this feature, and provides a graphical interface for creating and editing the XCCDF tailoring file.

4.7.1 Selecting/Deselecting Groups and Rules

To enable/disable groups or rules

1. Left click on the checkbox on the far left on any row listed with a Type "Rule".
2. Enabling or Disabling a group will cascade Enable/Disable to all sub-groups and rules.
3. Enabling/Disabling can also be performed by left clicking on the row, and clicking the check-box in the right hand window.

4.7.2 Rule Details

This section displays the internal rule ID, Title, Description and FixText, along with a Rule Remarks text entry field that can be used to document why a rule is enabled or disabled.

4.7.3 Editing Refine-Rule Properties

Field	Description
Selector	A dropdown that shows any selector that may be present. For NIWC Enhanced content it contains "Automated" or "Manual", and allows users to change any automated rule to a manual question, which could be useful if the automated rule does not meet the end users organization needs. Instead of disabling the rule, the rule can then be performed with a manual question. For DISA created content, this dropdown will be empty.
Weight	End user customizable field to change the weight of the rule, which is used in calculating the overall score.
Severity	Dropdown for end user customization of the severity. SCAP allows for "high", "medium", "low", "info" and "unknown"
Role	Rules can be set to 'full', which is the default (and recommended option), but can also be set to 'unscored', where the test is performed, but failing does not impact any compliance score, or unchecked, which is essentially the same as disabling a rule.
Refine	Enter any text that you would like to see in SCC reports, which

Remarks	explains any updates made to the refine-rule properties.
---------	--

4.7.4 Values and Refine-Values

To modify a value:

1. Click on any row in tree listed with a Type of "Value"
 - A. Enter the desired value in the 'Set Value To' field, and click "Apply Value"
 - OR
 - B. Select the Refined Value Selector

Note that the Refine Value Selector may only have a single row, unless the SCAP content contains multiple options to choose from.

Note: Values and refine-values may not be present in content.

4.7.5 Importing an existing XCCDF Tailoring File

To import an existing XCCDF Tailoring File, click the button and browse to the desired XML file. Note that importing will overwrite your existing tailoring information if present.

4.7.6 Exporting a XCCDF Tailoring File

To export your XCCDF Tailoring information to share with another installation of SCC, click the Export button and browse to the desired destination directory.

4.7.7 Saving XCCDF Tailoring

Click Save and Close. The selected profile on the main SCC form should be updated to reflect the new <Profile>_tailored profile.

4.7.8 Entering XCCDF Tailoring File Creator Info

After clicking Save, a form will appear with the following fields. These will all be saved to the resulting XML results and HTML/Text reports if enabled.

Field	Description
Agency/Organization	Enter your Agency/Organization Name
Full Name	Enter your full name
Notes	Enter any notes that help explain why this XCCDF Tailoring file was created.
Version	Enter a version for this XCCDF Tailoring file
Status	Select the status of this XCCDF Tailoring file (Draft, Accepted, Deprecated, Incomplete, Interim)

4.9 Cisco IOS / IOS XE Offline (Show Tech) File Scanning

This option instructs SCC to scan an existing Cisco IOS/IOS-XE "show tech" file. The file format required for this type of assessment is very specific. To create a configuration file suitable for using in SCC:

1. Logon to the device with full (level 15) privileges (enable mode)
2. Type "show tech" saved to a single text file
3. Save the file as "<devicename>_<date>.txt"
NOTE: Save this file as Text Only with Line Breaks using a plain text editor such as Notepad. Do not use Microsoft Word or other document editor. Only one device per file
4. Disable any Windows or UNIX SCAP/OVAL content, otherwise many content errors will be reported
5. Enable just Cisco IOS specific SCAP/OVAL content

4.9.1 Browse for Cisco IOS / IOS XE Show Tech File

This option allows the end user to select a single file, either a text file (.txt) containing single Cisco IOS/IOS-XE "show tech" file, or a zip file containing a collection of Cisco IOS/IOS-XE "show tech" configuration file.

4.9.2 Browse for a directory containing Cisco IOS / IOS XE show tech files

This option allows the user to select a directory which contains a collection of text (.txt) based Cisco IOS/IOS-XE show tech files.

4.9.3 Appending other 'show' commands not included in 'show tech'

In the event that one or more IOS line tests error due to unsupported "show" commands, take note of the list of unsupported commands that are listed at the end of the analysis. Most Cisco IOS content is likely to attempt collection of "show" command output that is not provided by the "show tech" or "show tech-support" commands by default. The best recourse is to manually execute any unsupported commands and append their output to the "show tech" command output, or to perform the scan via SCC's SSH based scanning mode and let SCC collect all of the required show commands automatically.

To manually append additional 'show' command to an existing 'show tech' file:

1. Logon to the device with full (level 15) privileges (enable mode)
2. Execute the listed commands
3. Copy the output of each command and append it to the original "show tech" output file with a leading header on its own line that specifies the command that generated the output
4. Ensure each header consists of a sequence of 18 hyphens, a space, full show command (no shorthand notation), another space, and 18 more hyphens for consistency with the default "show tech" command output

Example:

<existing output of show tech> then

```
----- show startup-config -----  
Using 3130 out of 524284 bytes, uncompressed size = 8791 bytes  
!  
<output of show startup-config>
```

4.10 Standalone OVAL Usage

Standalone OVAL content usage is designed primarily for advanced SCC users, such as content authors who wish to run OVAL content without creating an entire SCAP benchmark. It can also be used to perform vulnerability scanning using existing OVAL vulnerability content. No standalone OVAL content is currently bundled with SCC.

1. Launch the Graphical User Interface of SCC
2. Follow instructions for Installing & Configuring Content (Section 4.2)
3. If standalone OVAL content is found during the content install, an OVAL content tab will appear for enabling/disabling content.

4.10.1 External Variables

If standalone OVAL content uses an external variables file, the file will need to match the name of the base oval document, but ending with "external-variables.xml". When installing standalone OVAL with external variables, just include the file in the zip file to be installed.

Example:

- o MyContent.zip
 - o myOVALContent.xml
 - o myOVALContent_external-variables.xml

4.11 Standalone OCIL Usage

Standalone OCIL (Open Checklist Interactive Language) content usage is designed primarily for advanced SCC users, such as content authors who wish to run OCIL content without creating an entire SCAP benchmark. No OCIL content is currently bundled with SCC.

4.11.1 Installing OCIL content

1. Launch the Graphical User Interface of SCC
2. Follow instructions for Installing & Configuring Content (Section 4.2)
3. If standalone OCIL content is found during the content install, an OCIL content tab will appear for enabling/disabling content.

SCC will also search the <SCC Install>\Resources\OCIL_Content directory and subdirectories for OCIL XML files.

Note: OCIL content that is part of a SCAP Stream, such as DISA STIGS, should be installed into the SCAP Content, not in the OCIL content. OCIL content included in a SCAP Stream is designed to work with XCCDF, OVAL and CPE. SCC might be able to process the OCIL file as 'raw OCIL' but unexpected errors may occur. For more information please see section 4.4.

OCIL schema validation occurs during installation of the OCIL content. If the OCIL content is deemed invalid, SCC will inform the user via a dialog box but will continue the installation process. The user will be notified of the result of the installation process in a separate dialog box.

4.11.2 Performing a standalone OCIL analysis

4.11.2.1 Creating and Selecting OCIL Targets

Unlike the automated portions of SCC, which the target is always a computer, whose hostname is automatically populated by programmatic means during the scan, with OCIL, the target could be a computer/system or a user/person, and each will need to be manually entered.

To create a new system target, click on the 'add target system' in the in upper left corner of the form. Then enter in as much information as you have.

FIELD	DESCRIPTION
System Name	The name of the system, generally a computer, which is the target for the checklist. This field is mandatory.
IP Address	Optional field to report on the IP Address associated with this system.
Organization	Optional field to document the organization that the system is a part of.
Description	Optional field to enter any other information about this system that might be relevant.

To create a new user target, click on the 'add target user' in the in upper left corner of the form. Then enter in as much information as you have.

FIELD	DESCRIPTION
Full Name	The name of full name of the person being interviewed. This field is mandatory.
Email	Optional field to report on the email address of the person being interviewed.
Organization	Optional field to document the organization that the person is a part of.
Description	Optional field to enter any other information about this person that might be relevant.

After creating and selecting the desired targets, click "Continue".

4.11.2.2 Answering Questions on the OCIL Document Form

This form allows you to answer any questions in the OCIL questionnaire. All of the question titles will be listed in the tree on the left hand side. To answer a question, click at the leaf node, which has a grey circle. This will populate the associated question on the right hand side.

4.11.2.3 Entering Artifacts

For each question, the user can manually enter an artifact which may help explain the results. To enter an artifact, click on the Add Artifact button.

4.11.2.4 Entering Content Mandated Artifacts

The OCIL content can also ask specifically for artifacts, which will cause a new form to appear, which contains:

FIELD	DESCRIPTION
Title	Content provided title
Description	Content provided description, likely explaining what data the end user should provide
Artifact	Location for the end user to enter data as specified in the Description field.
Save	After entering Data, click Save to close this form.

4.11.2.5 Save and Close

When you have completed all of the questions, click "Save Questionnaire Session and Close". This will take you back to the OCIL Questionnaire Form, or to the next Questionnaire if more than one has been enabled.

4.11.3 Generating Reports

If you have answered all of the OCIL Questionnaire, click "Create Reports" to proceed to creating reports with the results.

4.12 Post Scanning Report Generation

SCC, by default, creates most of the commonly used reports during each scan. However, additional reports can be from previous scan results. These are completely optional depending on your desired usage.

4.12.1 Select Directories

OPTION	DESCRIPTION
Source Directory	Location for the application to scan for XCCDF and/or OVAL XML results from previous reviews. This option is recursive (all subfolders will be scanned for files to use).
Destination Directory	Location where summary reports are to be saved.
Open Destination Directory when processing is complete	This opens Windows Explorer to the directory containing the new summary reports, when the processing is complete.

4.12.2 Reports

4.12.2.1 Generate Summary SCAP Reports (from XCCDF results)

SCC can generate multi-computer summary reports from the XCCDF XML (SCAP) results created by the SCC or other SCAP Validated applications.
To generate summary reports from existing XCCDF XML files:

Note: To create reports based on a subset of computers in the organization, organize the consolidated data in a directory structure similar to the example listed below:

*/Entire Organization
 /Sub Organization 1
 /Sub-Sub Organization
 /Sub Organization 2
 /etc.*

If the SCC is pointed at the entire organization, or any subset, the summary reports will only contain the desired subset of computers.

4.12.2.1.1 Select Reports to Generate

OPTION	DESCRIPTION
Site Summary	This report provides a consolidated list of checks, with a single CCE reference and the Pass, Fail, Error, Unknown, Not Applicable, Not Checked, and Not Selected occurrences for each check.
Site Summary Non-Compliance	This report provides a consolidated list of checks, with a single CCE reference and the Fail, Error and Unknown occurrences for each check that had a fail or an error status.

Computer List	This report lists the latest results for all computers reviewed and the Review Time, Pass, Fail, Error, Unknown, Not Applicable, Not Checked, Not Selected, Total along with the Original and Adjusted Scores.
Computer List Historical	This report lists all results for all computers reviewed, and the Review Time, Pass, Fail, Error, Unknown, Not Applicable, Not Checked, Not Selected, Total along with the Original and Adjusted Scores.

4.12.2.2 Generate Detailed SCAP Reports (from ARF or XCCDF/OVAL results)

SCC can regenerate single computer detailed reports from the XCCDF XML and OVAL XML (SCAP) results created by the SCC or other SCAP Validated applications.

4.12.2.2.1 Select Reports to Generate

REPORT	DESCRIPTION
All Settings	This report contains detailed pass and fail results from each check performed. It is a large report and is not intended for printing.
All Settings Summary	This report contains a summary of pass and fail results from each check.
Non-Compliance	Non-compliance reports contain detailed results from each failed check. It is a large report and is not intended for printing.
Non-Compliance Summary	This report contains a summary of the failed checks.

4.12.2.3 Generate Detailed OVAL Reports (from standalone OVAL results)

SCC can regenerate single computer detailed reports from the standalone OVAL XML results created by the SCC or other SCAP Validated applications.

4.12.2.3.1 Select Reports to Generate

REPORT	DESCRIPTION
All Settings	This report contains detailed pass and fail results from each check performed. It is a large report and is not intended for printing.
All Settings Summary	This report contains a summary of pass and fail results from each check.
Non-Compliance	Non-compliance reports contain detailed results from each failed check. It is a large report and is not intended for printing.
Non-Compliance Summary	This report contains a summary of the failed checks.

4.12.3 Select File Format(s) of Generated Reports

FORMAT	DESCRIPTION
--------	-------------

HTML	HTML formatted reports for viewing with a web browser
Excel	Excel Spreadsheet versions with separate tabs per SCAP stream

4.12.4 Generate Reports

To create the summary reports, click Generate. The status window will display the progress.

Note: Summary reports can also be created with a command line parameter, based on the settings configured in the GUI. Please refer to the Using the Software via Command Line for additional information.

4.12.5 Viewing Reports

After Multi-Computer Summary Reports are created a Windows Explorer window may open (based on user preferences), and the reports specified to be created will be available in HTML and/or Excel spreadsheets. To view, double click on any of the files.

4.13 File and Help Menus

4.13.1 File Menu

MENU ITEM	DESCRIPTION
Manage Configuration Profiles	This will open a form which allows users to switch between SCC configuration profiles. See section 4.13 Configuration Profiles for more information.
Import Options	This will delete your current options file and replace it entirely with the file you specify. This is only applicable to options files that match the same version of the SCC application as you are running. If an options file from an older version of SCC is specified, it quite likely will fail schema will fail, and errors will be reported. This is not intended as an upgrade feature.
Export Options	This will make a copy of your options XML file for backup purposes or for sharing with another installation of SCC.
Upgrade Options	This will upgrade your current installation of SCC using an existing options xml from an existing installation of SCC. This will not delete your current options file, it will keep your current options, and overlay them with the matching configuration settings from the previous installation. There are exceptions to this, as specified below: - o Content is not upgraded, nor are the enabled/disabled status for content. - o appUpdateURL and contentRepository settings are intentionally excluded, as they may change in a future release of SCC and upgrading could cause issues. If you have your own SCAP content repository or SCC application update repository setup, these files can be manually configured after the upgrade is complete.
Restore Default Options	This will delete your current options XML file, and restore them using the installation defaults.
Exit	Exits the SCC application.

4.13.2 Help Menu

MENU ITEM	DESCRIPTION
Help	Opens the SCC compiled help application (Windows Only).
User Manual	Opens this PDF.
Online Tutorials	Opens a web browser to https://www.niwcatlantic.navy.mil/scap/videos/
Release Notes	Displays SCC's Release Notes file.
Check For Content Updates	Downloads a small XML file from the URL(s) specified in the Options -> Update Options -> Content Repositories option, and then compares current content to that in the XML file and prompts the user to install any new or updated SCAP content.

Check for SCC App Updates	Downloads a small XML file from the URL(s) specified in the Options -> Update Options -> SCC Update URL option, and then compares current content to that in the XML file and informs the user if there is an updated version of SCC available for download.
End User License	Displays SCC's End User License Agreement.
Third Party Licenses	Displays a listing of all Third Partly Licenses in use by SCC.
Rate SCC via Online Survey	Our survey is now online 24/7 and we get reports on a monthly basis, so you can rate what you think of the SCC application and our tech support anytime. The website is: https://usnavy.gov1.qualtrics.com/jfe/form/SV_4ZpXv8JkUIDs4lw
About	Lists SCC's version, build number and supported SCAP features, and contact information.

4.13 Configuration Profiles

SCC's configuration profiles save all configuration settings of SCC and make it easily available for switching between. Every configuration setting, and hosts enabled are kept as part of the configuration profile, so you can select a collection of targets, content, and have custom tailoring or custom manual questions for a group of hosts, and then switch to another profile that has different hosts, content etc...

4.13.1 Profiles

4.13.1.1 New Profile

To create a new profile, click "New Profile", this will display a text entry box in which you can enter the name of the profile you wish to create. Only letter and numbers are supported, as this will be used the filename of the new options file to be created.

Ex: TestProfile1

4.13.1.2 Select a Profile

By default, SCC only has one profile called 'default', and it cannot be renamed. To create your own profile, click the New profile button. After you have created one or more new profiles you can select it from the dropdown list.

4.13.1.3 Delete Profile

This deletes the options file related to your custom profile and removes it from the dropdown list.

Profile Metadata

This text box displays some of the key options that may help you decide which profile to use. This is not a complete list, as every option available in SCC is stored as part of your new profile.

Example:

Options File: C:\Program Files\SCAP Compliance Checker
5.12\options.xml
Last Updated: 2023-07-28T14:44:15
Content Enabled: 5 of 16
Scan Type: local
Results Directory: User Home Directory (dynamic)
Tailoring and Manual Questions: Installation Directory (default)

5. COMMAND LINE USAGE

SCC has a separate executable for command line usage which is included in the installation package as 'csccl.exe'. The Command-line SCAP Compliance Checker (CSCC) allows for scripted or automated reviews by other applications or scheduled tasks.

Any changes made via the SCC GUI such as content installation, or application preferences impact the command line interface and vice versa, as the options for both interfaces are saved to the same 'options.xml' file located in the SCC installation directory.

5.1 Basic Command Line Usage

Below is a quick overview of how CSCC works.

1. Open a Command Prompt with an account that has Administrator privileges.
2. Install any additional SCAP Content into CSCC.
3. Run the Configuration Menu option of CSCC (--config).
4. View available SCAP content included with CSCC.
5. Enable SCAP Content and Select the desired profile from each SCAP Content stream.
6. Scan Computer(s) with Enabled SCAP Content.
7. View reports.

To view all available command line options, use the -? parameter.

```
csccl.exe -? or csccl.exe --help
```

5.1.1 Open a Command Prompt

Open an command prompt (admin for any local scanning) and change directory to the SCC installation directory.

Example:

```
Start -> Run -> cmd
```

```
C:
```

```
cd "\\Program Files\\SCAP Compliance Checker 5.12"
```

5.2 Command Line Configuration Parameters

5.2.1 Configuration Parameters

Below are the parameters available for installing content and configuring application options. All of the following options must be used individually, and are not compatible with any other parameter.

For descriptions of each option available to be edited, refer to section 4.4, Editing Options

To see the help related to configuration parameters run '**csc.exe --help config**'

```
#####
```

CONFIGURATION PARAMETERS:

Below are the parameters available for installing content and configuring application options. All of the following options must be used individually, and are not compatible with any other parameter.

-lc, --listConfigProfiles

Prints a list of all available SCC configuration profiles

-nc, --newConfigProfile

Creates and enables a new configuration profile for all future usage

based on the current configuration. This only supports letters and numbers

all other characters will be removed

Example: `csc.exe --newConfigProfile myProfile2`

See `--listConfigProfiles` for a list of config profiles

-sc, --selectConfigProfile

Selects a configuration profile for all future usage

Example: `csc.exe --selectConfigProfile myProfile`

See `--listConfigProfiles` for a list of config profiles

--config

Open a command line menu which displays several configuration options

-eb, --enableBenchmark BENCHMARK_ID [version]

Enable a benchmark based on its ID, with optional benchmark version.

Example: csc.exe --enableBenchmark Windows_10_STIG

Example: csc.exe --enableBenchmark Windows_10_STIG 001.002

See --listAllBenchmarks for a list of benchmarks.

-db, --disableBenchmark BENCHMARK_ID [version]

Disable a benchmark based on its ID, with optional benchmark version.

Example: csc.exe --disableBenchmark Windows_10_STIG

Example: csc.exe --disableBenchmark Windows_10_STIG 001.002.

See --listAllBenchmarks for a list of benchmarks.

-ct, --createTailoringFile BENCHMARK_ID [version]

Create a XCCDF Tailoring file which allows for manual edits if user does not have access to SCC GUI for tailoring.

This feature creates the file, and selects that newly created tailored profile for use

Example: csc.exe --createTailoringFile Windows_10_STIG

Example: csc.exe --createTailoringFile Windows_10_STIG 001.002

See --listAllBenchmarks for a list of benchmarks.

-aa, --applicableToAll BENCHMARK_ID [version]

Set a benchmark applicable to all targets based on its ID, with optional benchmark version.

Example: csc.exe --applicableToAll Windows_10_STIG

Example: csc.exe --applicableToAll Windows_10_STIG 001.002

-na, --notApplicableToAll BENCHMARK_ID [version]

Remove the applicableToAll setting based on its ID, with optional benchmark version.

Example: csc.exe --notApplicableToAll Windows_10_STIG

Example: csc.exe --notApplicableToAll Windows_10_STIG 001.002

-ea, --enableAll

Enable all SCAP and OVAL content.

-da, --disableAll

Disable all SCAP and OVAL content.

-ub, --uninstallBenchmark BENCHMARK_ID [version]

Uninstall a benchmark based on its ID, with optional benchmark version.

Example: `csc.exe --uninstallBenchmark Windows_10_STIG`

Example: `csc.exe --uninstallBenchmark Windows_10_STIG 001.002`

See `--listAllBenchmarks` for a list of benchmarks.

-ua, --uninstallAll

Uninstall all SCAP and OVAL content.

--setProfile PROFILE BENCHMARK_ID

Set a profile to be applied to a specified content stream.

See `--listAllProfiles` for a list of profiles.

See `--listAllBenchmarks` for a list of benchmarks.

Example: `csc.exe --setProfile MAC-3_Sensitive Windows_10_STIG`

--setProfileAll PROFILE

Set a profile to be applied to all content installed in SCC, if applicable. If a profile cannot be applied to a content stream it is not applicable. See `--listAllProfiles` to obtain a list of profiles.

Example: `csc.exe --setProfileAll MAC-3_Sensitive`

--setOpt OPTION VALUE

Advanced user setting which allows command line configuration of any SCC option to a user specified value.

`--setOpt` can be called multiple times in a single command if needed, see second example.

Available options can be found in `--listOpt`, and need to be specified exactly. To set a value to an empty string, enter the value as all caps

NULL

Example: `csc.exe --setOpt dirSessionEnabled 0`

Example: `csc.exe --setOpt dirSessionEnabled 0 --setOpt debugEnabled 1`

--generateOptionsFile

Delete the options file, restore default settings, and reinstall all content. Note that this may take a few minutes.

--generateAutoAnswerTemplates

Delete and regenerate all of the Manual Question auto-answer template files found in

`Resources/Content/Manual_Questions/Templates`

--restoreDefault

Restore all options to installation default.

-is FILE PROFILE [--force], --installScap FILE PROFILE [--force]

Install one or more SCAP content streams from an XML file or ZIP archive. Specifying an XCCDF benchmark profile name after the filepath will enable that profile for the given SCAP stream. Use the optional --force switch to reinstall

Example: cscs -is C:\Users\User1\SampleScapContent.zip

Example: cscs --installScap C:\Users\User1\SampleScapContent.zip

Example: cscs -is C:\Users\User1\SampleScapContent.zip MAC-3_Sensitive

Example: cscs -is --force C:\Users\User1\SampleScapContent.zip

Example: cscs --installScap --force
C:\Users\User1\SampleScapContent.zip MAC-3_Sensitive

-isr FILE PROFILE [--force], --installScapRun FILE PROFILE [--force]

Install, enable, and conduct an analysis with a SCAP Content stream from a zip file. Specifying an XCCDF benchmark profile name after the filepath will enable that profile for the given SCAP stream. Use the optional --force switch to reinstall.

Example: cscs -isr C:\Users\User1\SampleScapContent.zip

Example: cscs --installScapRun C:\Users\User1\SampleScapContent.zip
MAC-3_Sensitive

Example: cscs -isr --force C:\Users\User1\SampleScapContent.zip

Example: cscs --installScapRun --force
C:\Users\User1\SampleScapContent.zip MAC-3_Sensitive

-iv FILE [--force], --installOval FILE [--force]

Install OVAL Content from a single xml file or a zip file containing multiple xml files. Use the optional --force switch to reinstall.

Example: cscs --installOval C:\Users\User1\sampleOval.xml

Example: cscs -iv --force C:\Users\User1\sampleOval.xml

-ivr FILE [--force], --installOvalRun FILE [--force]

Install, enable, and conduct an analysis with OVAL Content from a single xml file or a zip file containing multiple xml files. Use the optional --force switch to reinstall.

Example: cscs --installOvalRun C:\Users\User1\sampleOval.xml

Example: cscs -ivr --force C:\Users\User1\sampleOval.xml

--installTailoringProfile FILE

Install an existing XCCDF Tailoring Profile file from another installation of SCC, created by the SCC GUI Tailoring interface. Installing a tailoring profile will set the selected profile for the matching content to be the tailored profile in the selected tailoring file

Example: `cscs --installTailoringProfile <filepath to tailoring xml>`

--installDeviationFile FILE

Install an existing SCC Deviation file from another installation of SCC, created by the SCC GUI Deviation interface.

Installing a deviation file will automatically enable it to be used for future scans.

Example: `cscs --installDeviationFile <filepath to deviation xml>`

--checkForContentUpdates [--installUpdates | --installAll]

Check for SCAP content updates from an online content repository.

Additional settings may need to be pre-configured before usage.

refer to: `cscs --config -> Options -> Update Options`

Example: `cscs --checkForContentUpdates`

Example: `cscs --checkForContentUpdates --installUpdates`

Example: `cscs --checkForContentUpdates --installAll`

--installUnixPlugin FILE

Install the UNIX Plugin file, to allow SSH based scanning of remote UNIX hosts. The file below can be obtained from the same location you downloaded SCC: `SCC_5.5_UNIX_Remote_Scanning_Plugin.scc`

Example: `cscs --installUnixPlugin
<path>SCC_5.5_UNIX_Remote_Scanning_Plugin.scc`

Refer to `--installCredentialDB` if this computer does not have the ability to open the SCC GUI to update/maintain the hosts/credentials

--installCredentialDB FILE

If this computer is not able to open the SCC GUI to create/maintain the SCC Host Credential Database, which is used to enter hosts and credentials for SSH based remote scanning of UNIX and Cisco devices, this feature allows you to install a previously created Host Credential Database and use it for scanning.

*** NOTE 1: You will not be able to create new hosts, or edit credentials, just perform scans using the Master Password for the existing set of hosts/credentials. When host passwords expire, or the Master Password expires, you'll need to obtain an updated Host Credential DB and reinstall it via this command.

*** NOTE 2: This feature will always overwrite the existing host credential db (if found). You will need to manually copy the `hostCredentials.db` from another installation, by default it's found in `<your home directory>\SCC\Config`

--upgrade FILE

Upgrade the configuration options from a previous installation of SCC.

SCC will load the current settings, and overlay the configuration settings from the selected option file, excluding content, appUpdateURL and contentRepository.
Example: `csc -upgrade <filepath to existing SCC options xml>`

5.3 Command Line Scanning Parameters

5.3.1 Scanning Parameters

Below are the parameters available for performing scans. Many of the options can be used in combination, unless indicated below. Any configuration change from a Scanning Parameter is temporary, and does not get saved for future use.

To see the help related to scanning parameters run '**cscs.exe --help scan**'

```
#####
```

SCANNING PARAMETERS:

Below are the parameters available for performing scans. Many of the options can be used in combination, unless indicated below. Any configuration change from a Scanning Parameter is temporary, and does not get saved for future use.

no parameters

Review the local computer based on the configuration settings found in options.xml. If options.xml does not exist in the installation directory, it will be created based on application defaults

-d, --debug

Create a verbose debug log file in the Logs directory for troubleshooting purposes.

-ds, --debugToScreen

Debug to the Screen. This option will print a very large amount of data to the terminal, which can be captured and shared with our team, and should only be used to help diagnose crash type issues.

-dn, --doNotSaveChanges

Do not save changes back to SCC's options file.

This option prevent SCC from updating the option XML file being used and is primarily for usage with automation such as cron.

-ear, --enableAllRun

Enable all SCAP and OVAL content and run content.

-u DIRECTORY, --userDir DIRECTORY

Temporarily configure SCC to save user results and logs to the specified directory path.

Example: cscs -u C:\Users\User1

-cd DIRECTORY, --configDir DIRECTORY

Temporarily configure SCC to save application configuration files to the specified directory path.

Example: cscs -cd C:\Users\User1

-f FILE, --file FILE

Review all Windows computers specified in the new line delimited host file.

Example: cscs -f C:\Users\User1\sampleHosts.txt

-h HOSTNAME, --host HOSTNAME

Review the specified computer.

Example: cscs -h testcomputer1

--wmi --file FILE | --domain | --ou "OU=OU Name"

Review all computers specified with WMI method.

Note 1: Many command line parameters such as -d, -q, -r, -mr, -ear are not compatible with --wmi and should be configured via --setOpt or --config prior to calling cscs --wmi

Note 2: If the WMI alternate credentials feature is enabled, you will be prompted for the master password. For any automated scripting, the alternate credentials mode will need to be disabled, which is the default.

Note 3: If using the --ou method, make sure to use the following format (including quotes) "OU=Your Ou Name" If the OU is a child of another OU, use the LDAP comma separated format of "OU=OU Child,OU=OU Parent"

To scan multiple OU's, separate OU declarations with a semicolon ; (no spaces)

"OU=OU Child1,OU=OU Parent1;OU=OU Child2,OU=OU Parent2"

Example: cscs --wmi --file C:\Users\User1\sampleHosts.txt

Example: cscs --wmi --domain

Example: --wmi --ou "OU=Test Computers,OU=Lab"

Example: cscs --wmi --ou "OU=Test Computers1,OU=Lab;OU=Test Computers2,OU=Lab"

--scanDomain [nslookup ping]

Review the entire active directory domain using 'classic' mode.

Optionally verifying via nslookup and/or ping

--force32

Open the 32 bit version of SCC on 64 bit Windows

--ssh [cisco|unix]

Review all Cisco IOS/IOS-XE OR UNIX computers enabled in the Host Credential Manager, which is only available via the SCC GUI. You will be prompted to enter the SCC Host Credential Master Password in order to perform a remote command line SSH based scan.

Note that many command line parameters such as -d, -q, -r, -mr, -ear are not compatible with --ssh and should be configured via --setOpt or --config prior to calling cscs --ssh

'--ssh unix' can be used in combination with --wmi to scan both UNIX and Windows remotely at the same time.

Refer to --installCredentialDB for installing an existing SCC Host Credential Database from another system which is able to use the SCC GUI

--cisco FILE

Conduct an offline review against a Cisco IOS/IOS XE configuration file or ZIP archive of multiple configuration files located at the given file path.

**** Configuration files should be created with the 'show tech' command

Example: cscs --cisco C:\Users\User1\sampleConfigFile.txt

-o FILE, --options FILE

Review using the specified options file.

Example: cscs -o options.xml

Example: cscs --options C:\Users\User1\myOptions.xml

-q, --quiet

Review in quiet mode. No output will be displayed on the screen.

-r XCCDF_RULE_OR_OVAL_DEF, --rule XCCDF_RULE_OR_OVAL_DEF

Review a single Rule using the Rule ID from the XCCDF file or review a single definition from an OVAL document.

Example1: cscs -r account_lockout_duration

Example2: cscs -r oval:mil.disa.stig.adobe.reader:def:1

-mr RULE_COUNT RULE_ID RULE_ID .., --multipleRules RULE_COUNT RULE_ID...

Review multiple rules using the Rule ID from the XCCDF file or review X definitions from an OVAL document.

Example1: cscs -mr 2 account_lockout_duration logon_as_service

Example2: cscs --multipleRules 3
oval:mil.disa.stig.adobe.reader:def:1
oval:mil.disa.stig.adobe.reader:def:2
oval:mil.disa.stig.adobe.reader:def:3

**-rr XCCDF_RULE_ID_REGULAR_EXPRESSION --ruleRegex
XCCDF_RULE_ID_REGULAR_EXPRESSION**

Review all rule IDs in the selected profile that match the regular expression

Example1: `csc -rr rule_SV-213143r55.*`

Example2: `csc -ruleRegex mscp.content_rule_sysprefs_.*`

5.3.2 Command Line Examples

1. Review the local computer with customized report settings and do not display any data to the screen.

```
csc.exe -o myoptions.xml -q
```

2. Review several computers from a text file of hostnames with customized report settings.

```
csc.exe -f test.txt -o myoptions.xml
```

3. Review a single remote host 'computer1' in debug mode

```
csc.exe -h computer1 -d
```

5.4 Option Descriptions and Datatypes

Below are all of the options that can be configured via the --setOpt command line parameter, which is primarily designed for advanced users to automate command line reviews. This information (with the exception of the description) can be obtained by running the --listOpt command.

Example: `csc - --listOpt`

OPTION	DESCRIPTION	DATATYPE
SCC CONFIGURATION PROFILE OPTIONS		
sccConfigProfile	Name of this configuration	String
sccConfigProfileMetaData	Data about SCC configuration	String
selectedConfigProfile	Selected Configuration Profile to Load on launch, if not found, will revert to 'default'	String
GENERAL SCANNING OPTIONS		
reviewType	Type of review	String ('local', 'cisco', 'remote', 'multiremote' (remote/multiremote for Windows only)), cisco-remote, unix-remote, windows-unix-remote
offlineConfigPath	Target if 'reviewType' equals 'cisco', This is a fully qualified path to a valid CiscoIOS configuration text file	String
REMOTE SCANNING OPTIONS (WINDOWS ONLY)		
hostName	Target of SCC scan if 'reviewType' equals 'remote'	String
hostFile	Fully qualified path to a text file containing NetBIOS names of Windows computers, one per line	String
multiRemoteMode	Mode to determine how to create/select host file	String (Host File, Entire Domain, Selected OU)
remoteWindowsOU	Name of OU (or OU's), delimited by ;	String
remoteWMIEnabled	This option used to determine if remote Windows scans should be WMI based	Boolean (0/1)
alternateWMICredentials	Use other credentials for WMI bases scans	Boolean (0/1)
WMI/SSH REMOTE SCANNING OPTIONS		

remoteScanHostCooldown	Number of seconds between SSH/wmi connections. This will slow down refresh to the screen, but also decrease the number of auditable events for long running SSH scans	Integer
remoteBaseLocation_unix	Base directory to which SCC should create sub-directories for SSH based scanning. This directory must exist on the target system. Default is /opt	String
remoteStartupTimeout	Number of minutes to wait before assuming SCC is unable to start on target computer	Integer
remoteMaxThreads	Maximum number of WMI scanning threads to create on 64 bit Windows	Integer
remoteMaxThreads32	Maximum number of WMI scanning threads to create on 32 bit Windows	Integer
SCAP PROCESSING OPTIONS		
ignoreCPEOVALResults	Run SCAP content even if CPE OVAL applicability fails	Boolean (0/1)
answerOCILwithSCAP	Enable answering OCIL manual questions if found in a SCAP Content Datastream	Boolean (0/1)
filterSCAPContentPerFamily	Only display (and run) content that matches the family (windows/unix/cisco)	Boolean (0/1)
downloadExternalFiles	Download external SCAP 1.2 content files	Boolean (0/1)
forceOVAL510	Force OVAL results be compliant with 5.10.1 for SCAP 1.2 validation purposes	Boolean (0/1)
saveResultsForNotApplicableContent	Force saving of results for non-applicable content to pass SCAP 1.3 validation	Boolean (0/1)
OVAL PROCESSING OPTIONS		
ignoreRemoteFileSystems	Do not perform any file searches on remote file systems	Boolean (0/1)
ignoreCase	Ignore case specific content requirements for searching files, paths, registry keys, etc	Boolean (0/1)
itemCreationThresholdEnabled	Enable setting a maximum number of items to create, to save memory usage in SCC	Boolean (0/1)
itemCreationThreshold	Numeric value for item creation threshold if itemCreationThresholdEnabled equals 1	Integer between 0 and 999999.

OVAL PROCESSING OPTIONS (UNIX ONLY)		
maskPasswords	Do not display UNIX password hashes to reports or XML files, does not impact test accuracy.	Boolean (0/1)
useGetpwent	Use the system command of getpwent instead of parsing /etc/passwd	Boolean (0/1)
ignoreFileExtendedACL	Do not collect extended ACL information which can be time consuming, and may not be acutally used in SCAP content	Boolean (0/1)
OVAL PROCESSING OPTIONS (WINDOWS ONLY)		
onlyCollectSecurityPrinciples ThatHavePrivilegesAssigned	Only report on users/groups that have access token data assigned to them, used to save time when scanning large domain controllers	Boolean (0/1)
windowsRegistryUserAge	Obsolete, do not use.	n/a
OCIL PROCESSING OPTIONS		
sortOCILQuestionsByDISACAT	Sort OCIL Questionnaires by DISA CAT level (CAT I, CAT II, CAT III)	Boolean (0/1)
backupManualQuestionResults	GUI only option to backup manual questions results when SCC GUI saves manual question data to file	Boolean (0/1)
customManualQuestionAutoAnswerDirectoryValue	Enable user selected autoanswer result directory	Boolean (0/1)
customManualQuestionAutoAnswerDirectory	Directory specified by the end user to obtain autoanswer	
ignoreManualQuestionVersionChangesOnUpgrade	Ignore XCCDF rule version when upgrading manual question results. This prevents loosing data when DISA changes XCCDF rule versions.	String (Directory)
SCAN REPORTING OPTIONS		
allSettingsHTMLReport	Save the All Settings HTML report at the end of each scan	Boolean (0/1)
allSettingsTextReport	Save the All Settings Text report at the end of each scan	Boolean (0/1)
nonComplianceHTMLReport	Save the Non-compliance HTML report at the end of each scan	Boolean (0/1)
nonComplianceHTMLReport	Save the Non-compliance HTML report at the end of each scan	Boolean (0/1)
allSettingsSummaryHTMLReport	Save the All Settings Summary HTML report at the end of	Boolean (0/1)

	each scan	
allSettingsSummaryTextReport	Save the All Settings Summary Text report at the end of each scan	Boolean (0/1)
nonComplianceSummaryHTMLReport	Save the Non-compliance Summary HTML report at the end of each scan	Boolean (0/1)
nonComplianceSummaryTextReport	Save the Non-compliance Summary Text report at the end of each scan	Boolean (0/1)
sortReportsBySeverity	Sort reports by severity of each rule, grouping by CAT I, CAT II, CAT III etc..	Boolean (0/1)
limitCollectedItemsInReports	Limit the number of collected items in detailed HTML/Text reports based on user defined thresholds	Boolean (0/1)
maxOVALCollectedItemsDefPass	Threshold at which collected items that pass will be truncated from detailed reports.	Integer
maxOVALCollectedItemsDefFail	Threshold at which collected items that do not pass will be truncated from detailed reports.	Integer
XML RESULTS OPTIONS		
keepXCCDFXML	Save the XCCDF XML Results at the end of each scan	Boolean (0/1)
keepOVALXML	Save the OVAL XML Results at the end of each scan	Integer (0/1/2/3) 0 =Keep full oval 1= Keep oval without system characteristics 2 = Keep "thin" OVAL 3 = Do not keep OVAL XML
keepOCILXML	Save the OCIL XML Results at the end of each scan	Boolean (0/1)
keepARFXML	Save the ARF XML Results at the end of each scan	Boolean (0/1)
saveDISACKL	Save the DISA CKL XML Results at the end of each scan.	Boolean (0/1)
saveDISACKLB	Save the DISA CKLB XML Results at the end of each scan.	Boolean (0/1)
cklReportStyle	Choose CKL/CLB report style.) ; '0' = report per stream. '1' = Session report.	Boolean (0/1)
CKLMarking	String to be saved in the CKL/CKLB "Marking" field,	String less than 512 characters

	default is 'CUI'	
CKLClassification	String to be saved in the CKL/CKLB "Classification" field, default is 'Unclassified'	String less than 512 characters
primaryIPAddressPrefix	Optional IP address prefix string specified by the end user to determine the 'Primary' IP address for the CKL/CKLB report.	String less than 512
keepCPEXML	Save the CPE-OVAL results if CPE-OVAL results return false (not applicable to target)	Boolean (0/1)
SUMMARY VIEWER OPTIONS		
enableSummaryViewer	Enable the Summary Viewer HTML report to provide hyperlinks to all results from a scan	Boolean (0/1)
summaryViewerSort1	Set which field to sort the Summary Viewer Report by first	Case sensitive string (Session, Stream, Host)
summaryViewerSort2	Set which field to sort Summary Viewer Report by second	Case sensitive string (Session, Stream, Host)
summaryViewerSort3	Set which field to sort Summary Viewer Report by third	Case sensitive string (Session, Stream, Host)
LOGGING OPTIONS		
keepScreenLogs	Save screen logs from each application/scan session	Boolean (0/1)
debugEnabled	Save debug logs from each application/scan session	Boolean (0/1)
suppress_warnings	Don't print warnings to the error log	Boolean (0/1)
debugExcludeDateTime	Do not print date/time stamps on every line of debug, which allows for easier comparison between debug logs	Boolean (0/1)
debugTraceEnabled	Print Trace level debug, which is more than default, enable with caution	Boolean (0/1)
maxLogFileSize	Maximum size for logs (primarily debug) before creating a new file	Integer (MB)
SCAN SESSION OPTIONS		
enableScanSessionDB	Enable SCC to save a history of scan sessions, directory names and filenames for easier viewing.	Boolean (0/1)
openWithSCC	When double clicking on reports in the session viewer,	Boolean (0/1)

	open with SCC	
OUTPUT OPTIONS		
sharedOptions	Save options to SCC install directory?	Integer (1 = install to shared/install directory, 0 = install to users home directory)
userResultsDirectory	Path to which SCC will save results	String that is an absolute directory path
userResultsDirectoryValue	How 'userDataDirectory' is determined	Integer (0 = User's home directory, 1 = Running Application Directory, 2 = Custom Directory)
userConfigDirectory	Path to which SCC will save configuration information	String that is an absolute directory path
userConfigDirectoryValue	How 'userConfigDirectory' is determined	Integer (0 = User's home directory, 1 = Running Application Directory, 2 = Custom Directory)
customPathLocal	Path to which SCC will save temporary files. Default location will be <scc install>/Local directory. This setting can only be configured via command line with --setOpt	String (Absolute Directory Path, Empty String = default location"
OUTPUT SUBDIRECTORY OPTIONS		
dirApplicationLogsEnabled	Create a subdirectory called 'ApplicationLogs' for application logs	Boolean (0/1)
dirAllSessionsEnabled	Create a subdirectory called 'Sessions' for sessions	Boolean (0/1)
dirSessionEnabled	Create a date/time subdirectory of the scan	Boolean (0/1)
dirSessionResultsEnabled	Create a results subdirectory of the date/time of the scan	Boolean (0/1)
dirSessionLogsEnabled	Create a logs subdirectory of the date/time of the scan	Boolean (0/1)
dirContentTypeEnabled	Create a results subdirectory based on content type (SCAP/OVAL/OCIL)	Boolean (0/1)

dirTargetNameEnabled	Create a results subdirectory based on the target hostname	Boolean (0/1)
dirStreamNameEnabled	Create a results subdirectory based on the content stream name	Boolean (0/1)
dirXMLEnabled	Create an results subdirectory called 'XML' for saving XML results	Boolean (0/1)
OUTPUT FILENAME OPTIONS		
fileTargetNameEnabled	Include the target hostname in the result filenames	Boolean (0/1)
fileSCCVersionEnabled	Include the SCC version in the result filenames	Boolean (0/1)
fileTimestampEnabled	Include scan date/timestamp in the result filenames	Boolean (0/1)
fileContentVersionEnabled	Include the Content version in the result filenames	Boolean (0/1)
CONTENT OPTIONS		
newerContentInstall	What to do when installing a newer version of the same SCAP benchmark	0 = Do nothing. 1 = Disable older content, 2 = Archive older content, 3 = deleted older content. Integer (0/1/2/3)
contentDevModeSkipValidation	Allow for developer mode	Boolean (0/1)
validateContent	Perform XML schema validation before scanning	Boolean (0/1)
validateContentOnInstall	Perform XML schema when installing content	Boolean (0/1)
validateDigitalSignature	Perform XML digital signature validation on SCAP 1.2 content before scanning	Boolean (0/1)
failOnXMLDSig	Do not scan if XML digital signature validation fails	Boolean (0/1)
validateXMLResults	Perform XML schema validation on XML results generated by SCC	Boolean (0/1)
UPDATE OPTIONS		
httpProxyType	The type of proxy to be used for HTTP request. 0 for system proxy, 1 for env proxy, 2 for no proxy, and 3 for custom proxy	Integer (0-3)
httpProxyURL	URL to use for a custom proxy	String
appCheckForUpdates	Check for SCC application Updates	Boolean (0/1)
appUpdateCheckFrequency	Days between checking for updated SCC	Integer (>0)
appUpdateLastCheck	UNIX time of last SCC app	Integer

	update check	
appUpdateURL	URL to SCC update file	String
contentCheckForUpdates	Check for content Updates	Boolean (0/1)
contentUpdateCheckFrequency	Days between checking for updated content	Integer >0
contentUpdateLastCheck	NIX time of last content update check	Integer
includeDraftContent	Include pre-release (draft/test) content	Boolean (0/1)
contentRepository	List of content repository URL's	String(s)
DEVELOPER OPTIONS: MISC THRESHOLDS		
maximumRecentReports	Number of recent reports (or scan sessions) to save in the Recent Reports menu	Integer
freeSpaceThreshold	Minimum amount of free space before stopping scan	Integer (MB)
validationThreshold	Maximum file size of XML input or output to perform schema validation on	Integer (MB)
externalcmdTimeout	Number of seconds to wait for external commands to return data	Integer (Seconds)
SSH RESULT FILE TRANSFER OPTIONS		
sshEnabled	Enable sending of results via SSH after each scan	Boolean (0/1)
deleteOnTransferReports	Delete local results after sending to server via SSH	Boolean (0/1)
sshServer	Name of ssh server to send results to	String that is a hostname or ip address
sshPortNumber	Port number for SSH	Integer between 0 and 65536
sshConnectionType	SSH Connection Type	Integer (0 = username, private key and passphrase; 1 = username and password)
username	SSH Username	String that is a proper username
sshUserPassword	Encrypted by SCC, cannot be edited manually	String
sshPrivateKey	Full path to user's private key for SSH connections	String
sshUserKeyFile	SCC generated keyfile for use with SSH	String
sshKeyPassphrase	Encrypted by SCC, cannot be edited manually	String
sshServerDirectory	Directory on the SSH in which to send results	String

POST SCAN SCAP SUMMARY REPORTING OPTIONS		
summarySourceDirectory	Source directory for post scan SCAP summary reports	String that is an absolute directory path
summaryDestinationDirectory	Destination directory for post scan SCAP summary reports	String that is an absolute directory path
openSummaryDestinationDirectory	Open Windows Explorer to the directory containing the reports (Windows only)	Boolean (0/1)
computerListHistoricalHTMLReport	Generate the Computer List Historical HTML report	Boolean (0/1)
computerListHistoricalExcelReport	Generate the Computer List Historical Excel report	Boolean (0/1)
computerListHTMLReport	Generate the Computer List HTML report	Boolean (0/1)
computerListExcelReport	Generate the Computer List Excel report	Boolean (0/1)
siteSummaryHTMLReport	Generate the Site Summary HTML report	Boolean (0/1)
siteSummaryNonComplianceHTMLReport	Generate the Site Summary Non Compliance HTML report	Boolean (0/1)
siteSummaryExcelReport	Generate the Site Summary Excel report	Boolean (0/1)
siteSummaryNonComplianceExcelReport	Generate the Site Summary Non Compliance Excel report	Boolean (0/1)
POST SCAN SCAP DETAILED REPORTING OPTIONS		
detailSummarySourceDirectory	Source directory for post scan SCAP Detailed reports	String
detailSummaryDestinationDirectory	Destination directory for post scan SCAP Detailed reports	String
openDetailSummaryDestinationDirectory	Open Windows Explorer to the directory containing the reports (Windows only)	Boolean (0/1)
POST SCAN OVAL DETAILED REPORTING OPTIONS		
detailSummaryOVALSourceDirectory	Source directory for post scan OVAL Detailed reports	String
detailSummaryOVALDestinationDirectory	Destination directory for post scan OVAL Detailed reports	String
openDetailSummaryOVALDestinationDirectory	Open Windows Explorer to the directory containing the reports (Windows only)	Boolean (0/1)
POST SCAN SCAP/OVAL SHARED REPORT GENERATION OPTIONS		
allSettingsHTMLReportDetailSummary	Regenerate the All Settings HTML Report	Boolean (0/1)
allSettingsTextReportDetailSummary	Regenerate the All Settings Text Report	Boolean (0/1)
nonComplianceHTMLReportDetailSummary	Regenerate the Noncompliance HTML Report	Boolean (0/1)

nonComplianceTextReportDetailSummary	Regenerate the Noncompliance Text Report	Boolean (0/1)
allSettingsSummaryHTMLReportDetailSummary	Regenerate the All Settings Summary HTML Report	Boolean (0/1)
allSettingsSummaryTextReportDetailSummary	Regenerate the All Settings Summary Text Report	Boolean (0/1)
nonComplianceSummaryHTMLReportDetailSummary	Regenerate the Noncompliance Summary HTML Report	Boolean (0/1)
nonComplianceSummaryTextReportDetailSummary	Regenerate the Noncompliance Summary Text Report	Boolean (0/1)
DEVIATION OPTIONS		
deviationsEnabled	Use of deviations is enabled.	
THRESHOLD OPTIONS		
thresholdsEnabled	Enable usage of thresholds	Boolean (0/1)
thresholdsProfile	Allows usage of a different thresholds file value of 'default' translates to 'default-thresholds.xml'	String
thresholdsUnlockCode	Code generated by SCC Unlocker, to allow end users to modify thresholds	String
SCC SERVICE OPTIONS (WINDOWS ONLY)		
frequency	How frequently SCC Service should run	Integer (0 = Custom, 1 = Hourly, 2 = Daily, 3 = Weekly, 4 = Monthly)
schedule	Custom schedule if 'frequency' = 0	Integer (Hours)
randomizeSSHtransfer	Delay SCC Service (and SSH transfer) by a random amount of time to prevent DDoS if numerous computers are configured with the SCC service and SSH transfer	Boolean (0/1)
randomizeValue	Max amount of time to delay starting SCC	Integer (seconds)
timeout	Programmatically calculated by SCC, do not edit	Integer
lastServiceScan	Programmatically calculated by SCC, do not edit	Integer
nextServiceScan	Programmatically calculated by SCC, do not edit	Integer
INTERNAL OPTIONS - DO NOT EDIT		
version	Version of SCC, do not edit	String
timeStamp	date/timestamp of last write to options, do not edit	Integer

expignore	Over-ride software expiration for pre-release versions	Boolean (0/1)
scap12DBFilepath	Relative path to the SCAP 1.2 content database, do not Edit	String
iaControlDBLoaded	Is the IA Control database loaded	Boolean (0/1)
iaControlDBFilepath	Relative path to the IA Control mappings database	String
nvdcceFile	Filename (no path) to the NVD CCE xml file	String
cciFile	Filename (no path) to the DISA CCI xml file	String

5.5 Generating Post Scan Reports from the Command Line

If a large number of files are collected on a share that is accessed via a LAN or WAN, it may be most time effective to generate the reports via command line on the server that contains the collection of files. This allows for a scheduled task to be created that can be run on a user specified time frame.

For example, if 100,000 computers are reviewed, it will likely take many hours to generate the summary reports. Ideally, this could be run during an evening a day after all of the results are created.

This functionality requires configuring a custom options.xml file with the GUI, and calling the application via command line with specific parameters.

5.5.1 Post Scanning Report Generation Parameters

Below are the parameters available for creating reports after XML results have been created. All of the following options must be used individually, and are not compatible with any other parameter.

To see the help related to informational parameters run '**csc.exe --help info**'

```
#####
```

POST SCAN REPORT GENERATION PARAMETERS:

Below are the parameters available for creating reports after XML results have been created. All of the following options must be used individually, and are not compatible with any other parameter.

-s OPTIONS_FILEPATH, --summaryReports OPTIONS_FILEPATH

Generate SCAP summary reports using the specified options file.

Example: csc.exe -s options.xml

Example: csc.exe --summaryReports myOptions.xml

To set the source path for finding XML files to create summary reports from use the following

```
csc csc.exe --setOpt summarySourceDirectory <Path to XCCDF results>
```

To set the source path for finding XML files to create reports from use the following

```
csc csc.exe --setOpt summaryDestinationDirectory <Directory to save reports>
```

-ts OPTIONS_FILEPATH, --detailedSCAP OPTIONS_FILEPATH

Generate detailed reports for SCAP using the specified options file.

Example: `cscce.exe -ts options.xml`

Example: `cscce.exe --detailedSCAP myOptions.xml`

To set the source path for finding XML files to create detailed SCAP reports from use the following

```
cscce cscce.exe --setOpt detailSummarySourceDirectory <Path to
XCCDF and OVAL XML results>
```

To set the source path for finding XML files to create reports from use the following

```
cscce cscce.exe --setOpt detailSummaryDestinationDirectory
<Directory to save reports>
```

-tv OPTIONS_FILEPATH, --detailedOVAL OPTIONS_FILEPATH

Generate detailed reports for OVAL results using the specified options file.

Example: `cscce.exe -tv options.xml`

Example: `cscce.exe --detailedOVAL options.xml`

To set the source path for finding XML files to create detailed OVAL reports from use the following

```
cscce cscce.exe --setOpt detailSummaryOVALSourceDirectory <Path
to OVAL XML results>
```

To set the source path for finding XML files to create reports from use the following

```
cscce cscce.exe --setOpt
detailSummaryOVALDestinationDirectory<Directory to save
reports>
```

5.5.2 Informational Parameters

Below are the parameters available for information purposes only. No configuration changes or scanning occur. All of the following options must be used individually, and are not compatible with any other parameter.

To see the help related to informational parameters run '`cscce.exe --help info`'

```
#####
```

INFORMATIONAL PARAMETERS:

Below are the parameters available for information purposes only. No configuration changes or scanning occur. All of the following options must be used individually, and are not compatible with any other parameter.

--checkForSCCUpdates

Check to see if newer SCC releases exist via online query. Additional settings may need to be pre-configured before usage. refer to: csc.exe --config -> Options -> Update Options This does not download or update/install SCC, it just verifies it's current.

--getOpt OPTION

Advanced user setting to retrieve the value of any SCC option. Available options can be found with --listOpt, and need to be specified exactly.

Example: csc.exe --getOpt debugEnabled debugEnabled = 0

--listOpt

Advanced user setting to retrieve the configurable values for use with --getOpt and --setOpt

--listAllProfiles

List all profiles according to the installed content. Note that not all profiles are available to all content streams.

--listAllBenchmarks

List all benchmarks according to the content installed on the system. Useful when setting a profile for specific content.

-v, --version

Display one liner version information.

-V, --verboseVersion

Display version information.

-, --help [config scan post-scan info]

Display this help page, by default all sections will be displayed. With optional parameter(s) of config, scan, post-scan or info specific section(s) can be displayed.

5.5.3 Scheduling Command Line Generation of Summary Reports

The process for scheduling csc.exe via the Windows Task Scheduler is the same as any other application, except the parameters listed above will need to be included. Below is an example:

1. Click Start -> Programs -> Accessories -> System Tools -> Scheduled Tasks.

2. Click "New".
3. Click "Browse".
4. Select the command line version of the file (Example "C:\Program Files\SCAP Compliance Checker 5.12\csc.exe")
5. Choose the selected time frame (Daily, Weekly, Monthly, etc.).
6. Enter the credential for the software to run.
7. Click "Open Advanced Properties for this task when I click Finish".
8. Click "Finish".
9. In the Run line add the desired parameters after the double quotes.

Example: "C:\Program Files\SCAP Compliance Checker 5.12\csc.exe" -s options.xml

Test the scheduled task by right clicking and selecting "Run."

5.6 Multiple Computer Deployment

If the end user is automating the process of running the SCC software locally on multiple remote computers, below is the list of files that must be present for the application to run via command line.

- csc.exe
- bin directory
- lib32 directory
- lib64 directory
- scc.sig32
- scc.sig64
- options.xml (or any custom named options file)
- "Resources" directory, subdirectories and all files

5.6.1 Collecting Resulting Files

If the end user is pushing the command line version of the software out to the target computers, and would like to collect the results in a consolidated directory for generating multi-computer summary reports, below is documentation explaining which files to copy.

A directory structure will be created in the format (depending on user preferences), such as:

```
o SCC
  o Sessions
    o <Date Time Stamp>
      o Results
        o XML
```

The XML Directory will contain the resulting ARF, OVAL and XCCDF XML files based on user preferences.

The only file required for generating the multi-computer reports is the XCCDF file, which will be in the XML directory, in the format:

`<Computer>_SCC_5.12_<DateTime>_XCCDF-Results_<Stream>.xml`

After all of the XCCDF XML files have been collected and copied to a centralized share, multi-computer summary reports can be created. Please refer to "Generating Multi-Computer Summary Reports" section of the documentation for additional information.

5.7 Manual and Hybrid Question Answer Files

SCC supports answering of non-automated questions via Manual Question Auto-Answer Files. These are text based templates based on OCIL (Open Checklist Interactive Language) XML files. SCC includes 'enhanced' STIG SCAP content, which is programmatically created from STIG Manuals and SCAP Benchmarks, to provide 100% coverage of the STIG Manual.

5.7.1 Templates Directory

Files in this directory are programmatically created by SCC when content is installed, or when an end user changes the sort order of manual questions. Files in this directory should not be manually updated, as they will be overwritten, and ignored. Files from this directory should be copied to the Completed Files directory

Templates are located in SCC's Resources/Content/Manual_Questions/Templates directory

5.7.2 Completed Files Directory

This directory contains all of the completed autoanswer files, either create by SCC's Manual Questions GUI interface, or edited manually by any text editor.

The default directory is SCC's Resources/Content/Manual_Questions/Completed_Files, but can be changed by the end user via SCC Options -> Scanning Options -> Manual Questions Autoanswer Completed Files Directory

5.7.3 Usage

1. Ensure that SCC's option to process manual questions is enabled SCC Options -> Scanning Options -> Process Manual Questions if found in SCAP content
2. Copy desired template(s) from the Templates directory to your Completed Files directory
 - a. Important: Do not rename the file(s), or SCC will not find it.
3. Open template using a text editor
 - a. Mark answers with an X
 - b. Remove X from "Not Reviewed" line
 - c. Add any comments
 - d. Do not add/remove any other sections to this file. The only editable portions are selecting an answer with [X] and adding text based comments.
 - e. Save the file when complete
4. Run cscf to perform a scan, answers to manual questions should be applied to all future scans.

Example:

```
=====
Select One of the following by entering an X in the brackets
[ ] Finding
[ ] Not a Finding
[X] Not Applicable
[ ] Not Reviewed
Enter any comments : This is a sample comment that will be
included in XCCDF XML, HTML, Text and CKL reports.
```

5.7.4 Answering "Hybrid" Questions If found (Currently SQL Server Only)

For a background and examples of Hybrid Questions in use, refer to **Section 6** of this User Manual.

If SCC reports a failure for a Hybrid test, and if the results match your approved system documentation for that settings, you can document that approved value and SCC will use that data, if properly formatted, and use it to perform tests in future scans. Below is an example, and the user entered data is in **bold**.

```
QUESTION_TEXT      : Enter the list of database owners authorized
to be trustworthy and privileged

=====HYBRID
INSTRUCTIONS=====
Each line will be processed in order priority, and only the
first value found
that is applicable is applied to a target, so a target of 'ALL'
should only be used last in a list.
<SCOPE>:<TARGET>=<VALUE1>,<VALUE2>,<VALUE3>
SCOPE: COMPUTER, INSTANCE or DATABASE
If SCOPE is COMPUTER, TARGET can be one of: <ComputerName> or
'ALL' (where ComputerName is the NetBIOS name)
If SCOPE is INSTANCE, TARGET can be one of:
<ComputerName>\<InstanceName>, <InstanceName> or 'ALL'
If SCOPE is DATABASE, TARGET can be one of:
<ComputerName>\<InstanceName>\<DatabaseName>, <DatabaseName> or
'ALL'
VALUE FORMAT: Single value, or comma separated values (where
values are 'database_owner')
Refer to SCC User Manual Section 6 for more information and
detailed examples
=====END HYBRID
INSTRUCTIONS=====
ENTER AUTHORIZED VALUES BELOW AS
<SCOPE>:<TARGET>=<database_owner>,<database_owner>,<database_ow
ner>
DATABASE:ALL=dbo
ENTER AUTHORIZATION (Document Name, Page, ISSO/ISSM/DBA,
etc):Sample-SSP.docx, Page 2, Joe ISSM approved.
```

All future scans will you the value of 'dbo' and if that is what is what the system returns, the check will now pass.

SCC will also provide your Authorization information, entered after "ENTER AUTHORIZATION" as part of all SCC results, to justify your values to auditors.

5.7.5 Warnings if autoanswer files are not found

If SCC finds enhanced content that contains Manual Questions, and no Manual Question autoanswer file is found, SCC will print a warning during each scan, and all manual question results will be reported as not reviewed.

5.7.6 Interaction with SCC GUI

SCC's graphical Manual Questions feature imports files from the Completed_Files directory before making updates, so editing either manually via text editor or SCC GUI, or both should work as expected, unless both methods are used at the same time, and then the last one to save will win.

5.7.7 Upgrading manual question results from an older version of SCC

Starting with SCC 5.7 (on Windows) SCC will prompt to upgrade configuration and manual question results on first launch. If the manual question results are configured to be stored in the application installation directory (default), SCC will copy the files from 5.7 to 5.7.1 etc. If the manual question results directory has been configured to a custom directory, SCC 5.7.1 will use that custom directory, as a side effect of the configuration upgrade.

For Linux: When using package based installation, with the RPM or DEB upgrade method selected, SCC will install to the same directory as previous versions, upgrade the configuration to the current version, and have all of the manual questions results available.

5.7.8 Converting manual question results from one version of SCAP content to another

Starting with SCC 5.7.1, SCC now automatically converts manual question results to the latest SCAP content. Only questions that match the same XCCDF rule and version are updated, so if any change is made to an XCCDF rule in the updated STIG manual, answers will not be converted and those questions will need to be re-answered.

6. HYBRID TESTS

Hybrid Checks are a new feature of SCC 5.9 and later, and are only used in very specific use cases, primarily with MS SQL Server Instance and Database STIGs. Many of the requirements of the SQL STIGs require comparing the system configuration against user/system documentation. This means that the 'requirement' for automating a test is user defined, and historically have been manual checks, which require users to manually gather the system configuration and manually compare those settings against user/system documentation.

If Hybrid tests exists in content, they will be visible in the Manual Questions, both in the GUI and the command line (text file based) methods.

Hybrid tests take user input in an specific XML format and use it as the requirement to determine if the system configuration passes or fails. Refer to section 4.6.2 for instructions on using SCC's graphical variable builder, available on SCC's GUI, which guides the user to create variables described in the format below. In SCAP terminology, SCC takes this end user provided data and passes it in as an external variable, and that external variable is then used as an OVAL state to be compared against the system configuration.

The format of the data entered by the user for these checks is in the format of

```
<hybrid_variables>
  <hybrid_variable>
    <scope></scope>
    <target></target>
    <authorization></authorization>
    <authorized_values>
      <authorized_value></authorized_value>
    </authorized_values>
    <notes></notes>
  </hybrid_variable>
</hybrid_variables>
```

Where

- 'scope' can be one of COMPUTER, INSTANCE or DATABASE
- 'target' is what Computer\Instance\Database you want the value applied (see below for definitions, and examples)
- 'authorization' is documentation explaining where in system documentation the authorized values are located/approved by.
- authorized_value is usually a user, group, stored procedure, etc..
- 'notes' are completely optional, and are only for end users to explain why/how variables work, and are not included in any reports or SCAP processing.

This can be as simple as:

```
<hybrid_variables>
  <hybrid_variable>
    <scope>DATABASE</scope>
    <target>ALL</target>
```

```

    <authorization>System SSP, approved by ISSM on
    1/1/2024</authorization>
    <authorized_values>
        <authorized_value>dbo</authorized_value>
    </authorized_values>
</hybrid_variable>
</hybrid_variables>

```

Or a complex, and likely more real world like below.

```

<hybrid_variables>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>TestComputer1\parents-db</target>
        <authorization>Simpson Parents SSP, approved by Matt
        Groening on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>homer</authorized_value>
            <authorized_value>marge</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>TestComputer1\kids-db</target>
        <authorization>Simpson Kids SSP, approved by Matt Groening
        on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>bart</authorized_value>
            <authorized_value>lisa</authorized_value>
        </authorized_values>
    </hybrid_variable>
</hybrid_variables>

```

Note: Each line will be processed in order priority, and only the first value found that is applicable is applied to a target, so a target of 'ALL' (if needed) should only be used last in a list.

6.1 Hybrid Scope

The SCOPE portion of the user defined data is pretty straight forward, it's just one of the following, and it tells SCC how to apply the TARGET.

- COMPUTER
- INSTANCE
- DATABASE

6.2 Hybrid Target

The TARGET is where things can get a bit complex, depending on your needs. It can be as simple as ALL, where the value you are providing is applied as a requirement for all targets, but based on end user feedback, that is not always the case, and sometimes requirements need to be for All Instances/Database on a given computer, or all Instances, or all Databases, or a single Instance, or a single Database.

6.2.1 SCOPE = COMPUTER

Target can be one of:

- <ComputerName>, where <ComputerName> is the NetBIOS name or DNS FQDN of the server hosting SQL Server.
 - All Instances and Databases hosted on this computer will get the value you define.
- 'ALL'
 - All Instances and Databases hosted on any computer will get the value you define.

6.2.2 SCOPE = INSTANCE

Target can be one of:

- <ComputerName>\<InstanceName>, where InstanceName is the name of the SQL Instance on the computer specified
 - This specific Instance, on this specific computer, and all Databases hosted by this Instance will get the value you define.
- <InstanceName>
 - This specific Instance, on any computer, and all Databases hosted by this Instance will get the value you define.
- ALL'
 - All Instances and Databases hosted on any computer will get the value you define.

6.2.3 SCOPE = DATABASE

Target can be one of:

- <ComputerName>\<InstanceName>\<DatabaseName>
 - This specific Database in this specific Instance, on this specific computer, will get the value you define.
- <DatabaseName>
 - This specific Database, on any computer or Instance, will get the value you define.
- ALL'
 - All Databases hosted on any computer will get the value you define.

6.3 Hybrid Value(s)

The value(s) portion of the end user provided data is also pretty simple, it can be one of the following

- A single string/word
- A comma separated list of strings/words

6.4 XML Example using target 'ALL'

6.4.1 Single value of 'dbo' for all databases scanned.

Below is a sample hybrid question, and the example responses are in **bold**.

```
<hybrid_variables>
  <hybrid_variable>
    <scope>DATABASE</scope>
    <target>ALL</target>
    <authorization>System SSP, approved by ISSM on 1/1/2024</authorization>
    <authorized_values>
      <authorized_value>dbo</authorized_value>
    </authorized_values>
  </hybrid_variable>
</hybrid_variables>
```

6.5 More Complex XML Examples

The following examples are more real world style usage, based on the same Question Text from section 7.4.

6.5.1 Computer Scope

- All SQL Instances and SQL Databases on the computer 'TestComputer1' will have variables of 'dba1' and 'dba2'
- All SQL Instances and SQL Databases on the computer 'TestComputer2.testdomain.gov' will have variables of 'dba3'
- All SQL Instances and SQL Databases on all computers, except TestComputer1' and 'TestComputer2.testdomain.gov' will have a variable of 'dba4'

```
<hybrid_variables>
  <hybrid_variable>
    <scope>COMPUTER</scope>
    <target>TestComputer1</target>
    <authorization>TestComputer1 SSP, approved by ISSM on 1/1/2024</authorization>
    <authorized_values>
      <authorized_value>dba1</authorized_value>
      <authorized_value>dba2</authorized_value>
    </authorized_values>
  </hybrid_variable>
  <hybrid_variable>
    <scope>COMPUTER</scope>
```



```

    <target>TestComputer2.testdomain.gov</target>
    <authorization>TestComputer2 SSP, approved by ISSM on
    1/1/2024</authorization>
    <authorized_values>
        <authorized_value>dba3</authorized_value>
    </authorized_values>
</hybrid_variable>
<hybrid_variable>
    <scope>COMPUTER</scope>
    <target>ALL</target>
    <authorization>Network SSP, approved by ISSM on
    1/1/2024</authorization>
    <authorized_values>
        <authorized_value>dba4</authorized_value>
    </authorized_values>
</hybrid_variable>
</hybrid_variables>

```

6.5.2 Computer/Instance Scope

- The SQLInstance1 and SQL Databases in that instance on computer TestComputer1 will have variables of 'dba4' and 'dba5'
- The SQLInstance1 and SQL Databases in that instance on computer TestComputer2.testdomain.gov will have variables of 'dba6' and 'dba7'
- All other instances and databases will have an empty string variable, and would fail if results are returned from the instances/databases

```

<hybrid_variables>
    <hybrid_variable>
        <scope>INSTANCE</scope>
        <target>TestComputer1\Instance1</target>
        <authorization>TestComputer1 SSP, approved by ISSM on
        1/1/2024</authorization>
        <authorized_values>
            <authorized_value>dba4</authorized_value>
            <authorized_value>dba5</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>INSTANCE</scope>
        <target>TestComputer2.testdomain.gov\SQLInstance1</ta
        rget>
        <authorization>TestComputer2 SSP, approved by ISSM on
        1/1/2024</authorization>
        <authorized_values>

```

```

        <authorized_value>dba6</authorized_value>
        <authorized_value>dba7</authorized_value>
    </authorized_values>
</hybrid_variable>
<hybrid_variable>
    <scope>INSTANCE</scope>
    <target>ALL</target>
    <authorization></authorization>
    <authorized_values>
        <authorized_value></authorized_value>
    </authorized_values>
</hybrid_variable>
</hybrid_variables>

```

6.5.3 Instance Scope

- The instance of 'SQLInstance2' on any computer, and any databases in the 'SQLInstance2' would get the variable of 'abc' and '123'
- The instance of 'SQLInstance3' on any computer, and any databases in the 'SQLInstance3' would get the variable of def
- All other instances and databases on any computer would get the variable 'xyz'

```

<hybrid_variables>
    <hybrid_variable>
        <scope>INSTANCE</scope>
        <target>SQLInstance2</target>
        <authorization>Many System SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>abc</authorized_value>
            <authorized_value>123</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>INSTANCE</scope>
        <target>SQLInstance3</target>
        <authorization>Many System SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>def</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>INSTANCE</scope>

```

```

    <target>ALL</target>
    <authorization>Many System SSP, approved by ISSM on 1/1/2024</authorization>
    <authorized_values>
        <authorized_value>xyz</authorized_value>
    </authorized_values>
</hybrid_variable>
</hybrid_variables>

```

6.5.4 Database Scope

- The database of 'master' in any instance, on any computer, would get the variable of 'abc'
- The database of 'tempdb' in any instance, on any computer, would get the variable of 'def'
- The database of 'mytestDB' in any instance, on any computer, would get the variable of 'jkl'
- All other instances and databases on any computer would get the variable 'xyz'

```

<hybrid_variables>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>master</target>
        <authorization>Default DB SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>abc</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>tempdb</target>
        <authorization>>Default DB SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>def</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>mytestDB</target>
        <authorization>my test DB SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>jkl</authorized_value>
        </authorized_values>
    </hybrid_variable>
</hybrid_variables>

```

```

        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>ALL</target>
        <authorization>All DB Systems SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>xyz</authorized_value>
        </authorized_values>
    </hybrid_variable>
</hybrid_variables>

```

6.5.5 Computer\Instance\Database Scope

- The database of 'master' in TestInstance1 on TestComputer1 ' would get the variable of 'abc'
- The database of 'master' in TestInstance2 on TestComputer2.testdomain.gov' would get the variable of 'def'
- All other instances and databases on any computer would get the variable 'xyz'

```

<hybrid_variables>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>TestComputer1\TestInstance1\master</target>
        <authorization>Default DB SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>abc</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>TestComputer2.testdomain.gov\TestInstance2\master</target>
        <authorization>Default DB SSP, approved by ISSM on 1/1/2024</authorization>
        <authorized_values>
            <authorized_value>def</authorized_value>
        </authorized_values>
    </hybrid_variable>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>ALL</target>

```

```

    <authorization>All DB Systems SSP, approved by ISSM
    on 1/1/2024</authorization>
    <authorized_values>
        <authorized_value>xyz</authorized_value>
    </authorized_values>
</hybrid_variable>
</hybrid_variables>

```

6.6 What happens if I do not enter any data into Hybrid tests?

Short answer: Any test that returns any data from the target will fail.

SCC defaults all hybrid tests to

```

<hybrid_variables>
    <hybrid_variable>
        <scope>DATABASE</scope>
        <target>ALL</target>
        <authorization></authorization>
        <authorized_values>
            <authorized_value></authorized_value>
        </authorized_values>
    </hybrid_variable>
</hybrid_variables>

```

Which passes in a empty value which will be used as a comparison against any data returned from the system, based on the test. Depending on your system configuration, if no data is returned, the check will pass. If any data is returned, the check will fail and you will want to review the results, and compare against your system documentation. If the system documentation matches what was returned by SCC, you will want to update the hybrid test with the value in your documentation.

7. UNDERSTANDING SCAN RESULTS

7.1 Understanding Scan Reports

7.1.1 Summary Viewer Report

By default, with each scan session, a summary viewer HTML report is created which provides hyperlinks for easy browsing of the results created from that scan session. It's saved to the root of the scan session directory.

Ex: SCC_Summary_Viewer_2017-01-06_112807.html

This report can be sorted by clicking on any column heading, or filtered by typing a hostname, content stream etc. in the 'search' box.

7.1.2 Single Computer HTML and Text Reports

Depending on the user selected options, the following reports may be available in both HTML and/or text based formats:

REPORT	DESCRIPTION
All Settings Report	The <i><Computer>_SCC_5.12_All-Settings_<Content Name>.html</i> report contains the XCCDF results in a human readable format. The report is divided into five sections: Score, System Information, Stream Information, Results and Detailed Results. The Scores section contains the calculated scores for the target system. The System Information section contains information about the target system (CPE Information), such as the host name, IP addresses, operating system, processor, memory, manufacturer, model, serial number, BIOS version, and Ethernet Interfaces. The Content Information section contains information about the XCCDF benchmark, such as the XCCDF filename used, status (if officially accepted content along with the date it was officially accepted), content installation date, the profile used, the testing start and end times, and the identity of the user who ran the benchmark. The Results section contains the individual rule results, comprised of the CCE reference and the check title. To view the "Detailed Results" for an individual item, just click on the text. The Detailed Results section contains in-depth information on each rule performed in the benchmark. This section varies slightly between SCAP and standalone OVAL/OCIL, but contains fields such as Title, Result, CCE Identities, CVE Identities, severity, weight, definitions, tests, collected items
All Settings Summary Report	Contains the same information as the "All Settings Report", except excludes the Detailed Results, which allows for easier printing.
Non-Compliance	The <i><Computer>_SCC_5.12_Non-Compliance_<XCCDF Content</i>

Report	<i>Name>.html</i> report contains same results in the same format as the "All Settings Report", but only includes the Failed, Error, and Unknown checks.
Non-Compliance Summary Report	Contains the same information as the "Non-Compliance Report", except excludes the Detailed results, which allows for easier printing.

7.1.3 Understanding the Result Status Information

All of the reports show the number of checks performed, and the result for each. The result types are specified by the SCAP standards and are summarized below.

RESULT	EXPLANATION
Pass	The SCC was able to correctly interpret the check in the XML content, perform the check on the target system, and all check requirements were met. Example: Password Length Requirement 12 Characters, Target Computer: 12 Characters
Fail	The SCC was able to correctly interpret the check in the XML content, perform the check on the target system, and one or more of check requirements were not met. Example: Password Length Requirement 12 Characters, Target Computer: 8 Characters
Error	The SCC was able to correctly interpret the check in the XML content, however an error occurred while performing the check. This is typically due to a configuration of the target system, or insufficient permissions of the user running the software.
Unknown	The SCC was not able to interpret the check in the XML content. This could be due to a flaw in the XML content, or an incompatibility between the SCC and the XML content such as OVAL version.
Not Applicable	The SCC was able to interpret the check in the XML content, but it was not applicable to the target system.
Not Checked	The SCC was able to interpret the check in the XML content, however the XML content did not result in any evaluation to be performed. Also, if a probe is not supported, the check will show up as Not Checked.
Not Selected	The SCC was able to interpret the check in the XML content, however the XML content instructed the SCC not to perform this check.
Total	Numeric sum of Pass, Fail, Error, Unknown, Not Applicable, Not Checked, and Not Selected.

7.1.4 Understanding Color Coding in the HTML Reports

The HTML reports have color coding to assist in understanding what failed, and why it failed.

7.1.4.1 Color Coding in the 'Results' Section

COLOR	DESCRIPTION
Blue	The overall rule passed all of the required tests. Example: "Account Lockout Duration - (CCE-2928-0) - Pass
Red	The overall rule failed one or more of the required tests. Example: "Account Lockout Duration - (CCE-2928-0) - Fail

7.1.4.2 Color Coding in the 'Detailed Results' Section for Class = Compliance

Per OVAL specifications, for compliance checks, a test result of "True = Compliant", and "False = Not Compliant".

COLOR	DESCRIPTION
Blue	The individual test result was True, or the result was False but did not cause the overall test to fail.
Red	The individual test was False and contributed to the overall rule being marked as Fail.

7.1.4.3 Color Coding in the 'Detailed Results' Section for Class = Patch

COLOR	DESCRIPTION
Blue	SCC was able to verify that the patch was installed as required in the underlying tests. Result = Pass
Red	SCC was not able to confirm that the patch was installed as required, as one or more of the underlying tests failed. Result = Fail

7.1.4.4 Color Coding in the 'Detailed Results' Section for Class = Vulnerability

Per OVAL specifications, for Compliance checks, a test result of True = Vulnerable and False = Not Vulnerable.

COLOR	DESCRIPTION
Blue	The individual test result was False (meaning not vulnerable), or the result was Pass (vulnerable) but did not cause the overall test to fail.
Red	The individual test was True (Vulnerable) and contributed to the overall rule being marked as Fail.

7.2 Navigating the Results Directory

The User Data Directory, which contains both Application Logs and Scan Sessions, is configurable, see "Editing Options" for details. By default the data is stored a subdirectory called "SCC" in the user's home directory, but can be configured to store results to the installation directory, or any custom directory.

The default directory structure is as follows, but is user configurable.

ApplicationLogs

This directory contains SCC logs (screen, debug, error) related primarily to the SCC application itself, startup, forms, etc. and not specifically related to performing SCAP content scanning.

Sessions

<Date/Time Scan Session>

The Date/Time directory is created each time the Analyze Computer button is pressed, or a scan is completed via CSCC. This helps organize all scan related data for a single session.

SCC_Summary_Viewer_<Date/Time Scan Session>.html

The Summary Viewer report provides hyperlinks to all of the HTML, Text and XML based reports created from a single scan session.

Logs

This directory contains log files (screen, debug, error) specifically related to a scan session.

Results

<SCAP/OVAL/OCIL>

<Computer>_SCC_5.12_All-Settings_<Content>.html

<Computer>_SCC_5.12_Non-Compliance_<XCCDF Content>.html

XML

XML files (see table below)

Checklists

DISA Checklist (CKL) reports from any NIWC Enhanced SCAP Benchmark that contains Manual Questions

7.2.1 Contents of the XML Directory

The XML folder contains XML output generated by SCC. This output can be XCCDF results, OVAL results and OVAL variables files. Refer to the "Editing Options" for enabling or disabling saving the XCCDF and OVAL XML files after each review.

These files are not designed to be human readable, but are intended to be read into another SCAP, XCCDF or OVAL compatible software product to provide consolidated results.

Note: All filenames included in the table below are SCC's default result filenames

XML FILE	DESCRIPTION
NIST ARF 1.1	The <Computer>_SCC_5.12_<DateTime>_ARF_<XCCDF Content Name>.xml file contains the ARF results in a machine readable format. This high level summary of the review including the asset information from each system and the pass/fail status of each check performed. This results file is required for SCAP 1.2 compliance.
XCCDF Results	The <Computer>_SCC_5.12_<DateTime>_XCCDF-Results_<XCCDF Content Name>.xml file contains the XCCDF results in a machine readable format. This is a high level summary of the review including the asset information from each system and the pass/fail status of each check performed.
OCIL Results	The <Computer>_SCC_5.12_<DateTime>_ocil-res-Results_<XCCDF Content Name>.xml file contains the detailed OCIL in a machine readable format. This is a detailed report pass/fail results from each OCIL patch check performed during a review. This file only exists if SCAP content contains an OCIL questionnaire.
OVAL CPE Results	The <Computer>_SCC_5.12_<DateTime>_CPE-Results_<XCCDF Content Name>.xml file contains the CPE results in a machine readable format. This contains platform information about the target system including the operating system, network interfaces and processor type.
OVAL Patch Results	The <Computer>_SCC_5.12_<DateTime>_OVAL-Patch-Results_<XCCDF Content Name>.xml file contains the detailed OVAL patch results in a machine readable format. This is a detailed report of pass/fail results from each OVAL patch check performed during a review. This file only exists if the SCAP content contained an OVAL patch file.
OVAL Results	The <Computer>_SCC_5.12_<DateTime>_OVAL-Results_<XCCDF Content Name>.xml file contains the detailed OVAL results in a machine readable format. This is a detailed report of pass/fail results from each OVAL definition performed during a review.
OVAL Variables	The <Computer>_SCC_5.12_<DateTime>_OVAL-Variables_<XCCDF Content Name>.xml file contains a list of OVAL variables in a machine readable format.

7.3 Viewing Screen, Error or Debug Logs

The directory containing SCC Logs (if any exist) can be opened in the Windows Explorer by clicking:

Results -> Open Log Directory

Depending on the user selected preferences, the following log files may be present:

7.3.1 Application Logs

Application Logs are logs that are created when the application is started, and during application execution outside of any scan (when the analyze button is pressed). Application Logs are created in the Logs\ApplicationLogs directory (unless that directory option is disabled) and then they are saved in the root of the Logs directory. The ApplicationLogs directory is only created when logs exist, so may not be created depending on user preferences.

Some of the following logs might be present, depending if screen or debug logs are enabled, or if any application errors occurred.

REPORT	DESCRIPTION
Screen Log	<i>SCC_5.12_<DateTime>_Screen_Log.txt</i> This option saves the analysis log printed to the "Status" screen to a text file for viewing after the review. This file is not saved by default, but can be enabled in Options.
Error Log	<i>SCC_5.12_<DateTime>_Error_Log.txt</i> This report contains any errors that may have occurred while SCC is running, but not during a specific scan. This also contains any errors that may have occurred during command line usage. If this file exists, and the error log does not provide enough information to resolve the issue, please contact NIWC (see appendix G for technical support) and provide the error log for our analysis.
Debug Log	<i>SCC_5.12_<DateTime>_Debug_Log.txt</i> This option saves a large amount of additional information related to what occurred during a primary SCC operation, or when run via command line.. This option is disabled by default and should only be used when attempting to resolve errors in the application, as it will slow down the application and potentially use large amounts of disk space. Please refer to Appendix A.8 Debugging for additional information on SCC debug logs and their intended usage.

7.4.1 Scan Logs

Scan Logs are logs that are created when any SCAP/OVAL/OCIL content is used to scan a target computer. By default the logs are created in a date/timestamp 'session' directory within the Logs directory. Each time the Analyze button is pressed, a new scan log subdirectory is created. This directory name matches the same date/time session directory created in the Results directory.

REPORT	DESCRIPTION
Scan Screen Log	<i>SCC_5.12_<DateTime>_Screen_Log.txt</i> This option saves the analysis log printed to the "Status" screen to a text file for viewing after the review. This file is not saved by default, but can be enabled in Options.
Scan Error Log	<i>SCC_5.12_<DateTime>_scan<number>_Error_Log.txt</i> This report contains any errors that may have occurred during a GUI based scan. The scan<number>, such as scan001 or, scan002 corresponds to each review that is started by clicking the Analyze button. Normally this file will not exist. If this file exists, and the error log does not provide enough information to resolve the issue, please contact NIWC (see appendix G for technical support) and provide the error log for our analysis.
Scan Debug Log	<i>SCC_5.12_<DateTime>_scan<number>_Debug_Log.txt</i> This report contains any debug that occurred during a scan. The scan<number> such as scan001, scan002 corresponds to each review that is started by clicking the Analyze button. This option saves a large amount of additional information related to what occurred during a review. This option is disabled by default and should only be used when attempting to resolve errors in the application, as it will slow down the application and potentially use large amounts of disk space. Please refer to Appendix A.8 Debugging for additional information on SCC debug logs and their intended usage.

8. RUNNING SCC AS A SERVICE

This section of the manual is only applicable to installations of SCC on Windows. Automated command line usage of SCC on Linux, Solaris and Mac OS X is possible, but requires end users to script the command line interface by methods such as cron.

Installing SCC as a service, is completely optional, and only provided to allow easier automated scanning. It is recommended to configure sending results via SSH with running SCC as a service, to automate data collection as well. Refer to the SSH section for instructions on its usage.

8.1 Installing the SCC Service

During the installation of SCC on Windows, the installer provides an option to install the SCC Service component. If this was not selected at install, re-install the software and enable this option.

8.2 Configuring the SCC Service

The SCC application has a SCC Service Configuration component available, but only if the SCC Service was installed during application install.

Start -> Programs -> SCAP Compliance Checker 5.12 -> SCAP Compliance Checker 5.12 -> Options -> Show Options -> Service Options

8.2.1 Service Status

This option displays the current status of the service. However this does not indicate that the SCC application is actively scanning the computer, just that the service itself is running.

8.2.2 Scan Scheduling

This option allows the frequency at which SCC should perform scans, based on the options specified on this form.

Last Scan: Displays the date/time of the last scan performed by SCC as a service

Next Scan: Displays the date/time of the next scheduled scan, assuming the SCC service is Enabled and running.

Below are the user configuration options for scheduling scans:

FREQUENCY	DESCRIPTION
Daily	SCC will scan once per day
Weekly	SCC will scan once per week.
Monthly	SCC will scan once per month
Custom (Number of Hours)	Enter a numeric value say N (1-1000 etc.) for the frequency to at which SCC should scan. SCC will then scan the computer every N hours.

8.2.3 Other Options

All of the remaining options available for configuration are the same as SCC's options and include Reports, File Types, Logging, SCAP Content, OVAL Content, SSH Options. Refer to 'Using the Software via GUI' for documentation for each option.

Note: *The options saved for the SCC service will inherit all configuration settings from the SCC application, but only at the time the Save Service Configuration is pressed.*

8.2.4 Files/processes used in the SCC Service

The following process will appear in the process table, and will be running in the background:
SCC_Service32.exe

Depending on the scan frequency, periodically, the command line version of SCC will also be present in the process table: csc.exe

8.3 Update Service Configuration

8.3.1 Save Current Configuration to Service

This button saves the current configuration of SCC to the SCC service, then starts or restarts the SCC Service as needed.

8.3.1 Stop Service

This button stops and sets the startup of the SCC Service to 'Manual', effectively disabling the service. To re-enable the service, click the Save Current Configuration to Service button..

9. RUNNING SCC AS A NON-ADMINISTRATOR

This section of the manual is only applicable to installations of SCC on Windows. Starting with SCC 5.5, SCC allows for a limited operation of SCC without local Administrator rights.

9.1 Installing SCC

9.1.1 Installing as Administrator to Program Files

SCC can be installed by an Administrator in the normal location (C:\Program Files). If it is, then non-Administrators will not have write access to this installation directory and any files that need to be updated/added, etc. will be located in the home directory of the non-Administrator. See section 9.2 for more information.

9.1.2 Extracting Zipped version of SCC as a non-Administrator

SCC can be 'installed' to any directory by unzipping the zipped version of SCC. If the directory is writable by non-Administrators, it is not recommended to have this be a shared installation, because running the application as an Administrator will lock down a few of the subdirectories that contain sensitive data, and errors may occur when later attempting to run vis non-administrator. If the directory is not writable by non-Administrators, refer to Section 9.2.

9.2 Install directory non-writable by non-Administrators

If the non-Administrator who launched SCC cannot write to the installation directory, then a directory called "SCC" and a subdirectory called "SCC_5.12".will be created in the home directory of the non-Administrator.

Example: C:\Users\testuser\SCC\SCC_5.12

Launching SCC as a non-Administrator the first time may take a while as all of the SCAP content is copied from the SCC installation directory to the home directory of the non-Administrator..

9.3 GUI usage as non-administrator

SCC will launch as normal, and appear the same as when run as an Administrator, but will have limited scanning functionality. SCC will allow:

- o SSH based Unix scanning
- o SSH based Cisco IOS XE scanning
- o Offline Configuration Scanning of Cisco IOS XE
- o Post scan report generation

SCC will not allow any scanning of Windows (local or remote) because much of the data commonly queried in the Windows OS content is not available to standard users and the results would be inaccurate.

9.4 Command line usage as non-administrator

The command line interface of SCC will allow for the same limited scanning modes as the GUI, and the --help command will only show the command line parameters that are applicable when running as a non-administrator.

9.5 Uninstalling

If SCC was installed to C:\Program Files, then an Administrator will need to remove the software. For zip installations, just delete the directory that was extracted from the zip. Additionally, if a non-Administrator cannot write to the installation directory, then the home directory of the non-Administrator may need to be manually cleaned up by deleting the SCC\SCC_5.12 directory.

Example: C:\Users\testuser\SCC\SCC_5.12

10. WINDOWS UPDATE AGENT (WUA) TESTS

This section of the manual is only applicable to installations of SCC on Windows. SCC supports the OVAL test called 'wuaUpdateSearcher' which queries the Windows Update Agent. It works differently from other OVAL tests, and those differences are documented below.

10.1 How does SCC perform WUA tests?

SCC calls a Microsoft API, and that API in turn does all of the checking and returns results back to SCC. But SCC has to tell the API where to get its data. There are 3 different methods:

- Offline Cab File manually downloaded from Microsoft
- Query WSUS Server on network LAN/WAN
- Query Microsoft Windows Update via Internet

SCC performs them in the list above, in order

10.1.1 Method 1: Offline Cab File

SCC first looks to see if there is an offline cab file called 'wsusscn2.cab' located in the <SCC Install>\Resources\Content\WindowsUpdateAgent_Cab directory, it will use it to perform WUA tests. Microsoft posts updated versions of the wsusscn2.cab on Patch Tuesday every month, and it is recommended to download an updated version every month.

<https://catalog.s.download.windowsupdate.com/microsoftupdate/v6/wsusscan/wsusscn2.cab>

SCC will report an error if the file is found to be greater than 30 days old.

Note: The wsusscan2.cab file is currently over 700 MB, and will cause the scan to take longer than if SCC queries a WSUS server or Windows Update Online.

10.1.2 Method 2: Query WSUS Server

If SCC does not find the wsusscn2.cab file, SCC will check the target systems registry to determine if it's connecting to a WSUS server to receive updates.

Note: Results will be based on what WSUS categories are available on the WSUS server, and what updates are approved. If the WSUS server is out of date, or if it's missing certain categories of updates, such as SQL Server, Microsoft Office, etc., the results from this method will not be 100% accurate. On the flipside, this method will also not show false positives for updates that have not yet been approved.

10.1.2 Method 3: Query Microsoft Windows Update (via Internet)

If SCC does not find a wsscan2cab, and the target is not configured to obtain updates from a local WSUS server, SCC will attempt to query Microsoft Windows Update, if Internet connectivity is available.

APPENDIX A - FREQUENTLY ASKED QUESTIONS

A.1 Why can't I install a DISA STIG Manual XCCDF into SCC?

Below is a common question:

I tried to import a "Manual" STIG as a *.zip into the SCC and it gives me the following error:

```
"Unable to find OVAL document.*.zip Please ensure that all SCAP streams include a valid OVAL file that is named '<stream_name>-oval.xml'"
```

I then tried to import a manual STIG as a *-xccdf.xml into the SCC and gives me the following error:

```
"Error installing file, *-xccdf.xml: SCC did not find a valid SCAP 1.2 data-stream-collection"
```

Answer:

DISA STIG "Manual"s are not SCAP content. They contain an XCCDF XML file, with a xslt transform, meant to be viewed with Internet Explorer in order to perform a manual assessment of the system. They do not contain any OVAL xml, which is required for automation.

To obtain SCAP content from DISA, download "Benchmarks" from <https://www.cyber.mil/stigs/SCAP>

A.2 How can I scan CENTOS Linux, Rocky Linux, Alma Linux, Debian Linux etc. with an existing SCAP benchmark?

SCAP content is designed to be applicable to a specific OS or application version, but SCC has a feature to ignore this.

GUI:

```
Options -> Show Options -> Scanning Options -> SCAP Options -> Run all content regardless of applicability
```

CSCC

```
cscf --config -> 6. Configure Options -> Scanning Options -> 3. [X] Run all content regardless of applicability
```

A.3 I just installed new/updated SCAP content from DISA, why am I no longer seeing Manual Questions?

SCAP content from DISA does not contain any manual questions. The SCC team is maintaining a repository of "Enhanced" content, which contains all of the automated rules from the DISA content, along with manual questions from the DISA STIG Manual. In order to include manual questions in SCAP content, only install content from NIWC's repository located at:

<https://www.niwcatlantic.navy.mil/scap/scap-content-repository/>

Our "Enhanced" repository should be updated shortly after DISA posts new/updated benchmarks to their repository at cyber.mil.

A.4 Can I scan Linux/Solaris/Mac from Windows?

Yes. The SSH based scanning allows for UNIX (linux/solaris/mac) scans to be performed from Windows. Refer to section 4.3.

A.5 Is SCC officially SCAP validated?

Yes.

SCC version 4.1.1 was officially SCAP validated on August 26, 2016 against the SCAP 1.2 standards.

SCC version 5.6 was officially SCAP validated on November 4, 2022, against the SCAP 1.3 standards.

SCC currently supports SCAP versions 1.0, 1.1, 1.2 and 1.3.

<https://csrc.nist.gov/Projects/scap-validation-program/validated-products-and-modules/147-scc-scap-1-3-product-validation-record>

<https://csrc.nist.gov/projects/scap-validation-program/validated-products-and-modules/140-spawar-scap-1-2-product-validation-record>

A.6 Who should I report SCAP content issues to?

For content which lists the Publisher as DISA, please contact DISA:: disa.stig_spt@mail.mil

For content which lists the Publisher as NIWC LANT, please contact our team: scc.fct@navy.mil

A.7 Why does NIWC's 'Enhanced' content have a different version from DISA SCAP content?

In order to allow end users to be able to install both DISA content and NIWC Enhanced content at the same time, the benchmark version of our enhanced content needs to be different from DISA SCAP content. The Enhanced content will be based on the DISA STIG SCAP content version, but with an additional set of numbers for our Enhanced content release.

<DISA STIG SCAP Benchmark Version> + "." <NIWC Enhanced Content Version>, and the enhanced content is dropping the leading 00's for brevity.

Example:

DISA SCAP:

U_Adobe_Acrobat_Reader_DC_Continuous_V2R2_STIG_SCAP_1-2_Benchmark.xml
internally it's version 002.002

NIWC Enhanced SCAP:

U_Adobe_Acrobat_Reader_DC_Continuous_V2R2_STIG_SCAP_1-2_Benchmark-enhancedV1.xml internally it's version 2.2.1

If there were updates to the STIG Manual, but no updates to the DISA SCAP Benchmark after 2.2.1 was released, we would regenerate the content with updated manual questions and then the Enhanced content would be version 2.2.2.

A.8 Does SCC provide any remediation functionality?

No. This software only analyzes the system, it does not modify any setting.

A.9 Where can I learn more about creating my own SCAP content?

<http://csrc.nist.gov/publications/PubsSPs.html> - (SP800-126 and SP 800-117)

<http://ovalproject.github.io/> - OVAL

<http://scap.nist.gov/specifications/xccdf/> - XCCDF documentation

A.10 Why is SCC default SCAP content and application updates URL's hosted on a .com?

The default content repository URL included with SCC is an XML file maintained on github by DISA. The URL is part of DISA's <https://disa-stigs.github.io/> page, and updates are posted by an authorized DISA employee. It was decided to use DISA's github portal for the XML feed for two reasons.

1. To prevent any denial of service issues with their cyber.mil page, as we have thousands of end users, running SCC on millions of computers, and github has much more bandwidth, capability and reliability than cyber.mil.
2. Speed of updates. Any updated files posted to cyber.mil have to go through a lengthy approval process, and would delay content update availability by days or weeks.
3. Cyber.mil may be replaced with another website at some point, while github likely will remain a constant, and the XML file on github can then be updated to point to any cyber.mil replacement website.

Additionally, the XML file is just a pointer to cyber.mil, so if SCAP content or application updates are needed, they are downloaded directly from cyber.mil, but we assumed this would be much less frequent than end users checking for updates.

Below is an snip from the current XML feed located at:

<https://raw.githubusercontent.com/DISA-STIGS/DISA-STIGS.github.io/master/content-repository.xml>

```
<content-id>U_Adobe_Acrobat_Reader_DC_Continuous_V2R2_STIG_SCAP_1-2_Benchmark-enhanced.zip</content-id>
<location>https://www.niwcatlantic.navy.mil/wp-content/uploads/2022/12/U_Adobe_Acrobat_Reader_DC_Continuous_V2R2_STIG_SCAP_1-2_Benchmark-enhanced.zip</location>
<checksum
style="SHA512">6a8f8593d41c16807d483f272de24b5c7940348ce7ebf677928f4dadf887ac5247be8b996bdba74cf4a4b0ab32435e09b0e1aaa0d87820998ae5f15cf8a41276</checksum>
```

```

    <benchmarks>
      <benchmark>
        <benchmark-
id>xccdf_mil.disa.stig_benchmark_Adobe_Acrobat_Reader_DC_Continuous_
Track_STIG</benchmark-id>
        <title>Adobe Acrobat Reader DC Continuous
Track STIG SCAP Benchmark - NIWC Enhanced with Manual
Questions</title>
        <version>002.002</version>
        <style>SCAP_1.2</style>
        <status>accepted</status>
        <status-date>2021-06-22</status-date>
        <creator>DISA</creator>
        <publisher>DISA</publisher>
        <contributor>DISA</contributor>
        <source>STIG.DOD.MIL</source>
      </benchmark>
    </benchmarks>

```

As you can see, the actual content will be downloaded directly from cyber.mil, with SHA512 checksum being performed to ensure it's what is expected. SCC's application update XML feed works in the same manner.

A.11 Can I create my own offline SCAP content repository for my isolated network?

Yes, although you will need to have your own web server that you can write files to that doesn't require authentication.

Option 1: To make an offline copy of NWC's content repository;

1. Copy the following file from DISA's github repository to your computer:
<https://raw.githubusercontent.com/DISA-STIGS/DISA-STIGS.github.io/master/niwc-content-repository.xml>
2. Copy all of the SCAP Benchmark zip files from NIWC's content repository website:
<https://www.niwcatlantic.navy.mil/scap/scap-content-repository/>
3. Copy all of the SCAP Benchmarks to your own web server
4. Edit the content repository XML file URLs to match the URL's to your zip files
(ex: update
<location>https://www.niwcatlantic.navy.mil/wp-content/uploads/2022/12/U_Adobe_Acrobat_Reader_DC_Continuous_V2R2_STIG_SCAP_1-2_Benchmark-enhanced.zip</location>
to be
<location>https://your-webserver/yourDirectory/U_Adobe_Acrobat_Reader_DC_Continuous_V2R2_STIG_SCAP_1-2_Benchmark-enhanced.zip</location>
5. Copy the updated content repository to your server (ex: <https://your-webserver/content-repository.xml>)
6. Update SCC to use your webserver
 - a. via GUI:
SCC -> Options -> Update Options -> Right click on the existing URL, and click edit, then edit and save to <https://your-webserver/content-repository.xml>
 - a. via CSCC Config

```
csc -config -> 6. Configure Options -> 7. Update  
Options -> Delete Repository URL, then Add new Repository  
URL
```

a. scripted via CLUI:

```
csc --setOption contentRepository https://your-  
webserver/content-repository.xml
```

Options 2: To make an offline copy of DISA's content repository, that SCC can use for updating:

1. Copy the following file from DISA's github repository to your computer:
<https://raw.githubusercontent.com/DISA-STIGS/DISA-STIGS.github.io/master/content-repository.xml>
2. Copy all of the SCAP Benchmark zip files from DISA's Cyber.mil website:
<https://cyber.mil/stigs/scap/>
3. Copy all of the SCAP Benchmarks to your own web server
4. Edit the content repository XML file URLs to match the URL's to your zip files
(ex: update

```
<location>https://dl.dod.cyber.mil/wp-  
content/uploads/stigs/zip/U_Adobe_Acrobat_Reader_DC_Classic_V2R  
1_STIG_SCAP_1-2_Benchmark.zip</location>
```


to be

```
<location>https://your-  
webserver/yourDirectory/U_Adobe_Acrobat_Reader_DC_Classic_V2R1_  
STIG_SCAP_1-2_Benchmark.zip</location>
```
5. Copy the updated content repository to your server (ex: <https://your-webserver/content-repository.xml>)
6. Update SCC to use your webserver
 - a. via GUI:
SCC -> Options -> Update Options -> Right click on the existing URL, and click edit, then edit and save to <https://your-webserver/content-repository.xml>
 - a. via CCCC Config

```
csc -config -> 6. Configure Options -> 7. Update  
Options -> Delete Repository URL, then Add new Repository  
URL
```
 - a. scripted via CLUI:

```
csc --setOption contentRepository https://your-  
webserver/content-repository.xml
```

To create your own custom content repository xml file, not based on the existing DISA repository, we suggest using the DISA STIG content repository xml feed as a template. The following checksum 'styles' are supported, SHA3_256 is recommended, but not easily available via windows/linux command line.

- SHA3_256
- SHA256
- SHA512

The following may also work but are not recommended

- SHA1
- MD5

A.12 Can SCC run directly from a CD-ROM?

Yes

A.13 Can SCC be run as a non-Administrator or non-root user?

On Windows, SCC offers limited functionality for non-Administrator users, including SSH based UNIX and Cisco scanning and alternate credentials Windows WMI remote scanning. Refer to section 8 for more information.

On Linux, Mac, and Solaris, root is required to run SCC.

A.14 How can I use RunAs (Secondary Logon) with SCC?

Background: Why would a user need to do this?

If a user is logged on a non-administrative account or to one domain and wishes to review computers in a different domain, the use of the RunAs command can allow the user to complete this task without having to logon to the computer as domain administrator of the target domain.

Option 1: Run via Start Menu Shortcut as privileged user

1. Click Start -> Programs -> SCAP Compliance Checker 5.12 -> then Right click on SCAP Compliance Checker.
2. Click on "RunAs".
3. Type the "Domain\Username" and "Password", and then Click "OK".

Option 2: Run Command Prompt as a privileged user

1. Open the command prompt as another user and then run the application from the installation directory from the command prompt.

```
RunAs /env /user:domain_name\username "cmd"
```

A new command window will appear with a header of cmd (running as domain_name\username)

2. Change the directory to the installation directory.
c: (or drive application is installed to)
cd\
cd "Program Files\SCAP Compliance Checker 5.12" (Install Folder)
3. Run the application
SCC.exe
or
CSCC.exe

OS Requirements:

- The Secondary Logon Service must be running.

Known Issue:

- The feature in the SCC software which opens the results directory (Results -> Open Results Directory) will not function as expected when running via RunAs. The Windows Explorer cannot be called while running from RunAs, this is a limitation of Microsoft Windows.

A.15 What type of network traffic does a remote Windows SCC scan generate?

SCC performs a variety of system calls as it attempts to perform compliance checks from the SCAP XML content. If the target computer is a remote computer, the SCC will generate network traffic to perform the checks. The volume and variety of network traffic will be dependent on the XML content, however, below is a list of various types of network traffic you can expect to see as a result of running SCC with the default USGCB SCAP content included in the SCC installer.

Protocols:

- ICMP Echo (ping) requests/replies
- TCP NBSS session requests
- TCP SMB AndX requests/responses
- TCP WINREG OpenKey/CloseKey requests/responses
- TCP RPC/DCOM (for WMI queries)
- HTTP GET (for WindowsUpdateAgent content, which is rarely if ever used in SCAP content)

Ports

- RPC TCP 135
- SMB/NBSS/WINREG TCP 445, 139
- WMI/DCOM/RPC Randomly allocated high TCP ports¹ TCP random port number between 1024 - 4000 and random port number between 49152 - 65535²
- HTTP/HTTPS 80, 443 (but unlikely, unless SCAP content has WindowsUpdateAgent tests)

For more information on WMI/RPC/DCOM ports

<https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/service-overview-and-network-port-requirements>

A.16 Where can I download the DOD Root Certificates for Windows (InstallRoot)?

DISA provides the DOD Root Certificates for Windows via an installation package called InstallRoot. As of the release date of SCC 5.12, the URL is: <https://www.cyber.mil/pki-pke/>

A.17 Why does SCC review all cached profiles to confirm that User profile settings are accurate

User based GPO's are applied on a per-user basis, meaning that each user can have different configurations, so the only way to know if the system is compliant is to check all users. Some tools just check the currently logged in user, which may only provide a false sense of security.

SCC uses the OVAL tested/vetted and approved 'ntuser' test to complete this task. SCC reviews the currently logged in user, and then finds any cached profile for all 'human' users and checks their ntuser.dat files. SCAP content can determine which users are in scope such as local, domain, enabled, disabled, and number of days since last login.

Making domain users compliant is generally very straight forward, and likely already implemented in your environment. Local users do not obtain security settings from any domain based GPO, and can be more challenging to configure. Usually the easiest answer is to disable all local users. If you need local users, please see FAQ 18, and how to make them compliant.

A.18 How do I make local users compliant with user level GPOs?

The Windows OVAL test called 'ntuser' is designed to review cached user profile ntuser.dat files to validate that they are compliant with policies. This test will be used for the NIWC developed Microsoft Office 365 SCAP content, and likely others. For Local users, applying User based GPO's can be less intuitive than Domain Users.

The LGPO tool included in the Microsoft Security Compliance Toolkit 1.0 can be used to apply GPOs (user and machine) to systems which are not domain-joined. There are likely other ways to accomplish this task, proceed at your own discretion.

Below is an example usage scenario:

To apply the Microsoft Office 365 (MS O365) DISA STIG GPOs (user and machine):

- 1) Obtain a copy of the LGPO tool from <https://www.microsoft.com/en-us/download/details.aspx?id=55319> (LGPO.zip)
- 2) Obtain a copy of the MS O365 DISA STIG GPO from <https://www.cyber.mil/stigs/gpo/> (which can be found in the Group Policy Objects (GPOs) - October 2022 download file (U_October_2022_STIG_GPO.zip))
- 3) Obtain a copy of the MS O365 DISA STIG from <https://www.cyber.mil/stigs/downloads/> (Microsoft Office 365 ProPlus STIG - Ver 2, Rel 7 (U_MS_Office_365_ProPlus_V2R7_STIG.zip))
- 4) Login to the non-domain system using a local Administrator account
- 5) Extract the contents of LGPO.zip, U_October_2022_STIG_GPO.zip, and U_MS_Office_365_ProPlus_V2R7_STIG.zip (or most recent Office GPO)
- 6) Run Command Prompt as Administrator and navigate to the LGPO\LGPO_30 directory
- 7) Execute the LGPO tool:
`LGPO.exe /g '<path>\U_October_2022_STIG_GPO\DoD Office 2019-M365 Apps v2r7\GPOs{9D99357A-A462-40DB-A86E-AB358B4CB1AA}' /g '<path>\U_October_2022_STIG_GPO\DoD Office 2019-M365 Apps v2r7\GPOs{DE620AC8-43FF-4123-8C7A-B589D92AC5F0}' /v > lgpo_o365_out.txt 2> lgpo_o365_err.txt`
- 8) User GPO's are applied on login, so any Local user will need to logoff and log back in to apply settings
- 9) Compare the users HKEY_CURRENT_USER registry keys/values against the MS O365 DISA STIG and verify that all policy settings have been properly applied (the DISA STIG can be viewed using the STIG Viewer)

APPENDIX B - KNOWN ISSUES

B.1 Potential out of memory crashes with very large OVAL XML content files

It is not recommended to install OVAL source content larger than 30 MB in size. When loading OVAL XML content, it's common for SCC to use 20-30 times the XML file size in RAM. This means that a 20 MB source OVAL XML file could use 400-600 MB of RAM to load and use. When memory usage goes above 1-2 GB, SCC stability issues may occur.

Source OVAL XML files larger than 10 MB are not include in any SCAP content currently available, but it is possible to download raw OVAL files, such as the entire CIS OVAL repository that could cause stability issues with SCC.

B.2 Unable to scan RHEL8 systems via SSH with OS application white listing enabled (SCC failed to launch)

Starting with SCC 5.5, we have added a new option, disabled by default, which can allow SCC to automatically whitelist itself and allow it to run remotely via SSH. See the SSH Remote Scanning Option to enable it, and for more information on how the temporary configuration change works.

B.3 Host Key Check Failed when scanning RHEL7/8 and Ubuntu systems via SSH when changing between SCC 5.4 and 5.4.2 or later

SCC's internal libssh2 module was updated for SCC 5.4.1, and this adds support for more modern/secure ssh host key exchanges. This causes the host key saved in SCC 5.4 to be different from SCC 5.4.2 on most modern Linux systems. There is an auto-negotiate that occurs, and the highest security method by both server and client is used. So for RHEL6 and Solaris 10, the keys will not change, but for anything more modern host keys will change.

To resolve this issue, using the Host Credential Manager, do a test connection on all hosts using 5.4.2 or later and accept the new host keys.

B.4 Account lockout issues when scanning Ubuntu remotely via SSH with correct credentials

Refer to the SSH Troubleshooting section for details.

B.5 Command prompt closes when run with UAC

With the update to csc.exe to automatically elevate to a privileged user, if you run csc.exe as a standard user, and pass in command line arguments that are invalid, Windows Vista/7 etc. will spawn a new command prompt as administrator, in which csc.exe runs. When the application closes, Windows closes this command prompt, and the end user will not be able to see the usage statement.

Example:

Running as a non-administrator:

```
csc.exe -z
```

Windows will prompt the user with a User Account Control prompt, then open a new command prompt. Since the -z is not a valid parameter, csc.exe will print the usage statement to the screen, however Windows will close the elevated command prompt before you are able to read it.

Workaround:

- Open the command prompt as administrator prior to running csc.exe

B.6 Maximum directory/file length error when copying SCC results

SCC creates an organizational tree structure of the results from each scan, in order to allow the data to be easier to find by the end user. However, as the directory and file paths contain the SCAP Stream Name, this path can be very long, such as the following example:

```
C:\Program Files\SCAP Compliance Checker 5.12\Results\SCAP\LONG-
COMPUTERNAME\U_Windows_7_V1R6_STIG_Benchmark\2\2011-12-23_160049\XML\LONG-
COMPUTERNAME_SCC-5.12_2011-12-23_160049_OVAL-CPE-
Results_U_Windows_7_V1R6_STIG_Benchmark.xml
```

The path above is 240 characters, which is allowable by the Windows OS, however if a user tries to copy the results to a network share, with a longer starting path, such as

```
\\somenetworkpath\some-long-sub-directory\some-other-sub-directory\SCAP Compliance Checker
5.12\Results\SCAP\LONG-COMPUTERNAME\U_Windows_7_V1R6_STIG_Benchmark\2\2011-12-
23_160049\XML\LONG-COMPUTERNAME_SCC-5.12_2011-12-23_160049_OVAL-CPE-
Results_U_Windows_7_V1R6_STIG_Benchmark.xml
```

The new path is 290 characters long, which exceeds the limit of 260 set by Windows

<http://msdn.microsoft.com/en-us/library/windows/desktop/aa365247%28v=vs.85%29.aspx#maxpath>

Several workarounds exist for this:

- o In the Content and Options menu, on the Output Options tab you are able to reduce the folders created by SCC in the Output Subdirectory Options section and reduce the length of the filename using the Output Filename Options
- o Only copy the results to a directory path that has less length than the SCC install path
- o Only copy the results subdirectory that you need, such as SCAP
- o Write a script to copy just the files you need without any directory structure into a flat directory. Since SCC includes the computername, date timestamp and SCAP stream name in the resulting files, the likelihood of conflicts is low.

B.7 Reviewing Domain Controllers with numerous user accounts

Domain controllers with large numbers of users and computers (several thousand), may not be able to be reviewed using the current DISA STIG SCAP content. There are several checks that require .* operations, which take a long time to process and can generate GB's of results, and likely will cause SCC to crash before completion.

As a partial workaround, we have added an OVAL processing option to prevent Access Token (User Rights) from causing SCC to crash.

Other checks may still exist in the content that perform .* operations for User accounts and SID's, and the only workaround would be to manually disable these rules in the XCCDF XML file.

B.8 Issues with WMI scanning if GPO is configured to block execution of all .bat files

SCC's remote WMI scanning requires the ability to create an execute a batch file. Without it, the scan will fail. See the Requirements section for more details.

B.9 Issues with remote scanning (Classic and WMI) if scanning PC is not fully patched

At some point during 2022, Microsoft released a security patch that appears to have broken some remote API calls from outdated versions of Windows to patched computers. If you experience issues, and are also unable to connect to the target computer via Remote Desktop Protocol, the patch level of your scanning computer may be out of date. Updating the scanning computer to current appears to resolve this issue.

The error reported is

```
Failed to connect to <target computer> via WMI, Error  
Win32::OLE(0.1712) error 0x80070005: "Access is denied"      after  
character 0 in "WinMgmts:{impersonationLevel=impersonate}!//<target  
computer>" : Access is Denied
```

APPENDIX C - TROUBLESHOOTING

C.1 Troubleshooting UNIX SSH Remote Scanning

C.1.1 SSH Authentication Troubleshooting

C.1.1.1 For Password Authentication: Verify sshd_config is configured with "PasswordAuthentication yes"

This setting is disabled by default on SUSE Enterprise Linux 12, and could be configured to "PasswordAuthentication no" on any system. Some systems may have it commented out "# PasswordAuthentication yes". We recommend having it explicitly set, as each OS could have different default value.

SCC will not be able to SSH to the system, and the error returned from the system will appear the same as a bad username/password. To make debugging more challenging, manually SSH'ing outside of SCC will likely work as expected. This setting appears to make the connection mandate an interactive session, breaking any automation.

C.1.1.2 For Private Key Authentication: Verify private key is using RSA

Putty generated keys are not support. The file should look like the following:

```
-----BEGIN RSA PRIVATE KEY-----  
large text block that is your private key...  
-----END RSA PRIVATE KEY-----
```

or

```
-----BEGIN OPENSSH PRIVATE KEY-----  
large text block that is your private key...  
-----END OPENSSH PRIVATE KEY-----
```

SCC 5.12 should log a specific error regarding this issue.

C.1.1.3 For direct root login: Verify sshd_config is configured with 'PermitRootLogin yes'

This method is not allowed in DISA STIG's so it's not a recommended method for SCC.

C.1.2 SSH Escalation Troubleshooting

C.1.2.1 SUDO: Verify that sudo is installed on Solaris 10/11

'sudo' is not installed by default, and will need to be installed if sudo scanning methods are enabled.

SCC 5.12 should log a specific error regarding this issue.

C.1.3 Verify target partitions exist and have sufficient freespace

SCC uses /tmp for pushing files to before moving them to their final directory (usually /opt/scc-remote), but /opt can be changed by the end user to any directory. SCC will create the 'scc-remote' subdirectory on each scan, and remove it when it's complete.

SCC requires at least 200 MB free in /tmp and 2048 MB in /opt/scc-remote

C.1.4 Known issues with remote RHEL targets and Application Whitelisting with fapolicy

Starting with SCC 5.5, we have added a new option, disabled by default, which can allow SCC to automatically whitelist itself and allow it to run remotely via SSH. See the SSH Remote Scanning Option to enable it, and for more information on how the temporary configuration change works.

C.1.5 Known issues with remote Ubuntu targets and pam_tally2 and account lockout (when using the correct password)

The STIG for Ubuntu 18 has the user add pam_tally2 to the /etc/pam.d/common-auth file, with the lockout being 3 wrong passwords. This affects SCC because of an oversight in the pam_tally2 functionality, where, upon a successful SSH login, an unsuccessful login attempt will be tallied. Remote scanning appears as several SSH scans at once, causing the user to be locked out immediately and preventing the scan. According to the man page for pam_tally2, the login attempt counter is incremented, then the password is checked, and afterwards pam_setcred should be called to reset the counter if successful. On a default install of Ubuntu 18, this pam_setcred is not installed, nor is it called in the etc/pam.d/common-auth file, meaning the attempt counter never gets reset. A fix for this issue is adding the line **account required pam_tally2.so** to the **/etc/pam.d/common-account** file before any other account statements.

The user is also advised to reset the tally counter on their account used to scan.
pam_tally2 --user USERNAME --reset where USERNAME is the account name.

The STIG is scheduled to be updated to resolve this issue in October 2021.

C.2 Troubleshooting Windows Remote Scanning

C.2.1 Verify Scanning Computer is Fully Patched

If the scanning computer is not at least as patched as the target computer, it is quite likely that any WMI based query may fail with an "Access is Denied" error. There appears to be a patch released in 2022 that breaks WMI (and RDP) unless both the scanning and target computers are at the same patch level. The easy solution is to ensure that the scanning computer (the one you are logged into) is fully patched.

C.2.2 Ensure Necessary Services are Enabled and Running

Refer to Section 2 - Requirements to determine the required services depending on the scanning mode selected.

C.2.3 Ensure a Client Firewall is not Blocking RPC, Registry, Shares or WMI

If a client firewall is blocking LAN/WAN access to the Remote Procedure Calls (RPC), Remote Registry, File Shares or WMI, remote reviews with SCC will not be possible. Enabling these port exceptions will vary for each firewall product. Please refer to your firewall software documentation regarding opening specific ports.

C.2.4 Verify Administrative Rights

This issue should only occur for a user wanting to review a computer that is not part of a domain for which that person is a domain administrator. In order to attain local administrative rights on a single remote computer, the user may need to map an administrative share or connect to the remote registry of that system.

C.2.5 Verify "Manage auditing and security log" User Right contains the Administrators group

On Windows Vista and later, SCC uses the auditpol.exe application to obtain the system's audit configuration. In order to run this command, the user running the software must have the User Right, "Manage auditing and security log". By system default, the Administrators group is a member of this right. However, if the security setting is modified and the Administrators group is removed, errors will be reported, any check related to the Windows 'Audit Policy' will error out.

C.2.6 Verify "Access this computer from the network" User Right for target computer.

If the account you are logged into does not have access to the target computer over the network, remote scans will not be possible. In order to scan remotely, the Administrators or Domain Admins group(s) may need to have this user right. If you do not have this user right, you may receive 'access is denied' type errors.

C.2.7 Verify "Deny access to this computer from the network" User Right for target computer.

If the account you are logged into is a member of a group being denied access to the target computer over the network, then remote scans will not be possible. If the Administrators or Domain Admins group(s) are being denied access, this could be a reason for errors, especially 'access is denied' type errors.

C.2.8 Additional items for scanning non-domain joined computers

C.2.8.1 If the target is not in Active Directory, verify scanning credentials

For Classic scans, it is necessary to add to the target system a local admin account with the same credentials as being used to log into the SCC host.

For WMI scans, SCC's WMI Alternate Credentials feature can be used, or follow the same method as Classic with matching credentials on both systems.

C.2.8.2 If the target is not in Active Directory, verify UAC remote restriction are disabled for WMI based scans

To disable UAC remote restrictions, on the target computer, follow these steps:

1. Click Start, click Run, type regedit, and then press ENTER.
2. Locate and then click the following registry subkey:
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System
3. If the LocalAccountTokenFilterPolicy registry entry doesn't exist, follow these steps:
4. On the Edit menu, point to New, and then select DWORD Value.
Type LocalAccountTokenFilterPolicy, and then press ENTER.
Right-click LocalAccountTokenFilterPolicy, and then select Modify.
5. In the Value data box, type 1, and then select OK.
6. Exit Registry Editor.

Reference: Microsoft Windows 10 Security Technical Implementation Guide :: Version 2, Release: 7 Benchmark Date: 07 Jun 2023 Rule Title: Local administrator accounts must have their privileged token filtered to prevent elevated privileges from being used over the network on domain systems.

STIG ID: WN10-CC-000037 Rule ID: SV-220799r569187_rule Vul ID: V-220799

Check Content:

If the system is not a member of a domain, this is NA.

<https://learn.microsoft.com/en-us/troubleshoot/windows-server/windows-security/user-account-control-and-remote-restriction>

C.2.8.3 If the target is not in Active Directory, verify Limited User Account (LUA) settings for classic scans

When performing a non-domain Classic mode remote scan, it is necessary to disable Limited User Account (LUA) via the Registry at HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System by changing the EnableLUA value from "1" to "0".

After the change is made, it is necessary for the computer to be rebooted for the new setting to take effect.

Note: This configuration is not compliant with the DISA STIG's so proceed only if you have to use SCC's 'classic' Windows remote scanning mode.

C.2.8.4 Verify "Deny access to this computer from the network" User Right for target computer.

For non-domain scans, make sure that the local administrator account that you plan to scan with is not listed.

C.2.9 Testing Connections

If SCC is unable to perform a remote review of a system, please perform the following tests before reporting any issues.

C.2.9.1 Map an Administrative Share

To map an administrative share to a target computer:

1. Right Click on My Computer.
2. Click on Map Network Drive. The Map Network Drive window will open.
3. In the Folder field, type in the drive you are wishing to access, most likely the C drive.
\\servername\C\$
4. Deselect the “Reconnect at logon” option.
5. Click on Finish.
6. If prompted to enter Username and Password, do so using an account with administrative rights on that system.

C.2.9.2 Ensure Administrative Shares are enabled on the target computer

If the steps in C.6.1 do not allow you connect, but also do not prompt you for alternative connections, it's possible that the system has been configured to disable administrative shares.

1. On the target computer, login locally and open a command prompt
2. Type 'net share'
3. Verify that both the C\$ and ADMIN\$ shares appear

```
C:\>net share
Share name      Resource
-----
C$              C:\
IPC$            Remote IPC
ADMIN$          C:\Windows
Remote Admin
The command completed successfully.
```

If either the C\$ or ADMIN\$ shares do not appear, refer to the following MSDN article regarding the registry values in
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\LanmanServer\Parameters\AutoShareServer and
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\LanmanServer\Parameters\AutoShareWks

C.2.9.3 Remotely Connect to a Computer's Registry

For Classic remote scanning, verify that the Remote Registry is available, To Remotely Connect to Computer's Registry:

1. Click on Start – Run. The Run window will open.
2. Type in Regedit and click OK. The Registry Editor will open.
3. Select File – Connect Network Registry...
4. Type in the NetBIOS computer name in the window provided. Select OK.
5. If prompted to enter Username and Password, do so using an account with administrative rights on that system.

C.2.9.4 Testing the WMI Connection

WMI scanning requires WMI to operate at all, and even with classic mode WMI queries can be required to perform certain checks. To verify that WMI functional perform either of the following methods, where <Computername or IP is replaced with the computer you wish to test.

Powershell Method

1. Open an Admin Power Shell Prompt
2. Type `Get-WmiObject Win32_ComputerSystem -ComputerName <Computername or IP>`
3. If WMI is available remotely you should see the following:

Domain	: <your domain>
Manufacturer	: <Your manufacturer>
Model	: <your model>
Name	: <Computername>
PrimaryOwnerName	: <some user>
TotalPhysicalMemory	: <some number>

If you see the following, WMI is not going to work remotely, and it could be caused by several conditions. Refer to the next section, Testing RPC Issues.

"Get-WmiObject : The RPC server is unavailable."

MMC Snap In Method

1. Click Start -> Run.
2. Type `mmc.exe`.
3. Click File - > Add/Remove Snap-In.
4. Click the Add button.
5. Scroll Down to WMI Control and Click on it.
6. Click the Add button.
7. Click the radio button next to "Another Computer".
8. Type the computer you want to test.
9. Click Finish.
10. Click Close.
11. Click OK.
12. Right Click on WMI Control for <Computername or IP>.
13. Click Properties.
14. If the General Tab states "Successfully Connected to: \\<Computername>", then a WMI based review should be possible.
15. If any errors are listed on this tab, WMI is not going to work remotely, and it could be caused by several conditions. Refer to the next section, Testing RPC Issues.

C.2.9.4.1 Microsoft WMI Links

Below are some links maintained by Microsoft related to WMI.

- Troubleshooting and Tips <https://technet.microsoft.com/en-us/library/ee692772.aspx>

C.2.9.5 Testing RPC Issues

Several common issues can cause the RPC server unavailable error:

- The RPC service is stopped on the remote computer;
- The RPC Endpoint Mapper port TCP/135 is not accessible on the remote computer;
- The Dynamic RPC range is blocked by firewalls installed between your computer and the remote computer.

1. Make sure RPC Endpoint Mapper port 135 is listening on a remote computer.

- Login to the remote/target computer
- Use the following command: "netstat -ano | find "135"
- The following should be returned:

```
TCP      0.0.0.0:135          0.0.0.0:0
          LISTENING      924
TCP      [::]:135            [::]:0
          LISTENING      924
```

2. Verify that the RPC Service is enabled and running

- Login to the remote/target computer
- Run the following command: "sc query rpcss"
- The following should be returned:

```
SERVICE_NAME: rpcss
              TYPE                : 20  WIN32_SHARE_PROCESS
              STATE                 : 4   RUNNING
                                   (NOT_STOPPABLE,
NOT_PAUSABLE, IGNORES_SHUTDOWN)
              WIN32_EXIT_CODE       : 0   (0x0)
              SERVICE_EXIT_CODE    : 0   (0x0)
              CHECKPOINT            : 0x0
              WAIT_HINT             : 0x0
```

- If any errors are reported with the RPC Service , verify that the dependencies for RPC Service are running (DCOM Server Process Launcher and RPC Endpoint Mapper)
 - On the remote/target computer, run the following 2 command
 - sc query DcomLaunch
 - sc query RpcEptMapper
 - Both should return back

```
STATE                : 4   RUNNING
```

3. If the RPC Service is running, verify if remote connections report any errors

- From your local scanning computer (not the remote/target computer)

- Open an Admin Powershell Prompt
- Type Test-NetConnection <Computername or IP> -port 135

The command may take some time to return data, but should eventually return back

```
ComputerName      : <Computername>
RemoteAddress     : <Target IP>
RemotePort        : 135
InterfaceAlias    : Ethernet
SourceAddress     : <Your local IP>
TcpTestSucceeded  : True
```

4. Verify 'Restrictions for Unauthenticated RPC Clients Setting' is not set to 'Enabled: Authenticated Without Exception'

Review the following setting: Policies\Administrative Templates\System\Remote Procedure Call "Restrictions for Unauthenticated RPC Clients". This can be set to either "Not Configured" (default) or "Enabled: Authenticated". If it is set to the highest setting "Enabled: Authenticated Without Exception" we have seen instances where this will no longer allow review of a system via WMI.

5. Verify DCOM is enabled

- <http://msdn2.microsoft.com/en-us/library/ms687298.aspx>

C.2.9.7 Test ability to run auditpol.exe on target system

To test run the following command, which should return back all of the audit configuration of the local computer:

```
Auditpol /backup /file:<file>
```

If this returns any error messages, SCC will not be able to check the audit configuration of the computer. In order to allow SCC or most applications to view the audit configuration, the Administrators group will need to be added back to the "Manage auditing and security log" User Right.

C.2.9.8 Re-scan with SCC

If all of the above tests were successful, please re-scan the target computer with SCC.

APPENDIX D – SCC AND SCAP

D.1 SCAP Validations & Capabilities

- SCAP Versions Supported
 - SCAP Version: 1.0
 - Validation Date: February 25, 2009
 - SCAP Version: 1.1
 - SCAP Version: 1.2
 - Validation Date: August 26, 2016
 - SCAP Version: 1.3
 - Validation Date: November 4, 2022
 - SCAP Version 1.4
 - Self Assertion of compliance: May 5, 2025
- SCAP Capabilities
 - Authenticated Configuration Scanner (ACS)
 - Common Vulnerability Enumeration (CVE)
 - Open Checklist Interactive Language (OCIL)

D.2 Standards Supported

STANDARD	VERSION SUPPORTED
SCAP	1.0, 1.1, 1.2, 1.3, 1.4
OVAL	5.3 -> 5.12.1
OCIL	2.0
XCCDF	1.1.4 and 1.2
CPE	2.2, 2.3
CCE	5.0
CVE	
AI	1.1
ARF	1.1
TMSAD	1.0
Software Identification (SWID) Tags	2015

D.3 SCAP Implementation

SCAP (Security Content Automation Protocol) is a suite of standards used to determine the presence of vulnerabilities, patches and configuration issues on a target system. SCAP content consists of machine readable XML files that contain configuration data, checklist data and logic used to scan a system. The standards include CVE (Common Vulnerabilities and Exposures), CCE (Common Configuration Enumeration), CPE (Common Platform Enumeration), XCCDF (eXtensible Configuration Checklist Description Format), OVAL (Open Vulnerability and Assessment Language) and CVSS (Common Vulnerability Scoring System).

SCAP Compliance Checker processes SCAP content on a target system and produces HTML and text reports, XCCDF results and OVAL results. The HTML and text reports provide benchmark scores and information that a system administrator can use to make the target system more secure. The XCCDF results and OVAL results can be used by other tools in a variety of ways since they are generated using the industry standard XCCDF and OVAL results formats.

SCAP Compliance Checker reads in a SCAP stream which includes XML files written in the XCCDF, OVAL and CPE Dictionary schemas. SCAP Configuration Checker then generates XML results files using the XCCDF and OVAL results schemas. The HTML reports are generated by transforming the generated XCCDF and OVAL XML results files into human readable output. This output contains detailed scoring and results information, as well as CVE, CCE and CPE identifiers.

SCAP Compliance Checker is capable of validating SCAP streams against the industry standard XCCDF and OVAL schemas. All output generated by SCAP Configuration Checker can also be validated.

SCAP Compliance Checker 5.12 implements SCAP version 1.0, 1.1, 1.2, 1.3 and 1.4

D.3.1 How SCC Process SCAP 1.0/1.1 Data Streams

SCC follows the Use Case Requirements in NIST 800-126 which document the following:

COMPONENT	STREAM LOCATOR	REQUIRED/OPTIONAL
XCCDF Benchmark	xxxx-xccdf.xml	Required
OVAL Compliance	xxxx-oval.xml	Required
OVAL Patch	xxxx-patches.xml	Optional
CPE Dictionary	xxxx-cpe-dictionary.xml	Required
CPE Inventory	xxxx-cpe-oval.xml	Required

Where "xxxx" indicates the SCAP stream name, which must be consistent across all files in the SCAP Stream.

From 800-126: "The notation "xxxx" designates a locator prefix that SHALL be associated with a use case specific data source component stream."

The SCC order of operations with a SCAP stream is as follows, and the USGCB 2.0.0.0 Windows XP Stream is used as an example. SCAP Stream Name = "USGCB-Windows-XP"

1. SCC verifies if the XCCDF Benchmark, OVAL Compliance, CPE Dictionary and the CPE Inventory exist for the specified SCAP stream.

USGCB-Windows-XP-xccdf.xml
USGCB-Windows-XP-oval.xml
USGCB-Windows-XP-cpe-dictionary.xml
USGCB-Windows-XP-cpe-oval.xml

2. If all required files are present, SCC then loads the XCCDF file to gather platform information.

USGCB-Windows-XP-xccdf.xml

3. Based on the Profile that was selected in the options form, the SCC then finds the matching profile, and then checks to ensure the profile is not an abstract profile. (<Profile> element doesn't have an "abstract" attribute or the attribute is set to "false".)

united_states_government_configuration_baseline_version_2.0.0.0

4. Next the CPE Dictionary is processed. The platform element from the XCCDF is used to determine what CPE items the target system is part of.

USGCB-Windows-XP-cpe-oval.xml
USGCB-Windows-XP-cpe-dictionary.xml

5. If the content is applicable to the target computer based on the CPE OVAL tests, the XCCDF content is then traversed and loads the OVAL file and/or the OVAL patches files (from filename) and definitions are processed. The definitions that get processed come from the XCCDF rules found during the XCCDF traversal.

USGCB-Windows-XP-oval.xml
USGCB-Windows-XP-patches.xml

6. XML results are created, based on user settings in the options form of the GUI or the --config from the command line.

<Computer>_SCC_5.12_<Date-Time>_OVAL-CPE-Results_USGCB-Windows-XP.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-Patch-Results_USGCB-Windows-XP.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-Results_USGCB-Windows-XP.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-Variables_USGCB-Windows-XP.xml
<Computer>_SCC_5.12_<Date-Time>_XCCDF-Results_USGCB-Windows-XP.xml

7. HTML and/or text based reports are generated based on end user options

D.3.2 How SCC Processes SCAP 1.2, 1.3 and 1.4 Data Streams

SCAP 1.2 as defined in NIST 800-126 rev 2 introduces data-stream-collections, data-streams, and components which are used to combine SCAP 1.0/1.2 components into a single file. Each SCAP 1.2 stream must contain a single data-stream-collection which in turn must contain at least one data-stream and one component. A data-stream may contain dictionaries and checklists, but must contain at least one check.

Upon installation of a SCAP 1.2 stream, if the file contains multiple data-streams within the data-stream-collection SCC will create a new record in the SCAP Content options for each data-stream. The user is then able to select/de-select content based on the data-stream allowing the user to run one or more data-streams from the same data-stream-collection during any given analysis run.

The SCC order of operations with a SCAP 1.2 stream is as follows, and the USGCB 1.2.7.1 Internet Explorer 8 Stream is used as an example. SCAP 1.2 Stream Name = "scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip".

Note: *This is the data-stream name, not the data-stream-collection name*

1. SCC verifies if the XCCDF Benchmark, OVAL Compliance, OCIL Questionnaire, CPE Dictionary and the CPE Inventory components exist for the specified SCAP stream.

```
scap_gov.nist_comp_USGCB-ie8-xccdf.xml
scap_gov.nist_comp_USGCB-ie8-OCIL.xml
scap_gov.nist_comp_USGCB-ie8-oval.xml
scap_gov.nist_comp_USGCB-ie8-cpe-oval.xml
scap_gov.nist_comp_USGCB-ie8-cpe-dictionary.xml
```

2. If all required files are present, SCC then loads the XCCDF component to gather platform information.

```
scap_gov.nist_comp_USGCB-ie8-xccdf.xml
```

3. Based on the Profile that was selected in the options form, the SCC then finds the matching profile, and then checks to ensure the profile is not an abstract profile. (<Profile> element doesn't have an "abstract" attribute or the attribute is set to "false".)

```
xccdf_gov.nist_profile_united_states_government_configuration_baseline
_version_1.2.3.1
```

4. If SCC detects an OCIL Component, the user is prompted to fill out the questionnaire or skip the questions and continue analysis.

```
scap_gov.nist_comp_USGCB-ie8-OCIL.xml
```

5. Next the CPE Dictionary component is processed. The platform element from the XCCDF is used to determine what CPE items the target system is part of.

```
scap_gov.nist_comp_USGCB-ie8-cpe-oval.xml
scap_gov.nist_comp_USGCB-ie8-cpe-dictionary.xml
```


6. If the content is applicable to the target computer based on the CPE OVAL tests, the XCCDF content is then traversed and loads the OVAL file and/or the OVAL patches files (from filename) and definitions are processed. The definitions that get processed come from the XCCDF rules found during the XCCDF traversal.

```
scap_gov.nist_comp_USGCB-ie8-oval.xml
scap_gov.nist_comp_USGCB-ie8-patches.xml
```

7. XML results are created, based on user settings in the options form of the GUI or the --config from the command line. SCAP 1.2 specifies the use of the NIST Asset Reporting Format (ARF) 1.1 for results generation. SCC generates an ARF results file, but we also chose to include the old reports for our current user population.

```
<Computer>_SCC_5.12_<Date-Time>_ARF_ scap_gov.nist_datastream_USGCB-
ie8-1.2.3.1.zip.xml
<Computer>_SCC_5.12_<Date-Time>_OCIL-Results_
scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-CPE-Results_
scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-Patch-Results_
scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-Results_
scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip.xml
<Computer>_SCC_5.12_<Date-Time>_OVAL-Variables_
scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip.xml
<Computer>_SCC_5.12_<Date-Time>_XCCDF-Results_
scap_gov.nist_datastream_USGCB-ie8-1.2.3.1.zip.xml
```

8. HTML and/or text based reports are generated based on end user options.

D.3.3 CVE Implementation

The CVE (Common Vulnerabilities and Exposures) standard links unique identifiers with known security vulnerabilities and/or exposures. CVE identifiers are typically found in the OVAL patch definition content of a SCAP data stream. An OVAL patch definition may contain a reference element that associates the definition with a CVE identifier. Links to various websites containing more information about the vulnerability and/or exposure may also be provided in the reference element.

When the SCAP Compliance Checker processes a SCAP data stream against a target system, any CVE identifiers associated with entities in the stream will be found and provided in the results HTML and text files. It is important to distinguish that SCC does not contain any static CVE database and only imports CVE information from the content stream.

In the SCAP Compliance Checker results HTML files, CVE identifiers can typically be found in the OVAL results HTML file for the patch content. Detailed information on each definition processed can be found in the Definitions section of the HTML file. For each definition, there is a "CVE" row that displays any CVE identifiers that are associated with the definition.

It is important to note that when SCC finds a CVE identifier, it automatically creates a link in the CVE row to the NVD (National Vulnerability Database) webpage for that particular CVE identifier. This allows the user to determine the impact that a particular CVE has based on CVSS impact metrics. This also allows the user to prioritize different vulnerabilities found by comparing vulnerability scores with each other.

CVE Specification - <https://cve.mitre.org>

D.3.4 CCE Implementation

The CCE (Common Configuration Enumeration) standard links unique identifiers with known system configuration issues.

When the SCAP Compliance Checker processes a SCAP data stream against a target system, any CCE identifiers associated with Rules and/or definitions in the stream will be found and provided in the results HTML files. If no CCE identifiers are found within the SCAP data stream, SCC will not provide CCE information in the result files.

CCE identifiers are typically found in the OVAL definition content and the XCCDF content of a SCAP data stream. An OVAL definition may contain a reference element that associates the definition with a CCE identifier. A link to the CCE website containing more information about the system configuration issue is also provided in the reference element. An XCCDF Rule may contain an ident element that associates the Rule with a CCE identifier.

In the SCAP Compliance Checker results HTML files, CCE identifiers can typically be found in the HTML reports. For OVAL results HTML files, detailed information on each definition processed can be found in the Definitions section of the HTML file. For each definition, there is an "Identities" row that displays any CCE identifiers that are associated with the definition, in addition to the CCE identifier.

It is important to note that CCE identifiers in the Detailed Results section of the reports, provides a link to the CCE website to allow the user to gather additional information (e.g. attack vectory, dates, etc.) regarding the configuration issue.

SCAP Compliance Checker 5.12 implements CCE version 5.0, however the Detailed Results section of the reports displays the CCE version 4.0 as well.

CCE 5.0 Specification - <http://cce.mitre.org/>

D.3.5 CPE Implementation

The CPE (Common Platform Enumeration) standard is a structured naming scheme for hardware, operating systems and applications. It allows different tools to specify names for IT platforms in a consistent way. The XCCDF file included in a typical SCAP data stream contains one or more platform elements. The platform element contains a CPE identifier that associates an XCCDF Benchmark, Rule or Group with a target platform. If the target system is not an instance of the CPE identifier specified in a platform element, then the XCCDF Benchmark, Rule, or Group associated with that platform element is not applicable to the target system and will not be processed.

In order to determine if the target system is an instance of a CPE identifier, SCAP Compliance Checker processes the CPE dictionary and the CPE OVAL content in the SCAP data stream. The CPE dictionary contains one or more CPE identifiers, each associated with an OVAL definition that resides in the CPE OVAL content. If SCAP Compliance Checker processes the OVAL definition and the definition returns a result of "true", then the target system is said to be

an instance of the associated CPE identifier. A list of CPE identifiers that the target system is an instance of is compiled in this fashion from the CPE dictionary, then used when processing the XCCDF file. If the CPE identifier specified by a platform element in the XCCDF file is not in the compiled CPE instance list, then the Benchmark, Rule or Group associated with that CPE identifier is not applicable to the target system and will not be processed. Rules that are not applicable to the target system will have a result of "not applicable".

SCAP Compliance Checker 5.12 implements CPE version 2.2, 2.3.

CPE 2.3 Specification - <https://csrc.nist.gov/publications/detail/nistir/7695/final>

D.3.6 CVSS Implementation

The CVSS (Common Vulnerability Scoring System) standard is a system used to assign scores to vulnerabilities. By assigning a score to a vulnerability, one can determine its relative severity when compared to other vulnerabilities.

In the SCAP Compliance Checker the CVE identifiers can typically be found in the security patches section of the HTML reports. For each security patch check, there is a "References" row that displays any CVE identifiers that are associated with the definition. Each CVE identifier will have a link to the NVD database webpage for that CVE. Each link can then be used to obtain the CVSS information from the National Vulnerability Database (NVD) site, including the NIST-calculated CVSS score, the full CVSS vector, and the CVSS calculator.

CVSS 2.0 Specification - www.first.org/cvss

D.3.7 ARF 1.1 Implementation

The ARF (Asset Reporting Format) is a data model to express the transport format of information about assets and the relationships between assets and reports. The standardized data model facilitates the reporting, correlating, and fusing of asset information. SCC automatically generates the results of all SCAP 1.2 data streams into the ARF 1.1 format. The file will be included in the same folder as the other XML result files.

ARF 1.1 Specification - <http://csrc.nist.gov/publications/PubsNISTIRs.html#NIST-IR-7694>

D.3.8 AI Implementation

The Asset Identification (AI) 1.1 specification provides a standardized model for representing and identifying assets. The specification provides the necessary constructs to uniquely identify and correlate assets based on known identifiers and/or information about the assets. SCC identifies all assets utilizing the AI 1.1 specification in the ARF 1.1. result files.

AI 1.1 Specification - <http://csrc.nist.gov/publications/PubsNISTIRs.html#NIST-IR-7693>

D.3.9 TMSAD Implementation

The Trusted Model for Security Automation Data (TMSAD) is a common trusted model that can be applied to specification within the security automation domain (e.g SCAP). The TMSAD is composed of recommendations on how to use existing specifications to represent signatures, hashes, key information, and identify information in the context of an XML document and permits users to establish integrity, authentication, and traceability for security automation data.

SCC implements the TMSAD by verifying digitally signed SCAP 1.2 data streams. The XML digital signature (XMLDSig) implementation is based on requirements from the TMSAD, which includes requirements from W3C (<http://www.w3.org/TR/xmlldsig-core>), and the NIST SP800-126 (<http://csrc.nist.gov/publications/nistpubs/800-126-rev2/SP800-126r2.pdf>).

Supported algorithms include:

- Digests: SHA1, SHA256, SHA384, SHA512
- Encryption: DSA_SHA1, RSA_SHA1, RSA_SHA256, ECDSA_SHA256
- ECDSA Named Curves: prime256v1, secp256k1, secp384r1, secp521r1
- Transforms: C14N, C14N11, EC14N (with and without comments), enveloped signature transform
- Canonicalization: C14N, C14N11, EC14N (with and without comments)

Note: *The current implementation only supports reference that point to elements within the same document (enveloped signatures)*

TMSAD 1.0 Specification - <http://csrc.nist.gov/publications/PubsNISTIRs.html#NIST-IR-7802>

D.3.10 XCCDF Implementation

XCCDF (Extensible Configuration Checklist Description Format) is a language used for writing security checklists and benchmarks. SCAP Compliance Checker loads XCCDF content from a SCAP stream and determines if the Rules specified by the XCCDF content are satisfied by a target system.

SCAP Compliance Checker validates XCCDF content, imports it and allows the user to select a profile from the content. Rules are automatically selected and unselected based on the profile the user selects.

The SCAP stream's CPE dictionary and its associated OVAL definitions are then processed to determine which XCCDF Rules are applicable to the target system. Rules that are found to be inapplicable to the target system based on CPE identifiers are automatically unselected.

SCAP Compliance Checker then traverses the XCCDF content, processing all selected XCCDF Rules against a target system. Scores are calculated using all of the current XCCDF scoring models including the default, flat, flat unweighted and absolute models. Additionally two custom scoring methods are calculated, the spawar-original and spawar-adjusted.

A benchmark results XML document is generated using the XCCDF Results schema. This results file is then transformed into an HTML report, along with more in depth reports generated from the SCAP stream's OVAL content. The benchmark results XML document can be imported into other tools since it uses the industry standard XCCDF Results schema.

SCAP Compliance 5.12 implements XCCDF version 1.1.4 and 1.2.

XCCDF 1.1.4 - <https://csrc.nist.gov/publications/detail/nistir/7275/rev-3/final>

XCCDF 1.2 - <https://csrc.nist.gov/publications/detail/nistir/7275/rev-4/final>

D.3.11 OVAL Implementation

OVAL (Open Vulnerability and Assessment Language) is a language used to standardize the transfer of security content among different tools. SCAP Compliance Checker loads OVAL

content in conjunction with an XCCDF checklist and processes the OVAL definition content against a target system.

SCAP Compliance Checker is able to process all four of OVAL's schemas: the Definitions schema, the System Characteristics schema, the Results schema and the Variables schema.

The Definitions schema is used to define definitions that test a machine's state. This schema is used in SCAP streams to specify patch, vulnerability and configuration content. SCAP Compliance Checker imports OVAL Definitions files and processes the OVAL definitions against a target system.

The System Characteristics schema is used to store data collected from a system. SCAP Compliance Checker uses Object data from OVAL Definitions content and generates System Characteristics data that is later used for testing purposes. This data is stored in an XML file using the OVAL System Characteristics schema.

The Results schema takes State data from OVAL Definitions content along with System Characteristics data and produces Definition and Test results. These results are stored in an XML file that follows the OVAL Results schema. SCAP Compliance Checker then transforms this XML file and produces human readable HTML report documents.

The Variables schema is used to import external variable data into the OVAL engine during processing of an OVAL definition. SCAP Compliance Checker processes the XCCDF content of a SCAP stream and extracts any variables that need to be imported into the OVAL engine. It then creates an XML file using the OVAL Variables schema that contains these variables. The OVAL engine later uses this file during OVAL processing.

By using the industry standard OVAL schemas, SCAP Compliance Checker can share data with any tool that understands OVAL.

SCAP Compliance Checker 5.12 implements OVAL version 5.3 -> 5.12.1.

OVAL 5.12.1 Specification - <https://github.com/OVAL-Community/OVAL>

D.3.12 OCIL Implementation

The Open Checklist Interactive Language (OCIL) defines a framework for expressing a set of questions to be presented to a user and corresponding procedures to interpret responses to these questions. SCAP Compliance Checker loads OCIL content in conjunction with an XCCDF checklist and processes the OCIL questionnaires against a target system. SCAP Compliance Checker can also process OCIL outside of a SCAP 1.1 data stream.

SCAP Compliance Checker 5.12 implements OCIL version 2.0

OCIL 2.0 Specification - <http://scap.nist.gov/specifications/ocil/#resource-2.0>

D.3.12.1 OCIL CPE Implementation

SCAP validation requirement SCAP.V.1800.1 states:

“The vendor SHALL provide instructions on how the product indicates the applicability of the imported SCAP source data stream to a target platform. Instructions SHOULD also describe how the product indicates data streams are not applicable for a target platform. This requirement is testing the use of the OCIL questionnaire associated with a CPE name via the CPE dictionary and the platform id to determine applicability of the data stream.”

SCC allows the OCIL Questionnaire to be answered prior to running the CPE applicability check so that the end user does not have to answer the same questions multiple times if multiple systems are being scanned. This allows SCC to create an OCIL Results file into a temporary directory for each system. After finishing the OCIL Questionnaire and continuing the analysis, if a CPE applicability check is included in the SCAP stream, only the OCIL questionnaires deemed applicable will be included in the final ARF results file.

D.3.13 SWID Tags Implementation

SCAP 1.3 validation requirements state:

"SCAP.V.2850.1: The vendor SHALL provide instructions on how the product identifies SWID tags using OVAL inventory class definitions that are part of an SCAP source data stream.."

and

"SCAP.V.2860.1: The vendor SHALL provide instructions on how the product identifies SWID tags using OVAL inventory class definitions that are part of a standalone OVAL Definition file."

SCC has support for the OVAL independent XMLFileContent test, with which content authors can create SCAP/OVAL content to find and process SWID tag files and report them as OVAL inventory.

SWID 2015 specification: <https://csrc.nist.gov/Projects/Software-Identification-SWID/resources#resource-2015>

D.4 OVAL Probes Supported by SCC 5.12 for Windows

The following OVAL probes are supported in the Windows version of SCC. For probe support on other platforms, please refer to the platform specific documentation for each release of SCC.

- Apache
 - httpd
- Cisco IOS
 - global
 - interface
 - line
 - snmp
 - version
 - version55
- Independent
 - EnvironmentVariable
 - EnvironmentVariable58
 - Family
 - FileHash
 - FileHash58
 - LDAP
 - SQL
 - SQL57
 - SQLEXT (SCC specific OVAL test, was submitted to the OVAL Board for inclusion in OVAL 5.12, but hasn't been approved)
 - Shellcommand
 - TextFileContent
 - TextFileContent54
 - Variable
 - XMLFileContent
- Windows
 - AccessToken
 - ActiveDirectory
 - ActiveDirectory57
 - AppCmd
 - AuditEventPolicy
 - AuditEventPolicySubCategories
 - Cmdlet
 - DnsCache
 - File
 - FileAuditedPermissions
 - FileAuditedPermissions53
 - FileEffectiveRights
 - FileEffectiveRights53
 - Group
 - Group_SID
 - Interface
 - License
 - LockoutPolicy
 - Metabase
 - NTUser
 - PasswordPolicy

- PeHeader
- Port
- PrinterEffectiveRights
- Process
- Process58
- Registry
- RegKeyAuditedPermissions53
- RegKeyEffectiveRights
- RegKeyEffectiveRights53
- Service
- ServiceEffectiveRights
- SharedResource
- SID
- SID_SID
- SystemMetric
- UAC
- User
- User_SID55
- UserSID
- UserRight
- Volume
- WMI
- WMI57
- WuaUpdateSearcher

D.4.1 SQL Database Management System Support

SCC supports reviews against the following SQL database configurations:

DATABASE MANAGEMENT SYSTEM	WINDOWS 2003 AND LATER	SOLARIS	RED HAT ENTERPRISE LINUX	DEBIAN LINUX
Microsoft SQL Server 2000 and Later	Yes			
Oracle Database 10g and 11g, Enterprise Edition		Yes	Yes	Yes
Oracle Database 10g and 11g, Express Edition		Yes	Yes	Yes

Local review capability is available for supported Oracle Database installations while local and remote review capabilities are available for supported Microsoft SQL Server installations.

D.4.2 SCAP Content Author Note on SQL and SQL57 implementation in SCC

SCC can recognize several common representations of the SQL Server and Oracle Database versions it supports. Such representations include chronological (SQL Server: 2005, 2008, 2008 R2; Oracle DB: 10g, 11g), short numerical (SQL Server: 9.0, 10.0; Oracle DB: 10, 11), and long numerical (SQL Server: 9.00.x, 10.00.x, 10.05.x; Oracle DB: 10.1, 11.2.0.x). Declaring multiple versions in a pattern match operation (e.g. "2005|2008", "10g|11g", or ".*") will enable SCC to concurrently analyze instances from all matching and supported versions of SQL Server or Oracle Database installed on the target system.

SCC's handling of the "connection_string" element does not treat it as a literal connection string. Rather, it is treated as a form for specifying which instances and, if reviewing a SQL Server installation, databases on the target system should be inspected. Disregarding the quotation marks, it has one required field, "server=<instance>" where <instance> is a literal instance name or a regular expression, and one optional field, "database=<database>" where <database> is a literal database name or a regular expression. When both fields are declared, they are separated by a semicolon (;). When reviewing a SQL Server installation, declaring the "server" field as "server=MSSQLServer" will enable SCC to submit database queries against the default instance. Omitting the "database" field for a SQL Server review will cause all queries to be submitted against the default database of the specified instance(s). When reviewing an Oracle Database installation, any database declaration in the "connection_string" entity will be ignored since it would not be applicable to the Oracle Database review process. Leveraging the pattern match operation of the "connection_string" element allows SCC to analyze multiple instances and multiple matching databases, where applicable, on each instance with a single SQL or SQL57 OVAL probe.

Due to SCC's dependency upon the Oracle SQL*Plus utility for conducting Oracle Database reviews, any SQL queries specified by Oracle Database specific OVAL probes are limited to a length of 257 characters.

APPENDIX E - REFERENCES & DEFINITIONS

E.1 References

DISA STIG SCAP Benchmarks
<https://www.cyber.mil/stigs/SCAP>

NIWC SCAP Compliance Checker
<https://www.niwcatlantic.navy.mil/scap/>

NIST SCAP Specifications
<http://nvd.nist.gov/scap.cfm>

E.2 Definitions

ACRONYM	DEFINITION
ARF	The Asset Reporting Format (ARF) is a data model to express the transport format of information about assets and the relationships between assets and reports.
CCE	Common Configuration Enumeration CCE™ provides unique identifiers to system configuration issues in order to facilitate fast and accurate correlation of configuration data across multiple information sources and tools. For example, CCE Identifiers can be used to associate checks in configuration assessment tools with statements in configuration best-practice documents and security guides, are the main identifiers used for the settings in the U.S. Federal Desktop Core Configuration (FDCC) data file downloads, and are a key component for enabling security content automation.^[1]
CIS	The Center for Internet Security. Current managers of the open source project which maintains OVAL. [6]
CPE	Common Platform Enumeration CPE™ is a standardized method of describing and identifying classes of applications, operating systems, and hardware devices present among an enterprise's computing assets. CPE can be used as a source of information for enforcing and verifying IT management policies relating to these assets, such as vulnerability, configuration, and remediation policies. IT management tools can collect information about installed products, identify products using their CPE names, and use this standardized information to help make fully or partially automated decisions regarding the assets.^[1]
CVE	Common Vulnerability Enumeration CVE® International in scope and free for public use, CVE is a dictionary of publicly known information security vulnerabilities and exposures. CVE's common identifiers enable data exchange between security products and provide a baseline index point for evaluating coverage of tools and services.^[1]
DISA	Defense Information Systems Agency The Defense Information Systems Agency (DISA) is a United States Department of Defense agency that provides information technology (IT) and communications support to the President, Vice President, Secretary of Defense, the military Services, and the Combatant Commands.^[2] With respect to SCC and SCAP, DISA creates and maintains SCAP content for the DISA STIGS.
MITRE	MITRE is a not-for-profit corporation, chartered to work solely in the public interest. MITRE operates multiple Federally Funded Research and Development Centers (FFRDCs).^[1] With regards to SCAP, MITRE develops and maintains several standards such as CPE, CCE and CVE (and formerly OVAL).
NIST	National Institute of Standards and Technology

	NIST is a United States Government agency responsible for many government standards, including SCAP.
OCIL	Open Checklist Interactive Language The Open Checklist Interactive Language (OCIL) defines a framework for expressing a set of questions to be presented to a user and corresponding procedures to interpret responses to these questions. Although the OCIL specification was developed for use with IT security checklists, the uses of OCIL are by no means confined to IT security. Other possible use cases include research surveys, academic course exams, and instructional walkthroughs.^[3]
OVAL	Open Vulnerability and Assessment Language Open Vulnerability and Assessment Language (OVAL[®]) is an international, information security, community standard to promote open and publicly available security content, and to standardize the transfer of this information across the entire spectrum of security tools and services. OVAL includes a language used to encode system details, and an assortment of content repositories held throughout the community. The language standardizes the three main steps of the assessment process: representing configuration information of systems for testing; analyzing the system for the presence of the specified machine state (vulnerability, configuration, patch state, etc.); and reporting the results of this assessment. The repositories are collections of publicly available and open content that utilize the language.^[1]
SCAP	Security Content Automation Protocol SCAP (pronounced S-CAP) consists of a suite of specifications for standardizing the format and nomenclature by which security software communicates information about software flaws and security configurations.^[3] NIST is responsible for developing standards and guidelines, including minimum requirements, for providing adequate information security for all agency operations and assets; but such standards and guidelines shall not apply to national security systems.^[3]
SCC	SCAP Compliance Checker SCAP Validated Authenticated Configuration Scanner developed by NIWC Atlantic.
NIWC (formerly SPAWAR)	Naval Information Warfare Center NWIC Atlantic is a Department of the Navy organization. We meet our nation's demands for uninterrupted vigilance, fail-safe cybersecurity, adaptive response and engineering excellence by delivering secure, integrated and innovative solutions to many naval, joint and national agencies.^[4]
STIG	Security Technical Implementation Guides The Security Technical Implementation Guides (STIGs) and the NSA Guides are the configuration standards for DOD IA and IA-enabled devices/systems. Since 1998, DISA Field Security Operations (FSO) has played a critical role enhancing the security posture of DoD's security systems by providing the Security Technical Implementation Guides (STIGs). The STIGs contain technical guidance to "lock down" information systems/software that might otherwise be vulnerable to a

	malicious computer attack. ^[5]
SWID	Software Identification (SWID) Tags 2015 revision, a format for representing software identifiers and associated metadata ^[8] [SWID]; Version: ISO/IEC 19770-2:2015 published in October 2015
USGCB	United States Government Configuration Baseline The purpose of the United States Government Configuration Baseline (USGCB) initiative is to create security configuration baselines for Information Technology products widely deployed across the federal agencies. The USGCB baseline evolved from the Federal Desktop Core Configuration mandate. The USGCB is a Federal government-wide initiative that provides guidance to agencies on what should be done to improve and maintain an effective configuration settings focusing primarily on security. ^[3]
XCCDF	The Extensible Configuration Checklist Description Format XCCDF is a specification language for writing security checklists, benchmarks, and related kinds of documents. An XCCDF document represents a structured collection of security configuration rules for some set of target systems. The specification is designed to support information interchange, document generation, organizational and situational tailoring, automated compliance testing, and compliance scoring. The specification also defines a data model and format for storing results of benchmark compliance testing. The intent of XCCDF is to provide a uniform foundation for expression of security checklists, benchmarks, and other configuration guidance, and thereby foster more widespread application of good security practices.
XML	Extensible Markup Language (XML) is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable. ^[2]

[1] - <http://www.mitre.org>

[2] - <http://www.wikipedia.org>

[3] - <http://www.nist.gov>

[4] - <http://www.public.navy.mil>

[5] - <https://www.cyber.mil/stigs>

[6] - <http://cisecurity.org>

APPENDIX F - LICENSES

F.1 End User License Agreement

IN NO EVENT SHALL THE UNITED STATES NAVY (OR GOVERNMENT) OR ANY EMPLOYEES THEREOF BE LIABLE TO ANY PARTY FOR DIRECT, INDIRECT, SPECIAL, INCIDENTAL, OR CONSEQUENTIAL DAMAGES, INCLUDING LOST PROFITS, ARISING OUT OF THE USE OF THIS SOFTWARE AND/ OR ITS DOCUMENTATION, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE, NOR SHALL THE UNITED STATES NAVY (OR GOVERNMENT) OR ANY EMPLOYEES THEREOF ASSUME ANY LEGAL LIABILITY OR RESPONSIBILITY FOR THE ACCURACY, COMPLETENESS, OR USEFULNESS OF THIS SOFTWARE AND/OR ITS DOCUMENTATION.

THE UNITED STATES NAVY (OR GOVERNMENT) SPECIFICALLY DISCLAIMS ANY WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE SOFTWARE AND ACCOMPANYING DOCUMENTATION, IF ANY, PROVIDED HEREUNDER IS PROVIDED "AS IS". THE UNITED STATES NAVY (OR GOVERNMENT) HAS NO OBLIGATION HEREUNDER TO PROVIDE MAINTENANCE, SUPPORT, UPDATES, ENHANCEMENTS, OR MODIFICATIONS. ANY REPRODUCTION OF THIS WORK MUST INCLUDED THE ABOVE NOTICES AND THE FOLLOWING NOTICE: "PORTIONS OF THIS SOFTWARE ARE OFFICIAL WORKS OF THE U.S. GOVERNMENT. THE U.S. GOVERNMENT MAY PUBLISH OR REPRODUCE THIS SOFTWARE, OR ALLOW OTHERS TO DO SO, FOR ANY PURPOSE WHATSOEVER."

FOR MORE INFORMATION CONTACT:
OFFICE OF INTELLECTUAL PROPERTY
NAVAL INFORMATION WARFARE CENTER PACIFIC
SAN DIEGO, CA 92152

APPENDIX G - TECHNICAL SUPPORT & FEEDBACK

Technical support is available for government users and contractors to the federal government.

G.1 Technical Support

- o For assistance with the SCC application (installation, usage, errors, crashes) please email: scc.fct@navy.mil

G.2 Tutorials

There are a series tutorials for SCC which can be viewed at:
<https://www.niwcatlantic.navy.mil/scap/videos/>

G.3 Software Releases

The latest official release information can be obtained from our website:
<https://www.niwcatlantic.navy.mil/scap/>

To be notified via email with updates on SCC, release notifications, customer support surveys, please email: scc.fct@navy.mil

G.3.1 Download Location

DISA maintains the authoritative download of SCC, and starting with SCC 5.4, no longer requires a CAC to obtain: <https://www.cyber.mil/stigs/SCAP>

G.4 Provide Feedback on SCC

We ask for your feedback with an annual survey, but you can submit feedback anytime at the website below.

https://usnavy.gov1.qualtrics.com/jfe/form/SV_4ZpXv8JkUIDs4lw

APPENDIX H - CREDITS AND FUNDING

H.1 Credits and Funding

The development of SCC has been funded by several different agencies over the years:

- o 2008 - 2010 Internal Revenue Service (IRS)
- o 2010 - 2012 National Security Agency (NSA)
- o 2013 - 2022 Defense Information Systems Agency (DISA).
- o 2022 - 2025 A collaboration of two different groups of end users

H.2 SCC Funding for FY26 and Beyond

From FY23 to present, we have been graciously funded by two small groups of end users. Ideally, we would like to have 3 or 4 groups to help increase stability and decrease the financial burden for the contributing groups.

We are looking to find additional agencies/teams to share the costs of our labor (6 government GS-13 developers) along with materials and maintenance for the SCC development/test lab. Becoming a funding sponsor allows your group to influence the future of SCC and ensure it works correctly in your environment.

Please contact our team for more information.

SCC Email	scc.fct@navy.mil
SCC Website	https://www.niwcatlantic.navy.mil/scap/